

Steering a warp drive without exotic matter

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Abstract

A useful warp drive must accelerate, yet every *compact* positive-energy construction is static or constant-velocity; accelerating ones need exotic matter, abandon the dominant energy condition (DEC), or are non-compact (no Bondi four-momentum to balance). We construct a compact positive-energy object driven by photon-rocket recoil on a tidally shielded flat cavity. The principle is Bondi–Sachs balance read as propulsion: any asymptotically flat drive with a confined dominant-energy source changes its Bondi four-momentum only by radiating to null infinity, so it cannot steer without radiating. A Kinnersley photon-rocket exterior with prescribed passenger worldtube fixes the matching thin shell, whose flat interior makes it a warp drive. Bulk energy conditions collapse to $n^2 \geq 0$; steering obeys the closed-form law $-\dot{m} \geq 3m|a|$, paid by Bondi mass loss, and both budget and kinematic ceiling survive a positive Λ to all orders. The main result is existence, the accelerating positive-energy warp-drive solution the literature has lacked: admissible shells exist at small acceleration about a static anchor (surface DEC for $2m/R < 24/25$), and a slow fixed-axis burn assembles into one real-analytic spacetime, its constant- u cuts *characteristic* (shell fixed order-by-order, not Cauchy-evolved); the rotating axis stays open. The acceleration–compactness frontier is bracketed between a rigorous lower bound and the closed-form ceiling. Minimum-radiation steering is the gravitational-wave–silent Damour dipole, its velocity path a hyperbolic geodesic with Tsiolkovsky constant 3, universal in the news-silent class: the 3 is the collimation penalty of dipole exhaust over a floor of 1. The radiating equilibrium is linearly *unstable* (flux anti-damps the radial mode), yet growth $\sim$ luminosity is bounded over any finite burn by the fuel budget. The drive is causal, subluminal, and a *reaction* drive needing no exotic matter in the thin-shell idealization: steering a warp drive is a problem of energy budget, not negative energy.

1 Introduction

Alcubierre’s warp drive [1] (reviewed in [2]) showed that general relativity permits a compact region (a “bubble”) to be transported through an ambient spacetime while its interior remains locally flat. It required exotic matter: the construction violates the null energy condition (NEC), and with it the weak, strong, and dominant conditions. This is not specific to Alcubierre’s ansatz. Santiago, Schuster, and Visser [3] proved that *generic* warp drives, defined through the standard shift-vector ansatz, violate the NEC; Lobo and Visser [4] had earlier quantified the unavoidable exotic content, and Pfenning and Ford [5] showed that the required negative energy is further constrained by quantum inequalities [6].

Two design approaches dominate the field, and both are overconstrained. *Metric-first* designs [1, 7, 8] choose the geometry and let Einstein’s equations determine the matter, which then turns out to be exotic. *Source-first* or static designs [9, 10, 11, 12] prescribe a physically reasonable matter profile and tolerate the geometry that results; these reach positive energy or reduced exotic content, but are *static or constant-velocity*. The positive-energy claims of Lentz [13] and Fell–Heisenberg [14] have not survived scrutiny: for the zero-vorticity Lentz soliton the Eulerian energy density integrates to zero over each slice, so it cannot be everywhere positive [3], and a direct Eulerian-frame computation exhibits the negative regions [15] (Table 1). Bobrick and Martire state the gap directly: warp drives “do not have any natural way of changing their velocities,” and “no metric which describes an accelerating warp drive solution has so far been presented in the literature” [9]. Barzegar, Buchert, and Vigneron [16] (see also [17]) raise a

complementary structural requirement: a physical warp drive must possess *covariant* (not merely coordinate) velocity and acceleration, and a mechanism *derived from matter* rather than reverse-engineered from a chosen metric. Buchert and Frackowiak [18] report an instability of warp fields in that metric-first class (Sec. 13).

The most recent positive-energy constructions sharpen this gap rather than close it. Fuchs *et al.* [10] match an explicit positive-energy shell, but at constant velocity, and name radiation- and momentum-transfer steering as the open direction beyond simple photon rockets; Rodal [11] reduces the exotic requirement to a predominantly positive invariant density at constant velocity, and Garattini and Zatrnyaylov [12] obtain a strictly non-negative energy density for a warp bubble in a de Sitter *universe*, with the weak and null conditions holding only up to a total divergence that averages to zero, and only when the bubble moves radially at the speed of the cosmic expansion, a velocity fixed by the background rather than chosen by the drive. One construction does accelerate positive-energy matter: Huey’s membrane models [19], whose sub-luminal member satisfies the dominant energy condition on the induced surface stress-energy. Its branes, however, are of *infinite* extent, a non-compactness Huey himself identifies as the chief obstacle to realism and as an unsatisfying evasion of the no-go theorems. A non-compact wall is not an isolated system: it carries no Bondi four-momentum, so no momentum balance constrains its acceleration. On the propulsion side, Füzfa [20] already used the exact Kinnersley photon rocket as a model of relativistic interstellar travel, but as a bare accelerating point mass, without an excised cavity, a matched passenger shell, a surface energy-condition guarantee, or a kinematic frontier: we adopt his exterior and add the passenger load it carries. The criterion the present construction meets, and none of these do, is the observer-robust *pointwise* dominant energy condition, in bulk *and* shell and over all observers, on a *compact*, asymptotically flat drive, together with a covariant, matter-derived acceleration (Table 1, Sec. 13).

A second, structural objection is also met: the causal-control obstruction of Krasnikov [21] and Everett and Roman [22], that an Alcubierre bubble cannot be steered from inside because its wall lies outside the ship’s causal future, does not arise here, the shell being a *timelike* worldtube in causal contact with the passenger worldline at every retarded cut (Sec. 13).

In a companion paper [23] we tested whether the source-first route can be pushed to admissibility. Using the observer-robust energy-condition framework of Ref. [24], an exhaustive scan of compact-support shell configurations found none admissible: the dominant-energy failures localize at the smooth source–vacuum transition, persist in the static limit, and appear as Hawking–Ellis [25] Type IV stress-energy in the transition tail. The obstruction is geometric, tied to the transition itself rather than to the warp velocity, and we re-derive this boundary cost in closed form below (Prop. 6). That result reframes the problem: if the obstruction is the transition itself, the design should remove the smooth source–vacuum transition from the interior and abandon the requirement that the exterior be vacuum. The simplest positive-energy replacement for that vacuum is outgoing null radiation.

This paper. We propose a third approach, *worldtube-first*. We prescribe the passenger worldtube (a timelike curve Γ together with a regular cavity) and we accept positive null radiation as the exterior matter we are willing to carry; Einstein’s equations then determine the shell. The exterior is taken to be the exact Kinnersley–Robinson–Trautman photon rocket [26, 27, 28, 29, 30], a known exact solution whose source is pure outgoing null dust. Its energy conditions reduce to a single positivity, $n^2 \geq 0$ (Sec. 5), and its central worldline is covariantly accelerated, by the recoil of anisotropic radiation, exactly as a photon rocket. We excise the worldline singularity, replace $r < R(u, \Omega)$ by a regular cavity, and match across a timelike worldtube using the standard Lanczos junction together with flux-source balance laws [31, 32]: the *timelike*-shell balance laws with an external crossing flux, mandatory here because the exterior radiation is emitted at the shell, the cavity within being vacuum.

The resulting object, a *radiative momentum warpsell*, is causal, subluminal, self-gravitating, and built entirely from matter that satisfies the energy conditions in the bulk. Its acceleration is real and is paid for by Bondi mass loss. The organizing principle is not the metric but a conservation law: no asymptotically flat, positive-energy drive can change its velocity without radiating four-momentum to null infinity, and the construction below is the explicit positive-energy object that realizes it. This formalizes, as a Bondi–Sachs conservation law, the momentum-balance obstruction Bobrick and Martire could state only heuristically [9]: “whatever is the acceleration mechanism, it must obey the conservation of 4-momentum. This is because all warp drive spacetimes are asymptotically-flat.” That is exactly Theorem 1: in the asymptotically flat setting considered here, a confined drive acquires four-momentum only by radiating it, and we exhibit the

positive-energy object for which that cost is exactly $n^2 \geq 0$. The principle forbids reactionless *velocity* change, not Wisdom-type geometric swimming, which produces a finite *position* displacement at fixed Bondi four-momentum (Cor. 1). Two things are new. The balance itself is classical Bondi–Sachs; this paper contributes (a) reading it as a no-reactionless-steering *principle*, and (b) *proving the existence* of a positive-energy spacetime that realizes it, supplying the accelerating warp-drive solution Bobrick and Martire recorded as never having been presented. We summarize the contributions:

1. **A model-independent no-reactionless-steering principle** (Sec. 3, Theorem 1): for *any* asymptotically flat, dominant-energy drive with a confined material source (and Bondi–Sachs peeling), the Bondi four-momentum changes only through radiation to null infinity, so steering forces radiation. The balance is the Bondi–Sachs flux law read as a propulsion no-go; it specializes correctly across three independent exact spacetimes (the accelerating Kinnersley rocket, static Schwarzschild, and Vaidya). The contributions below are its faces in an explicit positive-energy instance that realizes it.
2. **Positivity and the cost of thrust**: the bulk energy conditions of the exterior collapse to the single scalar inequality $n^2 \geq 0$ (Sec. 5), and the local dipole face of the principle is the closed-form control law $-\dot{m} \geq 3m|a|$, tying covariant proper acceleration to Bondi mass loss with ma the thrust (Corollary 4, Sec. 6). The Tsiolkovsky budget survives a positive cosmological constant to all orders in the acceleration, as does the kinematic ceiling, exactly and in closed form, $aR < \frac{1}{2}(1 - x - y)$ with $y = (\Lambda/3)R^2$ (Lemma 2).
3. **Existence of admissible accelerating shells** (the principal result). Per cut, an admissible shell exists for every small $a > 0$ about the static Schwarzschild–Minkowski anchor, whose surface dominant energy condition holds strictly for $2m/R < 24/25$, by an implicit-function argument on the Fredholm-index-zero rigidity operator $\Delta_{S^2} + 2$, with matched surface stress of Hawking–Ellis Type I and a strictly positive observer-robust surface-DEC margin (Theorem 2, App. D). The snapshots assemble into a single *real-analytic* time-evolved worldtube for real-analytic $a(u)$ on a fixed axis (Theorem 4, Sec. 9.2, App. E). The mechanism is that the retarded shape velocity cancels identically from the induced-metric jump, so the constant- u cuts are *characteristic* and the shape is rigidly determined order-by-order rather than Cauchy-evolved; analyticity enters only to majorize the resulting series, Cauchy–Kovalevskaya serving as a convergence majorant and not as an evolution theorem. The rotating axis is left open.
4. An **acceleration–compactness frontier** $a_{\max}R \leq g(2m/R)$ for a positive-energy shell, bracketed between a rigorous lower bound $c(x)$ (Lemma 6) and the closed-form kinematic ceiling $g(x) \leq \frac{1}{2}(1 - x)$ (Prop. 5), with the explicit frozen-shape axial threshold $\lambda_{\text{DEC}}(x)$ locating the leading-order binding directions (Lemma 5, Sec. 8), *proved* (Sturm) to lie strictly below the kinematic ceiling yet come within 0.7% of it near $x \approx 0.63$, so on the axis the frozen shell is bound by the dominant energy condition just short of the regularity ceiling. The frozen rear pole gets *safer* under acceleration, so the matched frontier is set by the global shape response rather than the rear-pole redshift. The thin shell is in turn the limit of admissible finite-thickness matter: a *tangential-pressure* wall is strictly DEC-admissible across its whole width, recovers the anchor for $2m/R < 4/5$, and stays admissible once made radiative, at adiabatic linear order in the flux, whereas a generic *smooth* (radial-tension) thickening reproduces the boundary cost (Sec. 10); the fully time-dependent $\mathcal{O}(\mu^2)$ back-reaction is left open (App. F).
5. **Maneuver design as geometric control**: the exterior, being exactly Kinnersley, radiates no gravitational waves at the Damour dipole, and that news-silence selects the dipole as the minimum-radiation maneuver, *proved* to saturate the control law (Sec. 12); the complementary free-profile optimum is a *hyperbolic geodesic* on $\mathbb{H}^3 = SO^+(3,1)/SO(3)$ with Tsiolkovsky constant 3, universal within the news-silent class (Theorem 6).

The body follows this order. Sec. 2 fixes the worldtube-first definition and Sec. 3 proves the principle; Sec. 4 gives the canonical construction, Sec. 5 establishes its bulk energy conditions, and Sec. 6 derives the Bondi recoil, the control law, and its survival under a positive cosmological constant (Lemma 2). The existence theorems are in Sec. 7, the acceleration–compactness frontier in Sec. 8, the finite maneuver and its time-evolved worldtube in Sec. 9, and the finite-thickness wall and its back-reaction in Sec. 10; Sec. 11 establishes that the radiating equilibrium is linearly unstable, and Sec. 12 the minimum-radiation maneuvers as optimal control. Sec. 13 places the

construction against the literature, Sec. 14 sets out the limitations and scope, and Sec. 15 concludes. The appendices carry the derivations and the Supplement (App. S) the numerics. The construction is subluminal, not reactionless, and astronomically but *positively* costly. Warp propulsion is thus an energy-budget engineering problem, not an exotic-matter impossibility.

Throughout we use $G = c = 1$, signature $(-, +, +, +)$, and $\kappa = 8\pi$, matching Refs. [24, 23].

2 What is a warp drive? A worldtube-first definition

The recurring criticism of warp spacetimes is that their “motion” is a coordinate statement [16, 17]: a shift vector $\beta^i(t, x)$ in an ADM form $ds^2 = -dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt)$ encodes a coordinate velocity that need not correspond to any covariant acceleration of matter. We therefore adopt an operational, gauge-invariant definition.

Definition 1 (Warpshell). A *warpshell* is a spacetime $\mathcal{M} = \mathcal{M}_- \cup_{\Sigma} \mathcal{M}_+$ such that: (i) the interior $\mathcal{M}_-$ is a regular timelike worldtube (the cavity) with bounded tidal tensor along a central worldline Γ ; (ii) Σ is a timelike hypersurface across which the induced metric is continuous (the first junction condition) and the surface stress-energy S_{ab} is well defined; (iii) the exterior $\mathcal{M}_+$ satisfies the energy conditions; and (iv) the central worldline carries a prescribed covariant proper acceleration $a^\mu = u^\nu \nabla_\nu u^\mu$ with gauge-invariant magnitude $\sqrt{a_\mu a^\mu}$. The warpshell is *admissible* if, in addition, S_{ab} satisfies the surface energy conditions.

Definition 1 makes “acceleration” a scalar invariant of a physical worldline, not a coordinate velocity, answering the covariance demand of Refs. [16, 17]. It also inverts the design order demanded there: we will fix Γ and the admissible exterior matter *first*, and let the Einstein equations return S_{ab} . Alcubierre and Natário bubbles are warpshells in this sense only in a degenerate, non-accelerating limit; the constant-velocity positive-energy shells of Refs. [9, 10] realize (i)–(iii) but reach (iv) only in the non-accelerating limit $a^\mu = 0$. Our task is to satisfy all four *with the exterior energy conditions intact*.

In what sense a warp drive. The object we construct is, mechanically, a rocket: it expends mass as radiation, and its passengers feel proper acceleration. It differs from a bare rocket, and belongs to the warp-drive lineage, through the structure of Definition 1: a passenger cavity that is exactly flat, so the interior tidal tensor vanishes and the occupants sit in an accelerated laboratory, joined to an exterior that is positive-energy in every frame. It is the accelerating, positive-energy member of the family whose constant-velocity members are the shells of Refs. [9, 10]: it shares the protected flat interior with the Alcubierre program and adds a covariant, matter-derived change of state of motion. Unlike the Alcubierre bubble it carries *no* local expansion or contraction of space ahead of or behind the cavity (no York-time “warp field”): the cavity is rigidly Rindler-accelerated and the exterior is the photon rocket, so the warp-drive connection is precisely the decoupled, tidally protected passenger laboratory, not a spatial-expansion mechanism.

A positive-energy, accelerating, compact object is not barred by the Santiago–Schuster–Visser no-go [3]: the construction lies outside that theorem’s prescribed-shift hypotheses (Sec. 13.1).

We claim no superluminal travel and no reactionless propulsion (Sec. 14); the drive is strictly subluminal, so the causality pathologies of superluminal warp drives [33] do not arise. The covariant *acceleration* long missing from positive-energy warp drives, and the conservation law that sets its cost, are the subject of Sec. 3 onward.

3 The no-reactionless-steering principle

The organizing principle of the paper is a statement stronger than “this particular construction radiates”: *no* confined, positive-energy drive can change its velocity without radiating four-momentum to null infinity. We state it first, as a model-independent theorem whose proof does not use the construction, and only then build the saturating object: the explicit worldtube-first construction of Sec. 4 below (whose control law (14) (positivity of the null-dust amplitude, Prop. 3) and momentum balance (12) we invoke as needed) is its local, positive-energy *saturating* face. We state the law, identify the one hypothesis the construction must supply, and record how the Kinnersley photon rocket of Sec. 4 saturates it.

Theorem 1 (No reactionless steering, model-independent). *Assume:*

(H1) Asymptotic structure. $(\mathcal{M}, g)$ is asymptotically flat at future null infinity $\mathcal{I}^+$, admitting a Bondi–Sachs expansion with the standard Newman–Penrose peeling behavior [34, 35, 36].

(H2) Confinement. *The material (timelike) source is supported in a worldtube of uniformly bounded spatial diameter (or, more generally, it suffices that it decay faster than the Bondi falloff outside such a worldtube), while outgoing null radiation may reach $\mathcal{I}^+$ (this is the matter-flux channel n^2 below). The hypothesis bounds the source's diameter, not its position: the drive is free to move, and to accelerate, but it may not split, so no massive exhaust separates from it and escapes to timelike infinity carrying momentum that never registers at $\mathcal{I}^+$. This is the one hypothesis that distinguishes a drive from an ordinary rocket ejecting massive propellant, and the construction of Sec. 4 satisfies it exactly.*

(H3) Positivity. *The total stress-energy satisfies the dominant energy condition.*

Then the Bondi four-momentum $P_B^\mu(u)$ evolves only through the four-momentum radiated to $\mathcal{I}^+$,

$$\frac{dP_B^\mu}{du} = -\frac{1}{4\pi} \oint \left[|\mathcal{N}(u, \Omega)|^2 + 4\pi n^2(u, \Omega) \right] \hat{\ell}^\mu(\Omega) d\Omega, \quad (1)$$

with $\mathcal{N}$ the Bondi news¹ and $n^2 \geq 0$ the outgoing matter flux. Consequently, on any retarded-time segment with vanishing news and no matter flux, P_B^μ is constant: the confined drive cannot change its total Bondi four-momentum, and hence, between stationary Bondi frames, its velocity (Prop. 1), without radiating.

Proof. The argument composes three classical ingredients; only the framing is new, and three hypotheses do distinct work. The central no-go, constancy of P_B^μ on a flux-free segment (Corollary 1), is the flux balance (1) evaluated at zero flux and needs *no* sign hypothesis; the dominant energy condition enters only to sign the matter integrand $n^2 \geq 0$ (the news integrand $|\mathcal{N}|^2 \geq 0$ automatically, being a modulus), orienting the radiated budget future-directed; and Bondi-mass positivity enters one level deeper, in the monotone-mass refinement (Corollary 3) alone, the only step conditional for an arbitrary distributional shell. (i) *The four-momentum is well-defined at a cut.* Under peeling the mass aspect $m_B(u, \Omega)$ and the $\ell = 0, 1$ momentum aspect are finite on each cut of $\mathcal{I}^+$, and their lowest two moments define $P_B^\mu(u) = \frac{1}{4\pi} \oint m_B(u, \Omega) \hat{\ell}^\mu(\Omega) d\Omega$, the Dray–Streubel/Geroch–Winicour supermomentum charge, supertranslation-covariant at a fixed cut [37, 38, 39]. (ii) *The flux-balance law.* Equation (1) is the Bondi–van der Burg–Metzner–Sachs four-momentum-loss formula [40, 41], with the gravitational channel $|\mathcal{N}|^2$, whose radiative symplectic flux structure at $\mathcal{I}^+$ is that of Ashtekar–Streubel [42], and the matter channel $4\pi n^2$, with $n^2 := \lim_{r \rightarrow \infty} r^2 T_{uu}$ the coefficient of the outgoing null flux $T_{ab} \sim (n^2/r^2) \ell_a \ell_b$ ($\ell_a = -(\text{du})_a$), non-negative by the dominant energy condition. (The matter reaching $\mathcal{I}^+$ is massless, so its leading outgoing radiative flux $T_{uu} \geq 0$ is along ℓ_a and the single scalar $n^2 \geq 0$ sets both the energy and the momentum moment of the matter flux; any massive component is confined by (H2) to the worldtube of bounded diameter, so it reaches timelike infinity with the drive rather than reaching $\mathcal{I}^+$, and (H2) is precisely what forbids it from separating and carrying momentum away unseen.) Both integrands are non-negative and contracted with the future-directed null $\hat{\ell}^\mu$. (iii) *Positivity bounds the budget.* Constancy of P_B^μ on a flux-free segment is now immediate and uses no sign hypothesis (Corollary 1). Bondi-mass positivity enters one level deeper: with $m_B \geq 0$ (Lemma 1; in closed form $m_B(u) = m(u) > 0$ for the explicit construction) the four-momentum radiable on $[u_0, u_f]$ is bounded by $m_B(u_0)$ (Corollary 3), so one cannot gain momentum by driving the mass negative. Positivity thus closes the negative-Bondi-mass bookkeeping loophole; it does *not*, by itself, constrain the redistribution of momentum among matter, shell, and field at fixed P_B^μ , the supertranslation-sensitive content addressed separately by Proposition 1. $\square$

Corollary 1 (Propulsion no-go). *On any segment with $\mathcal{N} \equiv 0$ and $n^2 \equiv 0$, P_B^μ is constant: a confined positive-energy drive cannot change its velocity (the Bondi-frame ratio P_B^i/P_B^0 between stationary cuts, Prop. 1) without radiating four-momentum to $\mathcal{I}^+$.*

The balance (1) bounds the radiated four-momentum from below by a single inequality, independent of the drive, the exhaust, and the mechanism.

Corollary 2 (Universal rocket bound: the radiated four-momentum is causal). *Under (H1)–(H3) the four-momentum carried away at every cut, $-dP_B^\mu/du$, is a future-directed causal vector, null only for a perfectly collimated emission and zero only for none. Equivalently, in every Bondi frame,*

$$-\frac{dP_B^0}{du} \geq \left| \frac{d\vec{P}_B}{du} \right| : \quad (2)$$

¹ $|\mathcal{N}|^2 = \mathcal{N}\bar{\mathcal{N}}$ is the modulus of the complex (spin-weight -2) Bondi news; only the matter channel n^2 enters the explicit construction below.

no asymptotically flat, dominant-energy drive gains momentum faster than it sheds Bondi energy. In the drive's momentary rest frame, where $P_B^\mu = (m_B, \vec{0})$ and the control law below is read, the Bondi energy is the Bondi mass and (2) states that mass loss dominates momentum gain, $-\dot{m}_B \geq |\dot{\vec{P}}_B|$.

Proof. Write the non-negative integrand of (1) as $I(u, \Omega) := |\mathcal{N}|^2 + 4\pi n^2 \geq 0$, and recall $\hat{\ell}^\mu(\Omega) = (1, \hat{n}(\Omega))$ is future-directed null with $\hat{\ell}^0 = 1$ and $|\hat{\ell}| = 1$. Then $-dP_B^\mu/du = \frac{1}{4\pi} \oint I \hat{\ell}^\mu d\Omega$ is an average of future-directed null vectors with non-negative weights, hence lies in the closed future cone, which is convex: it is future-directed causal, and it is null only if I concentrates on a single direction $\hat{n}$ and zero only if $I \equiv 0$. In components this is $-dP_B^0/du = \frac{1}{4\pi} \oint I d\Omega$ against $|d\vec{P}_B/du| = \frac{1}{4\pi} |\oint I \hat{n} d\Omega| \leq \frac{1}{4\pi} \oint I |\hat{n}| d\Omega = \frac{1}{4\pi} \oint I d\Omega$, by the triangle inequality and $I \geq 0$, which is (2). Causality of $-dP_B^\mu/du$ is frame-independent, so (2) holds in every Bondi frame; the rest-frame reading follows by evaluating it where $\vec{P}_B = 0$. The argument uses only $I \geq 0$, secured by (H3) through its matter part $n^2 \geq 0$ (the news part $|\mathcal{N}|^2 \geq 0$ automatic), and the nullity of $\hat{\ell}^\mu$, which is (H1). $\square$

Equation (2) is the general-relativistic form of the relativistic-rocket inequality, and it sets the scale against which the constants below are to be read. A nonzero emission saturates it only if perfectly collimated; any exhaust distributed over the sphere incurs a collimation factor above this bound. The Tsiolkovsky constant 3 of the control law (14) is that factor for the present drive: the Kinnersley exhaust is a *dipole* on the sphere, of collimation efficiency $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$, so it radiates three times the momentum-optimal mass for a given velocity change. The factor 3 is thus not a free constant of the construction, but the collimation cost of the least-radiating news-free exhaust pattern relative to the bound 1 of (2) (Sec. 12, Theorem 6).

Lemma 1 (Bondi-mass positivity). *Under the confinement and dominant-energy hypotheses of Theorem 1, the drive's mass is non-negative in each case the construction uses: (a) the Bondi mass $m_B(u) = m(u) > 0$ at every cut for the explicit Kinnersley rocket in closed form (Cor. 5); (b) the Bondi mass, for any smooth confined dominant-energy source, by the positive-mass theorem; and (c) the scalar ADM mass $m_{\text{ADM}} \geq 0$ at spatial infinity for the admissible distributional shell of Theorem 2.*

Proof. Case (a) is direct. For (b) the Nester–Witten integrand is a non-negative bulk integral under DEC on a complete asymptotically flat/null slice with no naked inner boundary, established spinorially by the Witten–Nester argument [43] and independently by the Schoen–Yau minimal-surface method [44], and carried to the Bondi mass at $\mathcal{I}^+$ on an asymptotically null slice [45, 46, 47]; the present cavity is regular, so those standard hypotheses hold. For (c) the surface DEC contributes to that integrand a non-negative surface term (Lemma 10, App. B), the strictly-positive Type-I margin of Theorem 2, entering the Sen–Witten identity with the explicit positive coefficient 8π of the distributional Israel junction (App. B). The published low-regularity positive-mass theorem of Lee–LeFloch [48], in the Lipschitz ($W_{\text{loc}}^{1,\infty}$) class to which the C^0 shell belongs and whose corner form is due to Miao and to Shi–Tam [49, 50], then closes the scalar ADM mass, $m_{\text{ADM}} \geq 0$, across the C^0 junction (the Witten spinor lies in $W^{1,2}$, carrying no codimension-one jump, so the junction delta enters the identity linearly against the continuous spinor current, with no δ^2). The stronger statement, that the full ADM *four-momentum* is future-causal, follows by running the same estimate over every asymptotic future-null translation *provided* the shell meets the corner hypotheses of the Lorentzian distributional positive-mass theorem [51]; the verification of that hypothesis, and the Bondi refinement at $\mathcal{I}^+$, we leave open (Sec. 14). $\square$

Open refinement. The Bondi refinement of case (c) at $\mathcal{I}^+$, the existence and decay of the asymptotically null Witten spinor transmitted across the C^0 shell (where the Ludvigsen–Vickers transport argument replaces the elliptic one), is left open (Sec. 14). It is not needed downstream: every use of Lemma 1 in this paper is carried by case (a), where $m_B(u) = m(u) > 0$ holds in closed form at every cut.

Corollary 3 (Monotone mass; bounded radiated momentum). *The energy component of (1) is $\dot{m}_B = -\frac{1}{4\pi} \oint [|\mathcal{N}|^2 + 4\pi n^2] d\Omega \leq 0$; with $m_B \geq 0$ (Lemma 1) the four-momentum radiable on $[u_0, u_f]$ is bounded by $m_B(u_0)$, so momentum cannot be acquired by driving the mass negative. In particular a mass-preserving ($\dot{m}_B \equiv 0$) drive cannot steer: with both integrands of (1) non-negative, $\dot{m}_B = 0$ forces $\mathcal{N} \equiv 0$ and $n^2 \equiv 0$, hence zero thrust, so the would-be reactionless*

dipole $n^2 \propto \cos \vartheta$ (zero monopole, nonzero momentum moment) is exactly what the bulk positivity $n^2 \geq 0$ forbids. The sign $m_B \geq 0$ is Lemma 1: unconditional for the Kinnersley rocket, conditional on the distributional theorem left open in Sec. 14 for an arbitrary shell.

Corollary 4 (Steering cost; reduction to the control law). *Any $\Delta P_B^i \neq 0$ requires a nonzero $\ell = 1$ flux, which strictly costs Bondi mass. For the Kinnersley rocket two facts hold separately. First, the gravitational channel vanishes (the Damour dipole is news-free, $\mathcal{N} \equiv 0$), so the four-momentum loss (1) is carried by the matter channel alone. Second, bulk positivity $n^2(\vartheta) \geq 0$, binding at the forward pole, is exactly the control law $-\dot{m} \geq 3m|a|$ (14), whose saturated integral is the Tsiolkovsky budget (15). The global law (1) thus reduces, in the rocket limit, to the dipole control law.*

Confined sources only. The theorem governs the *total* Bondi four-momentum of a confined source, and the confinement hypothesis gives the no-go its force. A device that expels *unconfined* massive reaction mass can change its payload’s velocity while the total Bondi four-momentum stays fixed, the momentum being carried to timelike infinity i^+ by the propellant rather than to $\mathcal{I}^+$; but that is an ordinary rocket with a material exhaust, not a reactionless drive, and its propellant leaves every bounded region, violating the hypothesis. For a positive-energy source that stays confined the only momentum channel to infinity is the null radiation reaching $\mathcal{I}^+$, so steering forces that radiation.

The remaining hypothesis needing care is the identification of P_B^i with the *kinematic* steering of the passenger worldline.

Proposition 1 (Center of momentum at null infinity). *The integrated momentum change of a confined drive between two stationary Bondi frames is the integrated $\ell = 1$ flux ΔP_B^i of (1), unambiguously and supertranslation-covariantly; the corresponding velocity change is read from the Bondi-frame ratio P_B^i/P_B^0 between those frames. The instantaneous identification of a center-of-mass worldline with the $\ell = 1$ mass-aspect moment is, by contrast, supertranslation-ambiguous in general: the relativistic center of mass has no invariant curved-space analog, the same ambiguity that lets a deformable body “swim” through curved spacetime by cyclic shape change with no momentum exchange [52, 53, 54]. We therefore state the instantaneous COM $\leftrightarrow$ scri identification as a hypothesis (a choice of Bondi rest frame); it is discharged exactly by the present construction, whose exterior is the Kinnersley rocket with the closed-form worldline $P_B^\mu(u) = m(u)v^\mu(u)$ [28], leaving no center-of-mass ambiguity. The two effects are complementary: the no-reactionless-steering principle forbids reactionless velocity change (ΔP_B^i), whereas swimming produces a finite position displacement at fixed P_B^μ .*

Corollary 5 (Unambiguous steering velocity for the Kinnersley rocket). *For the Kinnersley photon-rocket exterior of Sec. 4 the outgoing Robinson–Trautman congruence is shear-free, so its canonical cuts are good cuts ($\sigma^0 = 0$) on which the $\ell = 1$ mass-aspect moment is supertranslation-unambiguous [30] (a choice of this canonical shear-free cut family with the explicit Bonnor four-momentum, not a frame-invariant center of mass), and there the Bondi four-momentum is exactly the Bonnor value $P_B^\mu(u) = m(u)v^\mu(u)$ [28] at every retarded cut (v^μ the unit four-velocity, $m(u) > 0$). This closed-form four-momentum discharges the hypothesis of Proposition 1: the instantaneous steering velocity $v^i(u) = P_B^i(u)/P_B^0(u)$ is read directly off P_B^μ with no center-of-mass choice, so the no-reactionless-steering principle applies to the instantaneous velocity, not merely between stationary Bondi frames. The residual freedom in these good cuts is the four-parameter BMS translation subgroup, under which P_B^μ is invariant (a translation a^μ shifts only the mass-dipole moment, $J^{\mu\nu} \rightarrow J^{\mu\nu} + a^\mu P_B^\nu - a^\nu P_B^\mu$, not P_B^μ itself); P_B^μ transforms as a four-vector under the complementary Lorentz factor, which relates distinct Bondi rest frames. The steering velocity $v^i = P_B^i/P_B^0$ is therefore unambiguous: translation-invariant, and Lorentz-covariant between rest frames. (We claim the unambiguous velocity, not a fully invariant center-of-mass worldline, which the no-go does not use.)*

Relation to the classical Bondi–Sachs balance. The flux-balance law (1) is classical Bondi–Sachs, and the recoil of a source by anisotropic radiation is older still [55, 56]; we claim no novelty for it. The content of Theorem 1 is (i) reading it as a propulsion *no-go* (reactionless steering is forbidden in general relativity, Corollary 1); (ii) closing the negative-mass momentum loophole with Bondi-mass positivity (Corollary 3); (iii) distinguishing it from the center-of-mass swimming effect (Proposition 1); and (iv) exhibiting an *explicit, fully positive-energy* spacetime that *saturates* it. It is the positive-energy complement of Santiago–Schuster–Visser [3], supplying the source-first drive that they forbid metric-first (Sec. 13.1). It supplies the covariant accelerator

that Refs. [16, 17] demand ($dP_B^i/du \neq 0$ is a geometric Bondi-frame momentum flux at null infinity), and places the constant-velocity positive-energy shells [9, 10, 11] as the $dP_B^i/du = 0$ boundary case. The Tsiolkovsky budget (15) is its integrated form, making the construction the general-relativistic counterpart of the relativistic rocket [20]. Geometrically the law is a moment-map/symplectic statement on the Ashtekar–Streubel radiative phase space (Sec. 12.1): the asymptotic-translation Hamiltonian is the radiated four-momentum flux, P_B^μ the cut charge for the same asymptotic translation (linked by the balance law $dP_B^\mu/du = -F^\mu$), and the covariant steering is the frame-invariant projected recoil, which vanishes on the constant-velocity face.

Concretely, the explicit construction realizes the matter channel n^2 of (1) as the outgoing Kinnersley dipole of Sec. 6, whose monopole is the mass loss $-\dot{m}$ and whose $\ell = 1$ moment is the thrust ma ; the following proposition records that this instance saturates the law through the matter channel alone.

Proposition 2 (Photon-rocket instance, saturating the flux law). *Let $\mathcal{M}_+$ be the exact Kinnersley–Robinson–Trautman photon-rocket exterior of Sec. 4, matched to a confined cavity across a timelike Σ . Then the Bondi four-momentum obeys (12), whose monopole and dipole are the moment identities $\oint n^2 d\Omega = -\dot{m}$ and $-\oint n^2 \cos\vartheta d\Omega = ma$ (the thrust, App. B). Consequently (i) the worldline accelerates, with instantaneous-rest-frame thrust $dP_B^i/du = ma \neq 0$ (covariantly $dP_B^\mu/du = \dot{m}v^\mu + ma^\mu$), only if $n^2 \neq 0$, so steering requires radiation; and (ii) if $n^2 \equiv 0$ (the Schwarzschild limit (9)) then $\dot{m} = 0$ and $a = 0$, so P_B^μ is constant and the cavity cannot self-accelerate. The exterior DEC ($n^2 \geq 0$, Prop. 3) makes the radiated momentum future-directed, and the dipole emission has vanishing news, so this instance saturates (1) through the matter channel alone.*

The law can be read directly off the curvature, and its asymptotic hypotheses hold on the explicit spacetimes. The balance (12) closes two independent ways (kinematically from $P_B^\mu = m(u)v^\mu(u)$ and from the integrated radiated flux of the *curvature-derived* n^2) as an identity of the field equations, the bulk dipole (13) delivering the thrust ma (Sec. 6). The law specializes across three exact spacetimes: on the accelerating Kinnersley rocket the radiated flux equals the kinematic recoil and the news vanishes (the Damour dipole is gravitational-wave silent); on static Schwarzschild the flux vanishes; and on a Vaidya [57] monopole (the $G_0 \rightarrow 0$ straight-worldline limit of the same rocket) the energy flux equals $-\dot{m}$ with *zero* radiated momentum, so *isotropic mass loss does not steer*. The peeling hypothesis holds analytically on the algebraically special Kinnersley exterior, with the Damour dipole’s news silence ($\mathcal{N} \equiv 0$) physical rather than a coordinate artifact (App. H). On a saturating maneuver the Bondi mass is monotone and strictly positive with $\dot{m}_B = -F^0$. The no-reactionless-steering law thus holds for every confined, positive-energy drive in an asymptotically flat spacetime with Bondi–Sachs peeling, with the photon rocket as its saturating instance.

4 Canonical construction

4.1 The three regions

The spacetime has the nested structure

$$\mathcal{M} = \mathcal{M}_- \cup_\Sigma \mathcal{M}_+, \quad \Sigma : r = R(u, \Omega), \quad (3)$$

with $\mathcal{M}_-$ a regular cavity (Minkowski, or weakly curved ordinary matter satisfying the energy conditions), Σ a timelike worldtube, and $\mathcal{M}_+$ the photon-rocket exterior (Fig. S1, Supplement).

4.2 The exterior: an exact photon rocket

In retarded coordinates $(u, r, \zeta, \bar{\zeta})$ the Kinnersley–Robinson–Trautman line element is

$$\boxed{ds_+^2 = \frac{2r^2}{P^2} d\zeta d\bar{\zeta} - 2du dr - 2H du^2} \quad (4)$$

$$2H = K - 2r(\ln P)_{,u} - \frac{2m(u)}{r}, \quad K = 2P^2(\ln P)_{,\zeta\bar{\zeta}}, \quad (5)$$

where $m(u)$ is the Bondi mass and the steering is carried by

$$P(u, \zeta, \bar{\zeta}) = A(u) + B(u)\zeta + \bar{B}(u)\bar{\zeta} + C(u)\zeta\bar{\zeta}, \quad (6)$$

normalized so that $AC - B\bar{B} = \frac{1}{2}$ gives the unit-curvature ($K = 1$) round sphere. The functions $(A, B, C)(u)$ encode the instantaneous Lorentz frame of the source [26, 29]; a time-dependent boost

is a covariant acceleration of the central worldline (an arbitrary $K = 1$ conformal factor not of this boost form does not solve the field equations, App. A).

The stress-energy is pure outgoing null dust,

$$T_{ab}^+ = \frac{n^2}{r^2} \ell_a \ell_b, \quad \ell_a = -(\mathrm{d}u)_a \text{ (null)} \quad (7)$$

with the Einstein equations fixing the amplitude

$$4\pi n^2(u, \zeta, \bar{\zeta}) = -\dot{m} + 3m(\ln P)_{,u}. \quad (8)$$

Because ℓ_a is null, every contraction of T_{ab}^+ with a causal vector is proportional to n^2 ; the bulk energy conditions therefore reduce to the single positivity $n^2 \geq 0$, proved in Sec. 5.

4.3 The static limit is Schwarzschild

Setting $m = \text{const}$ and (A, B, C) constant ($K = 1$, $(\ln P)_{,u} = 0$), Eqs. (4)–(5) reduce to

$$\mathrm{d}s_+^2|_{\text{static}} = -\left(1 - \frac{2m}{r}\right)\mathrm{d}u^2 - 2\mathrm{d}u\mathrm{d}r + r^2\mathrm{d}\Omega^2, \quad (9)$$

the Schwarzschild metric in retarded Eddington–Finkelstein form, with r the areal radius and $n^2 = 0$ (vacuum). This is the anchor of the existence theorem (Sec. 7).

4.4 The shell and its surface stress

Across Σ the first junction condition $[h_{ab}] = 0$ (continuity of the induced metric) fixes $R(u, \Omega)$, and the Lanczos equation returns the surface stress

$$S_{ab} = -\frac{1}{8\pi}([K_{ab}] - [K]h_{ab}) \quad (10)$$

supplemented (because the null radiation crosses Σ) by the *timelike*-shell flux-source balance laws derived in Sec. 7.2 (the standard Israel relations with an external crossing flux). The cavity interior is vacuum ($T_{ab}^- = 0$), so the jump $[T_{ab}]$ that sources the shell is carried entirely by the exterior radiation. The drive is specified by the pair (Γ, m) : a steering curve and a Bondi mass budget. Einstein’s equations then fix the radiation (8) that carries it, and the junction returns the shell that confines it.

4.5 Absence of gravitational-wave news

The Kinnersley rocket emits *no* Bondi news: for a purely dipolar photon anisotropy the gravitational-wave amplitude generated by the exhaust exactly cancels the one generated by the accelerated motion of the source [28, 58], and Kinnersley is the *unique* axisymmetric Robinson–Trautman pure-radiation solution free of gravitational radiation [59]. This does not conflict with Theorem 1. The flux in (1) carries *two* channels, and the balance constrains their sum: here the entire four-momentum flux is carried by the matter (Ricci) channel n^2 , the gravitational (Weyl) channel $\mathcal{N}$ vanishing identically [30, 60]. The theorem constrains the total radiated four-momentum, not its gravitational-wave content, so a news-free exterior is not exempt: the drive still sheds mass at the rate (14), and Corollary 2 applies unchanged. That the news-free exhaust is also the least-radiating one is established below (Theorem 5).

5 Bulk energy conditions of the null-dust exterior

Proposition 3 (Bulk admissibility). *The exterior stress-energy (7) satisfies the null, weak, strong, and dominant energy conditions at every point of $\mathcal{M}_+$ if and only if $n^2 \geq 0$, equivalently*

$$-\dot{m} + 3m(\ln P)_{,u} \geq 0. \quad (11)$$

Proof. $T_{ab}^+ = (n^2/r^2)\ell_a\ell_b$ is algebraically rank-one null (Hawking–Ellis Type II, a defective null eigenvector; the Type-I $\leftrightarrow$ II classification subtlety is recorded once in App. A). For any causal v^a , $T_{ab}^+v^av^b = (n^2/r^2)(\ell_av^a)^2$, which is ≥ 0 for all v^a iff $n^2 \geq 0$; this is the WEC (and, restricting to null v^a , the NEC). The flux vector $-T^a{}_bv^b = -(n^2/r^2)(\ell_bv^b)\ell^a$ is proportional to the null ℓ^a ; with $\ell_a = -(\mathrm{d}u)_a$ a future-directed v^a gives $\ell_bv^b < 0$, so for $n^2 \geq 0$ the flux $-(n^2/r^2)(\ell_bv^b)\ell^a$ is a non-negative multiple of the future-null ℓ^a , hence causal and future-directed: the DEC. The trace vanishes ($\ell_a\ell^a = 0$), so the SEC reduces to the WEC. Conversely $n^2 < 0$ makes $T_{ab}^+\ell'^a\ell'^b = (n^2/r^2)(\ell_a\ell'^a)^2 < 0$ for null ℓ' transverse to ℓ , violating the NEC. $\square$

Proposition 3 reduces the bulk energy conditions to a single scalar, established in all frames by the observer-robust Hawking–Ellis criterion (Lemma 9). The choice of *pure* outgoing null dust keeps this positive: an imperfect-fluid reading of the photon rocket instead carries negative energy density on its trailing side, so that only its leading side satisfies the weak and dominant conditions [61], whereas the rank-one null source (7) is admissible at every point where $n^2 \geq 0$. Unlike Alcubierre-class drives, the exterior *satisfies* the NEC identically, one of the two independent reasons the construction lies outside the Santiago–Schuster–Visser hypotheses (Sec. 13).

6 Bondi recoil, the steering control law, and a positive Λ

6.1 Covariant acceleration and Bondi momentum

The exterior (4) is asymptotically flat at null infinity, with Bondi four-momentum $P_B^\mu(u)$ whose monopole is the Bondi mass $m(u)$ (the rest-frame energy P_B^0 ; the full four-momentum is $P_B^\mu = m v^\mu$) and whose evolution is fixed by the radiated null momentum [60],

$$\frac{dP_B^\mu}{du} = - \oint n^2(u, \Omega) \hat{\ell}^\mu(\Omega) d\Omega, \quad (12)$$

$\hat{\ell}^\mu$ the asymptotic null direction (with n^2 normalized as in Eq. (8), so $\oint n^2 d\Omega = -\dot{m}$; this is the matter channel of the model-independent law (1), whose $4\pi n^2$ inside the $-\frac{1}{4\pi}$ prefactor is the same quantity). The spatial part of Eq. (12) is the recoil: the central worldline accelerates because the radiation carries away anisotropic momentum. This is a coordinate-free statement ($dP_B^i/du \neq 0$), so the acceleration in Definition 1(iv) is covariant, answering the coordinate-artifact objection [16, 58]. Equation (12) is the local, dipole form of the conservation law of Sec. 3.

6.2 The axisymmetric control law

Write the steering dipole as $(\ln P)_{,u} = -\alpha(u) \cos \vartheta + (\text{monopole})$, with ϑ measured from the acceleration axis.² With the Kinnersley normalization of Sec. 4 the coefficient α equals the covariant proper-acceleration magnitude, $\sqrt{a_\mu a^\mu} = |\alpha|$, directed along $\vartheta = 0$ (App. A); we set $a = \alpha$ and use the two interchangeably hereafter. Throughout, ϑ is the angle from the proper-acceleration axis *in the momentary rest frame* of Γ at each retarded cut, so (13) and the control law below are per-cut rest-frame statements: at finite rapidity the fixed-frame angular pattern is the aberration of this rest-frame dipole, and “pure dipole” refers to the rest-frame form (App. A). Equation (8) becomes

$$4\pi n^2(u, \vartheta) = -\dot{m} - 3m \alpha(u) \cos \vartheta, \quad (13)$$

peaking at the rear pole $\vartheta = \pi$ (the exhaust, opposite the acceleration) and least at the forward pole (Fig. S1b). Positivity $n^2 \geq 0$ for all $\vartheta \in [0, \pi]$ requires the minimum, attained at the forward pole $\cos \vartheta = +1$, to be non-negative, giving the control law

$$\boxed{-\dot{m} \geq 3m |\alpha|}. \quad (14)$$

For general steering, $-\dot{m} \geq 3m \sup_\Omega [-(\ln P)_{,u}]_+$. The flux (13) is the exact Kinnersley dipole, so (14) holds to all orders in a , not as a small-amplitude truncation.

Proposition 4 (Steering positivity). *For any bounded acceleration history $\alpha(u)$ there exists a positive, monotonically non-increasing Bondi mass $m(u) > 0$ satisfying (14) on any finite interval $[u_0, u_f]$; explicitly any $m(u) = m_0 \exp[-3 \int_{u_0}^u |\alpha| du']$ saturating (14) works, and the retained mass fraction is*

$$\frac{m(u_f)}{m_0} = \exp\left[-3 \int_{u_0}^{u_f} |\alpha(u')| du'\right] \quad (15)$$

(so the fraction radiated away is $1 - m(u_f)/m_0$).

Equation (15) is the relativistic-rocket (Tsiolkovsky) budget, in the Bondi-mass-loss form Füzfa [20] used for radiation-powered interstellar travel: acceleration is paid for by Bondi mass loss. It replaces Alcubierre’s negative-energy requirement with a *positive*-energy inequality. The factor 3 (against the ideal collimated photon rocket’s $e^{-\Delta\eta}$) is the cost of the mandatory broad Kinnersley dipole exhaust $4\pi n^2 = 3ma(1 - \cos \vartheta)$ at saturation, which delivers only 1/3 of its radiated energy as net momentum (the $\langle \cos^2 \vartheta \rangle = 1/3$ projection): a property of the news-silent

²Bulk positivity $n^2 \geq 0$ binds at the forward pole, where the dipole subtracts, while the surface dominant-energy frontier (Sec. 8) binds at the distinct rear pole, where the effective compactness $x + 2ar$ is deepest.

class rather than a fundamental floor (App. G.1). The cost is thus severe (order-unity fractional mass loss for order-unity changes in rapidity), but it is positive and finite (Sec. 14).

The budget survives a positive cosmological constant, where the static energy estimate fails.

Lemma 2 (Steering cost and kinematic ceiling under a positive cosmological constant). *Let $\Lambda = 3H^2 > 0$ (admissible in the Robinson–Trautman class), and take from Ref. [62] the Kastor–Traschen conformal-time charge $Q_c = am$, with a the de Sitter scale factor and Q_c non-negative under the DEC (this normalization and its positivity are cited inputs, not re-derived here). In the exact Robinson–Trautman–de Sitter rocket the retarded coordinate u is proper time on Γ ($g_{uu}|_{r=0} = -1$ on the $m = 0$ background worldline, where the Coulomb term is absent; see proof) and a is the proper acceleration, and the aligned radiation is a pure acceleration-dipole to all orders, so the control law (14) carries no higher-order correction. Write*

$$H_u := \frac{d \ln a}{du} = \gamma H, \quad (16)$$

the expansion rate measured in the source’s own retarded time, with $\gamma = -u_\Gamma \cdot u_{\text{com}}$ the Lorentz factor of Γ relative to the comoving Hubble flow ($\gamma = 1$ for a comoving source). Then, along any burn,

$$\dot{Q}_c = a(\dot{m} + H_u m) \leq am(H_u - 3|a|), \quad (17)$$

so $\dot{Q}_c < 0$ whenever

$$|a| > \frac{1}{3}H_u = \frac{1}{3}\gamma H : \quad (18)$$

a steering burn strictly depletes the positive conformal charge, so the steering cost survives in de Sitter, to all orders in the acceleration. At $a = 0$ the residual $\dot{Q}_c = H_u am > 0$ is the cosmological expansion, not steering. The kinematic ceiling on the acceleration survives as well, and in closed form. The exterior areal gradient of the Robinson–Trautman–de Sitter rocket is

$$g^{rr} = 1 - \frac{2m}{r} - H^2 r^2 + 2ar \cos \vartheta, \quad (19)$$

exactly and to all orders in a , still affine in $\cos \vartheta$, so the areal foliation degrades first at the rear pole exactly, and regularity there ($g^{rr} > 0$) gives

$$aR < \frac{1}{2}(1 - x - y), \quad y := H^2 R^2 = \frac{\Lambda}{3} R^2. \quad (20)$$

Setting $\Lambda = 0$ recovers the flat kinematic ceiling $aR < \frac{1}{2}(1 - x)$ of Prop. 5 below: the cosmological term tightens it by exactly the compactness the de Sitter horizon contributes.

The Lorentz factor in (18) cannot be dropped. The condition $|a| > \frac{1}{3}H$, obtained by setting $\gamma = 1$ in (18), is not sufficient. On the hovering family at fixed de Sitter static radius, where the peculiar velocity is $v = H\rho_0$ and $a = H\gamma v$, that condition reads $\gamma v > \frac{1}{3}$ whereas (18) reads $v > \frac{1}{3}$. At $v = 0.32$ one has $|a|/H = 0.338 > \frac{1}{3}$, yet at DEC saturation $d \ln Q_c / du = \gamma H(1 - 3v) = +0.042 H > 0$, so the charge *increases*. The physical conclusion is unaffected: for any macroscopic maneuver $3|a| \gg \gamma H$ with $H \sim 10^{-33} \text{ eV}$, and the depletion holds by a wide margin unless $\gamma \gtrsim 3|a|/H$, which is never approached.

Proof. The static-time Killing energy is indefinite (its Killing field is spacelike beyond the cosmological horizon). For the exact Robinson–Trautman–de Sitter photon rocket [63] in retarded coordinates, the sole nonvacuum Einstein component is the aligned outgoing radiation $T_{uu} = n^2/r^2$ with, exactly and to all orders in the acceleration,

$$4\pi n^2(\vartheta, u) = -\dot{m}(u) - 3a(u)m(u) \cos \vartheta, \quad (21)$$

in the convention of Eq. (13) (ϑ from the acceleration axis, forward pole $\vartheta = 0$); the exact Robinson–Trautman–de Sitter chart of Ref. [63], from which this is derived, orients its polar axis oppositely ($\vartheta_{\text{RT}} = \pi - \vartheta$), giving the reversed dipole sign $+3am \cos \vartheta_{\text{RT}}$ there. The acceleration enters n^2 only through the $\cos \vartheta$ dipole, which integrates to zero over the cut, so it neither sources the monopole mass aspect nor shifts the Kastor–Traschen conformal charge $Q_c = am$ [62], non-negative by its spinor positivity under the DEC. The dominant energy condition $n^2 \geq 0$ at every ϑ , minimized at the forward pole $\cos \vartheta = \text{sgn } a$ exactly as in Eq. (13), is therefore the *exact* control law $-\dot{m} \geq 3|a|m$, carrying no higher-order coefficient correction. It remains to normalize the time in which the charge is differentiated. At $m = 0$ the chart is exactly de Sitter ($R_{abcd} = H^2(g_{ac}g_{bd} - g_{ad}g_{bc})$ identically, for arbitrary $a(u)$), so Γ is a worldline of the background;

on it $g_{uu}|_{r=0} = -1$ exactly and to all orders, so u is *proper time* along Γ and $\dot{} = d/du = d/d\tau$. The scale factor, however, is a function of *cosmological* time, so $d \ln a/du = (d \ln a/dt)(dt/d\tau) = \gamma H = H_u$ with $\gamma = dt/d\tau$ the Lorentz factor of Γ against the comoving flow; $\gamma = 1$ only for a comoving source. Hence $\dot{Q}_c = a(\dot{m} + H_u m) \leq am(H_u - 3|a|)$, which is (17) to all orders, and the threshold is (18). The same inequality controls the boosted mass-energy $d(\gamma m)/d\tau \leq -2m|a|$, so the depletion is not an artifact of neglecting the kinetic term.

For the ceiling (20), take the Robinson–Trautman–de Sitter rocket in the retarded chart of Ref. [63], $ds^2 = -\Phi du^2 - 2 du dr + 2ar^2 \sin \vartheta du d\vartheta + r^2 d\Omega^2$ with $\Phi = 1 - 2m/r - H^2 r^2 - 2ar \cos \vartheta - a^2 r^2 \sin^2 \vartheta$ (its polar axis oriented as ϑ_{RT} , cf. (21)). Inverting the (u, r, ϑ) block, whose determinant is $-r^2$, gives $g^{rr} = \Phi + g_{u\vartheta}^2/r^2 = \Phi + a^2 r^2 \sin^2 \vartheta$: the two a^2 terms cancel identically and $g^{rr} = 1 - 2m/r - H^2 r^2 - 2ar \cos \vartheta_{RT}$, which is (19) in the convention of (13). Being affine in $\cos \vartheta$ with nonzero coefficient $2ar$, it attains its angular minimum at the rear pole exactly, at every a , and $g^{rr} > 0$ there is (20). Setting $H = 0$ recovers the flat exterior gradient and its ceiling (Prop. 5). $\square$

7 Existence of admissible accelerating shells

The surface result is that admissible accelerating shells exist: the matched shell satisfies the *surface* energy conditions while the worldline accelerates. We prove it by deforming a strictly admissible static anchor.

7.1 The static anchor

Match a flat interior to the Schwarzschild exterior (9) across a static shell at areal radius R , with $x := 2m/R \in (0, 1)$. The extrinsic-curvature jump [32] gives the standard surface density and pressure

$$\sigma_0 = \frac{1}{4\pi R} (1 - \sqrt{1-x}), \quad (22)$$

$$p_0 = \frac{1}{8\pi R} \left(\frac{1-x/2}{\sqrt{1-x}} - 1 \right). \quad (23)$$

Both are strictly positive on $(0, 1)$, so the WEC ($\sigma_0 \geq 0, \sigma_0 + p_0 \geq 0$) holds throughout. The DEC requires in addition $\sigma_0 \geq |p_0| = p_0$. Writing $s := \sqrt{1-x}$,

$$8\pi R (\sigma_0 - p_0) s = -\frac{5}{2}s^2 + 3s - \frac{1}{2} = -\frac{1}{2}(5s-1)(s-1), \quad (24)$$

which is positive iff $1/5 < s < 1$, i.e.

$$\boxed{\text{surface DEC} \iff x = \frac{2m}{R} < \frac{24}{25}}. \quad (25)$$

Lemma 3 (Strictly admissible anchor). *For every $x \in (0, 24/25)$ the static Schwarzschild–Minkowski shell satisfies the surface WEC and DEC strictly. On any compact subinterval $[x_1, x_2] \subset (0, 24/25)$ the worst-case margin $\delta_0 := \min_{[x_1, x_2]} (\sigma_0 - p_0) > 0$ is bounded away from zero.*

The window (25) is wide: a flat-interior shell tolerates compactness almost up to $x = 1$.³ This $24/25$ window is the flat-interior thin-shell case of the sharp anisotropic Buchdahl-type bounds of Andréasson [64, 65] and Horvat–Ilić [66]: a purely tangential thin shell ($p_r = 0$, surface DEC $p_t \leq \rho$) is the $\Omega = 2$ member $\sup 2m/R = 24/25$ of the ladder $((2\Omega + 1)^2 - 1)/(2\Omega + 1)^2$ under the anisotropic bound $p_r + 2p_t \leq \Omega\rho$. We center the deformation at intermediate compactness, where the anchor margin δ_0 is large. This is precisely the regime that the smooth source-first shells of Ref. [23] could not reach: there the dominant-energy failure lived in the radial gradients of a finite-width transition (Hawking–Ellis Type IV in the tail), whereas the distributional junction concentrates the transition into S_{ab} over a *non-vacuum, positive-radiation* exterior. Section 7.4 makes the reconciliation explicit.

³Four compactness thresholds govern distinct objects and must not be conflated. The thin-shell surface DEC holds for $x = 2m/R < 24/25$ (the anchor, this section); a finite-thickness *tangential-pressure* wall is admissible for local $2m/r < 4/5$ (Sec. 10); the stiffened radiating wall stays subluminal for $x < x_\eta \approx 0.844$ (App. 1); and the kinematic ceiling $\frac{1}{2}(1-x)$ (Prop. 5) caps the acceleration *amplitude* $\lambda = aR$, not the compactness.

7.2 The junction and flux-source balance laws

Because the null dust (7) streams radially outward, its projections on the timelike Σ (unit outward normal N^a , intrinsic triad $\{e_{\hat{a}}\}$) are

$$T_{NN}^+ = \frac{n^2}{r^2}(\ell_a N^a)^2 \geq 0, \quad T_{N\hat{a}}^+ = \frac{n^2}{r^2}(\ell_b N^b)(\ell_c e_{\hat{a}}^c), \quad (26)$$

whose energy-flux component $T_{N\hat{0}}^+ \neq 0$ (the radially streaming dust carrying no lateral flux, $T_{N\hat{\vartheta}}^+ = T_{N\hat{\varphi}}^+ = 0$), so matter crosses the shell and the static Israel relation must be supplemented by the flux-source balance laws [31, 32] (the standard *timelike*-shell junction with an external crossing flux; *not* the lightlike Barrabès–Israel limit [67, 68], since Σ is timelike)

$$D_b S_{\hat{a}}^b = -[T_{N\hat{a}}], \quad (27)$$

$$S^{ab} K_{ab}^- = -[T_{NN}], \quad K_{ab}^- := \frac{1}{2}(K_{ab}^+ + K_{ab}^-). \quad (28)$$

Physically the shell is a *radiating* layer, the regularized rocket nozzle: the construction's constitutive hypothesis is that this surface layer supplies the prescribed outgoing null flux at Σ (the radiation Kinnersley's solution emits from the now-excised worldline singularity) while carrying the admissible S_{ab} , and the recoil (27) is the thrust. The budget the flux carries is fixed at null infinity, where the amplitude n^2 of (13) is the Bondi flux density itself,

$$\oint n^2 d\Omega = -\dot{m}, \quad -\oint n^2 \cos \vartheta d\Omega = m a : \quad (29)$$

the monopole of the radiated flux is the luminosity $-\dot{m}$ and its $\cos \vartheta$ -moment, the axial momentum flux, is the thrust $m a$, which is the Bondi budget (12). These are the moments used throughout, derived in Appendix B. At the shell itself the *same* flux is seen through the local projection: the radially streaming null dust carries no lateral component ($T_{N\hat{\vartheta}}^+ = 0$) but its normal energy flux $T_{N\hat{0}}^+$ carries angle-dependent normal and redshift factors (through the exact lapse (41)), so the finite-radius moments of $T_{N\hat{0}}^+$ are *not* the moments of n^2 and mix monopole with dipole at $\mathcal{O}(\lambda)$. Nothing needs to be imposed to reconcile them: the balance laws (27)–(28) hold *identically* once $[h_{ab}] = 0$ (they are the contracted Gauss–Codazzi identities, App. B), so the crossing flux and the Bondi budget agree automatically, with the projection factors carried by the shell's own S_{ab} rather than by a separate consistency condition.

7.3 Existence by openness

Introduce the dimensionless deformation parameters

$$\lambda := |a|R, \quad \mu := |\dot{m}|R/m, \quad (30)$$

and expand the surface stress about the anchor,

$$S_{ab} = S_{ab}^{(0)} + \lambda S_{ab}^{(a)} + \mu S_{ab}^{(m)} + \mathcal{O}(\lambda^2, \mu^2, \lambda\mu), \quad (31)$$

where $S_{ab}^{(a)}$ carries the dipole ($\ell = 1$) steering response (including the induced surface energy flux $S_{\hat{0}\hat{\vartheta}}$ from (27)), and $S_{ab}^{(m)}$ the monopole ($\ell = 0$) luminosity/tension shift.

Lemma 4 (Darmois solvability at the anchor). *At the static anchor the geometry is exactly Schwarzschild–Minkowski and the shell's induced 2-metric is the round sphere, so the linearized first-junction (Darmois) map DF_0 acts in its $\ell \geq 2$ shape sector as the round-sphere rigidity operator $\Delta_{S^2} + 2$, whose kernel is the $\ell = 1$ rigid translations. Off that kernel DF_0 is a bounded isomorphism in the $\ell \geq 2$ shape and $\ell = 0$ monopole blocks and is surjective onto the $\mathcal{O}(a)$ steering source in $\ell = 1$. Hence the first junction is locally solvable at the anchor for every small $a > 0$.*

Proof, the full argument in App. D. The $\ell = 1$ kernel is the rigid translations (Cohn–Vossen/Weyl rigidity [69]); off it the operative bound is the *Schauder* estimate $\|u\|_{C^{2,\alpha}} \leq C_S \|(\Delta_{S^2} + 2)u\|_{C^{0,\alpha}}$ (App. D), which delivers the C^2 shell embedding and hence the continuity of $[K_{ab}]$ and S_{ab} (the cruder L^2 inverse $\|(\Delta_{S^2} + 2)^{-1}\|_{L^2} = \frac{1}{2}$, smallest off-kernel eigenvalue magnitude 2, controls only the spectral gap that sets C_S). The $\ell = 0$ monopole block is invertible in closed form through the redshift Jacobian $\partial_R[\sqrt{f}] = m/(R^2\sqrt{1-x}) \neq 0$. In the $\ell = 1$ steering sector the foliation-tilt row $(\partial[h_{\tau\vartheta}]^{\text{sin}}/\partial w_1 = 1)$ and the closed-form $\cos \vartheta$ redshift row (77) are independent for every $x \in (0, 1)$, spanning the two-dimensional source $([h_{\tau\tau}]^{\text{cos}}, [h_{\tau\vartheta}]^{\text{sin}})$, while the constant part of $[h_{\tau\tau}]$

carries *no* $\mathcal{O}(a)$ source (a $\cos \vartheta$ shape feeds the $\ell = 0$ jump only through the $\langle \cos^2 \vartheta \rangle$ self-coupling, hence at $\mathcal{O}(a^2)$). Exact solvability for $a > 0$ then follows with no residual scalar obstruction: the $\mathcal{O}(a^2)$ $\ell = 0$ constant jump is absorbed by the radial breathing mode ρ_0 , whose derivative $\partial_{\rho_0}[h_{\tau\tau}]^{\ell=0} = -x/s^2 \neq 0$ (80) carries no one-dimensional cokernel (App. D). $\square$

Remark 1 (The shape dipole ρ_1 is pure gauge). Consider the anchor sphere $r = R_0$ in the flat interior, with induced metric h_{ab} , outward unit normal $\hat{n}$, and extrinsic curvature $K_{ab} = h_{ab}/R_0$. Translate it rigidly by a constant vector $\mathbf{a}$. The displacement field decomposes into a normal and a tangential part,

$$\mathbf{a} = Y \hat{n} + V, \quad Y = \mathbf{a} \cdot \hat{n}, \quad V^A = h_{ab}^{AB}(\mathbf{a})_B, \quad (32)$$

and $Y = \mathbf{a} \cdot \hat{n}$ is precisely an $\ell = 1$ harmonic. A translation is an isometry of the flat interior and carries the sphere to a congruent sphere, so the induced metric is unchanged. The first variation of h_{ab} under a displacement with normal part Y and tangential part V is $\delta h_{ab} = 2Y K_{ab} + (\mathcal{L}_V \hat{h})_{ab}$, whence

$$2Y K_{ab} + (\mathcal{L}_V \hat{h})_{ab} = 0 \quad \text{for every } \ell = 1 \text{ harmonic } Y. \quad (33)$$

An $\ell = 1$ normal displacement is therefore not a deformation of the shell but a rigid translation of it, its metric effect canceled by the accompanying tangential reparametrization; equivalently, the $\ell = 1$ normal modes of the round sphere are Killing-generated (Cohn–Vossen and Weyl rigidity [69], and consistently $(\Delta_{S^2} + 2)Y_1 = 0$). The angular block accordingly carries *no* $\ell = 1$ shape content, and the shape dipole ρ_1 is not determined by the junction: it is the residual rigid-translation gauge, the shell’s *center*, fixed by the convention that the interior embedding be centered on Γ .

The tangential part V is not optional. If the two sides are matched by rigidly identifying their polar angles, that is, by setting $V \equiv 0$, then the cancellation (33) is destroyed by the choice of parametrization, and the surviving $\ell = 1$ row of the induced-metric jump determines ρ_1 algebraically. That determination is spurious. Any value so obtained is an artifact of the frozen parametrization rather than a physical prediction, and different rigid identifications yield mutually inconsistent values. The tangential reparametrization is part of the matching data, and once it is retained no $\ell = 1$ obstruction arises. Nothing downstream depends on ρ_1 : the Fredholm index, the existence theorem, the kinematic ceiling, and the frontier bracket are all statements off the $\ell = 1$ gauge.

Theorem 2 (Existence of admissible accelerating warps). *Fix $x_* \in (0, 24/25)$ with anchor margin $\delta_0 > 0$. By Lemma 4 the first junction is solvable for $a > 0$: its shape (angular-defect) sector linearizes to the round-sphere rigidity operator $\Delta_{S^2} + 2$, elliptic and self-adjoint, whose kernel is exactly the three $\ell = 1$ rigid translations of the embedding, while the full linearized first junction is block-diagonal in spherical harmonics with each block square and of nonzero determinant (App. D), hence Fredholm of index zero; because a rigid translation is an isometry of the flat interior, the angular defect $[h_{AB}]$ vanishes identically on that kernel, so the lone candidate cokernel carries no solvability condition and the map is invertible on the $\ell \neq 1$ complement. The matched surface stress $S_{ab}(\lambda, \mu)$ is C^1 in (λ, μ) , uniformly in $\Omega \in S^2$. Then there exist $\epsilon_a, \epsilon_{\dot{m}} > 0$ such that for all $\lambda < \epsilon_a$, $\mu < \epsilon_{\dot{m}}$ satisfying the bulk law (14), the matched warps is admissible: its surface stress S_{ab} is Hawking–Ellis Type I and satisfies the surface WEC and DEC strictly. A sufficient explicit condition is*

$$C_a \lambda + C_{\dot{m}} \mu < \delta_0, \quad C_a := \|S_{ab}^{(a)}\|, \quad C_{\dot{m}} := \|S_{ab}^{(\dot{m})}\|. \quad (34)$$

The parameters are not independent: the bulk law (14) forces $\mu = |\dot{m}|R/m \geq 3\lambda$, so the binding case is the minimum-fuel slice $\mu = 3\lambda$, for which (34) reads $(C_a + 3C_{\dot{m}})\lambda < \delta_0$.

Proof, the full argument in App. D. Admissibility is decided by the worst-observer surface dominant-energy margin, the Hawking–Ellis Type-I eigenvalue margin (which for a Type-I stress also signs the WEC),

$$\Phi(\lambda, \mu) := \min_{\Omega \in S^2} \left(\tilde{\sigma} - \max\{|\tilde{p}_{\hat{\vartheta}}|, |p_{\hat{\varphi}}|\} \right), \quad (35)$$

built from the Type-I eigenvalues of the $(\hat{\sigma}, \hat{\vartheta})$ stress block, $\tilde{\sigma} = \frac{1}{2}[(\sigma - p_{\hat{\vartheta}}) + \sqrt{\Delta_s}]$ and $\tilde{p}_{\hat{\vartheta}} = \frac{1}{2}[(p_{\hat{\vartheta}} - \sigma) + \sqrt{\Delta_s}]$, with discriminant $\Delta_s = (\sigma + p_{\hat{\vartheta}})^2 - 4q^2 \geq 0$ (the stress is Type I iff $\Delta_s \geq 0$); the recoil flux $q = S_{\hat{\theta}\hat{\vartheta}}$ enters only through this boosted block and vanishes at the binding rear pole $\vartheta \rightarrow \pi$, where $\Phi = \sigma - p$. The extrinsic curvatures are algebraic-plus-first-derivative functionals of the smooth metric data $m(u)$ and $P(u, \zeta, \bar{\zeta})$, and the balance laws (27)–(28) are linear in the smooth crossing flux; hence $S_{ab}(\lambda, \mu)$ is C^1 near the anchor (no denominator $\sqrt{1 - x_*}$

vanishes). The maps $S_{ab} \mapsto \{\sigma, \sigma \pm p_A, \dots\}$ are continuous, and $\min$ over the compact S^2 preserves continuity, so Φ is continuous. $\Phi(0, 0) = \delta_0 > 0$ by Lemma 3 (at the anchor all flux terms vanish). Continuity gives an open neighborhood on which $\Phi > \frac{1}{2}\delta_0 > 0$; the bound (34) follows from the linear Lipschitz estimate $\|S_{ab} - S_{ab}^{(0)}\| \leq C_a \lambda + C_m \mu + \mathcal{O}(\lambda^2, \mu^2, \lambda\mu)$. By Lemma 4, $[h_{ab}] = 0$ is solvable for the interior / R at fixed (λ, μ) , the $\ell = 1$ steering rows nondegenerate (cos-row (77)) and the $\ell = 0$ monopole absorbed by the breathing mode (80), and the implicit function theorem returns the matched shell (Proposition 8, App. D). The same anchor solvability underlies the dynamical Theorem 4. $\square$

The closure is not tied to the axis. The anchor is rotationally invariant and the $\ell = 1$ dipole block transforms in the vector representation of $SO(3)$, so the same argument runs for an arbitrary steering direction.

Corollary 6 (Per-cut existence for any acceleration axis). *Fix $x_* \in (0, 24/25)$. At each retarded cut, in the momentary rest frame, a locally unique admissible accelerating shell exists for an arbitrary small proper-acceleration three-vector $a^\dagger$, not only an axisymmetric one: the $\ell = 1$ closure of Theorem 2 holds for every direction by $SO(3)$ -equivariance of the rotationally invariant anchor (App. D).*

This permits the three-dimensional treatment of the maneuver problem in Sec. 12: only the *assembly* of the per-cut shells into a single worldtube is restricted to a fixed axis (Sec. 9.2), never the per-cut existence.

Theorem 2 is the existence half of the paper's claim: *accelerating, positive-energy, energy-condition-respecting warpsells exist*. With Lemma 4 it is unconditional *in its hypotheses* on the small-amplitude window: the closure, carried by the nondegenerate cos-row (77) and the breathing-mode-absorbed monopole (80), leaves no residual solvability condition to assume. What is not explicit is the *sharp size* of that window, bracketed below by a rigorous positive lower bound (Lemma 6) and above by the closed-form kinematic ceiling (Prop. 5), and mapped numerically in the Supplement (Sec. S).

7.4 Reconciliation with the boundary-cost theorem

Reference [23] found *no* admissible smooth shells, the dominant-energy failure localizing at the source–vacuum transition (the closed-form radial-tension obstruction of Prop. 6 below). This does not contradict Theorem 2: the radiative warpsell removes both ingredients, replacing the finite-width matter-into-vacuum tail by a measure-zero distributional junction into a *non-vacuum* positive-radiation exterior, so the obstructing geometry is replaced by an admissible one rather than hidden.

8 The acceleration–compactness frontier

How fast can a positive-energy shell accelerate before DEC fails? Decompose the deformed surface stress in spherical harmonics about the anchor. The monopole ($\ell = 0$) is the luminosity/tension shift; the dipole ($\ell = 1$) is the steering, which produces a front–back asymmetry in σ, p and an equatorial surface flux $q_\vartheta = \lambda q_1 \sin \vartheta + \mathcal{O}(\lambda^2)$. The flux-inclusive Type-I surface-DEC margin is the boosted eigenvalue form (35), equal to $\sigma - p$ at the flux-free rear pole and implied by the conservative $\sigma \geq \sqrt{p^2 + q^2}$; the recoil flux $q = \mathcal{O}(\lambda)$ enters it only through the Type-I discriminant $\Delta_s = (\sigma + p)^2 - 4q^2$, hence at $\mathcal{O}(q^2) = \mathcal{O}(\lambda^2)$, and vanishes at the anchor, so the binding angular value to linear order is set by the cos-dipoles alone,

$$\Phi_{\text{DEC}}(\Omega) \geq (\sigma_0 - p_0) - \lambda(|\sigma_1| + |p_1|) + \mathcal{O}(\lambda^2), \quad (36)$$

and $\Phi_{\text{DEC}} \geq 0$ yields a frontier

$$a_{\text{max}} R \leq g(x), \quad (37)$$

which *defines* $g(x)$ as the largest $\lambda = aR$ admitting a *regular* admissible matched shell, i.e. the lesser of the λ at which the observer-robust worst-observer surface DEC margin first vanishes (App. C) and the λ at which the exterior areal foliation degenerates ($f - 2aR \rightarrow 0$, Prop. 5); the latter caps $g(x)$ below the kinematic ceiling $\frac{1}{2}(1 - x)$ *by construction*, since admissibility of a regular shell presupposes a regular foliation. A naive L^1 estimate $g \simeq (\sigma_0 - p_0)/(|\sigma_1| + |p_1|)$ *over-counts* the response (it sums the dipoles as if co-binding, whereas the matched DEC binds at the rear pole) and is superseded by the margin; the recoil flux q_1 peaks at the equator and enters the margin only at $\mathcal{O}(q^2)$, not linearly. Two facts then control g (App. C). The linear surface response is *locally Schwarzschild*, $\sigma_1 = \mathcal{O}(x^2)$, and the recoil flux does not enter the margin at

linear order (it enters only as q^2 , whose λ -derivative vanishes at the anchor); the $\mathcal{O}(\lambda)$ maneuver envelope is therefore *wide* and shape-dominated, scaling as $\mathcal{O}(1/x)$ at small compactness. The operative frontier is thereby trapped between a rigorous lower bound $c(x)$ (Lemma 6) and the closed-form kinematic ceiling (Prop. 5),

$$\boxed{c(x) \leq g(x) \leq \frac{1}{2}(1-x)}. \quad (38)$$

The value $g(x)$ itself has no known elementary closed form; the numerical sharpening $g(x) \sim 0.2$ that locates it inside the band is deferred to the Supplement (Sec. S), where Fig. S2 plots the bracket alongside the heuristic envelope, the thick-wall realizability bound, and a representative saturating burn.

The upper inequality is non-strict. Every *admissible* amplitude satisfies $\lambda < \frac{1}{2}(1-x)$ *strictly* (Prop. 5): at equality the exterior areal foliation degenerates, the shell worldtube becomes null at the rear pole, and the junction data cease to exist. The ceiling is therefore an open condition, attained by no admissible amplitude, and $g(x)$, being the *supremum* of admissible amplitudes, satisfies only $g(x) \leq \frac{1}{2}(1-x)$. Whether that supremum reaches the ceiling is a finite-amplitude question the perturbative matching does not settle. One route to strictness fails. If the rear-pole surface margin were driven to $-\infty$ by the diverging redshift as $h = f - 2aR \rightarrow 0$, the DEC would fail strictly below the ceiling; and $[K^{\hat{\tau}}_{\hat{\tau}}]$ does diverge as $h^{-1/2}$ there, independently of the shape. The shape, however, enters $[K^{\hat{\vartheta}}_{\hat{\vartheta}}]$ at the rear pole with the *same* $h^{-1/2}$ weight and an unrestricted sign, so it can cancel the divergence rather than merely rescale it. A dichotomy in the single dimensionless number $w := R''(\pi)/R$ then fixes the rear-pole condition: the rear-pole DEC survives $\lambda \rightarrow \frac{1}{2}(1-x)$ precisely when $w \geq w_*(x) := \max\{(2x-1)/6, (1-2x)/2\}$. The *frozen* shape has $w = 0 < w_*(x)$ for every $x \neq \frac{1}{2}$, so the frozen frontier does lie strictly below the ceiling; whether the *Darmois-matched* shape attains w_* at finite amplitude is open, and we state the bracket with $\leq$ accordingly. As a heuristic comparison, treating the *rigid-shape* rear pole as a static shell at effective compactness $x_{\text{eff}} = x + 2ar$ reaching the surface-DEC threshold $x_{\text{eff}} = 24/25$ gives

$$g(x) \lesssim g_{\text{env}}(x) := \frac{1}{2}\left(\frac{24}{25} - x\right), \quad (39)$$

empirical and not a rigorous inequality, tighter than the proven *ceiling* $\frac{1}{2}(1-x)$ but respected by the computed values. The two rigorous bounds are established next, the shape-independent closed-form ceiling first.

Proposition 5 (Rigorous kinematic ceiling). *A regular exterior areal foliation at the shell (spacelike areal-radius gradient) requires the rear pole to stay outside its acceleration-induced effective horizon, $f - 2aR > 0$. This gives a shape-independent kinematic ceiling*

$$a_{\text{max}}R < \frac{1}{2}(1-x), \quad x_{\text{eff}} = x + 2aR < 1, \quad (40)$$

with R the rear-pole areal radius, entering both $x = 2m/R$ and $\lambda = aR$. It is a regularity ceiling of the Kinnersley exterior foliation, exact and shape-independent (a property of the exterior geometry, not of S_{ab}), not a universal dominant-energy frontier. Every admissible amplitude obeys (40) strictly, so the operative DEC frontier obeys $g(x) \leq \frac{1}{2}(1-x)$; whether that supremum reaches the ceiling at finite amplitude is open (Sec. 8).

Proof. The $K = 1$ normalization makes $(\ln P)_{,u} = -a \cos \vartheta$ exactly, so the inverse metric is exactly

$$g^{rr} = 2H = f + 2ar \cos \vartheta \quad (41)$$

at every angle, with no $\mathcal{O}(a^2)$ remainder. Being affine in $\cos \vartheta$ with positive coefficient $2ar$, g^{rr} attains its angular minimum at the rear pole exactly: exterior regularity is pole-dominated at all angles, not merely to leading order. Coordinate and areal radius coincide on the axis, so the binding value is $f - 2aR$ at the rear-pole areal radius, and positivity of g^{rr} there is (40) (Eq. (60)). The ceiling is strictly tighter than the interior Born-rigid (Rindler-horizon) bound $\lambda < 1$ (Sec. 14), and the rigorous lower bound lies strictly inside it, the gap closing as $x \rightarrow 1$. $\square$

Lemma 5 (Frozen-shape axial surface dominant-energy threshold). *Match the flat, rigidly accelerated (Rindler) interior to the Kinnersley exterior across a rigid (spherical, shape-frozen) shell at $r = R$. The interior lapse $N = 1 + ar \cos \vartheta$ carries a nonzero pole redshift, $RK^{\hat{\tau}}_{\hat{\tau}} = \mp \lambda/(1 \mp \lambda)$, so the worst-observer surface-DEC margin at the two poles is*

$$8\pi R s_{\pm}(\sigma - p) = 3s_{\pm} - 3 + \frac{5}{2}x \pm 7\lambda \mp \frac{s_{\pm}\lambda}{1 \mp \lambda}, \quad s_{\pm} = \sqrt{1 - x \mp 2\lambda}, \quad (42)$$

$\lambda = aR$ (upper sign at the rear pole), the last term the Rindler interior redshift. The surface DEC holds strictly for $\lambda < \lambda_{\text{DEC}}(x)$, where $\lambda_{\text{DEC}}(x)$ denotes the exact threshold, the smaller of the two pole thresholds (the smallest positive margin-zero of (42)); it has no elementary closed form. Dropping the $\mathcal{O}(\lambda)$ interior term gives its static-interior estimate, which does,

$$\tilde{\lambda}_{\text{DEC}}(x) := \frac{6\sqrt{16-14x} - |24-35x|}{98} \approx \lambda_{\text{DEC}}(x), \quad (43)$$

a distinct object from λ_{DEC} , and it is the exact threshold that is used throughout. Both are strictly positive on the open interval $(0, \frac{24}{25})$, vanish at both endpoints, and lie strictly below the kinematic ceiling $\frac{1}{2}(1-x)$ (40). At small compactness both decay linearly, and the exact threshold does so with a closed-form slope,

$$\lambda_{\text{DEC}}(x) = \frac{1}{3}x + \mathcal{O}(x^2), \quad \tilde{\lambda}_{\text{DEC}}(x) = \frac{1}{4}x + \mathcal{O}(x^2), \quad \frac{\tilde{\lambda}_{\text{DEC}}}{\lambda_{\text{DEC}}} \longrightarrow \frac{3}{4} \quad (x \rightarrow 0), \quad (44)$$

so the static-interior approximation costs only the coefficient: it understates the front-binding threshold by a factor approaching $\frac{3}{4}$. The decay itself is a property of the frozen shape, and it is the matched frontier $g(x)$ that does not share it: the matched linear dipole slope is $\sigma_1 - p_1 = \mathcal{O}(x^2)$ against an anchor margin $M_0 = \mathcal{O}(x)$ (App. C), so the Darmois shape response removes the rigid front-pole penalty and the linear envelope $M_0/A \sim 1/x$ inherits no such decay, the numerically mapped $\underline{g}$ showing none down to the lowest sampled compactness, $\underline{g}(0.1) \approx 0.19$ (S1), while both frozen thresholds fall to zero (the $x \rightarrow 0$ asymptotics of g itself are not established; the rigorous lower bound $c(x)$ of Lemma 6 vanishes there). To leading order the poles are the binding directions: the $\mathcal{O}(\lambda)$ correction to the s -weighted margin (interior term included) is a pure dipole in $\cos\vartheta$, extremal at $\vartheta = 0, \pi$, so it fixes the binding angle, not which pole.

Proof. On the symmetry axis the Kinnersley exterior reduces exactly to the static form $-h dt^2 + h^{-1} dr^2 + r^2 d\Omega^2$ with $h = f \mp 2aR$ at the rear/front pole, the axial value of the exact lapse (41). This is a coordinate change on the *exterior* only: the interior is the rigidly accelerated flat cavity, whose Rindler lapse gives the nonzero pole redshift $K^{\hat{r}}_{\hat{r}} = \mp a/(1 \mp \lambda)$, that is $RK^{\hat{r}}_{\hat{r}} = \mp \lambda/(1 \mp \lambda)$, used already in (61). Matching the two across $r = R$ gives (42) by a direct expansion (App. C, Eq. (62)). For the estimate, drop the $\mathcal{O}(\lambda)$ interior redshift. The margin-zero of (42) is then $3s_{\pm} = 3 - \frac{5}{2}x \mp 7\lambda$, and squaring it against $s_{\pm}^2 = 1 - x \mp 2\lambda$ clears the radical and gives, at both poles, the quadratic

$$49\lambda^2 \mp (24 - 35x)\lambda + (\frac{25}{4}x^2 - 6x) = 0, \quad (45)$$

whose discriminant is $(24 - 35x)^2 - 49(25x^2 - 24x) = 36(16 - 14x) > 0$ on $(0, \frac{24}{25})$. The smallest positive root of the pair is (43), the absolute value $|24 - 35x|$ collecting the two pole branches into one expression.

For the small- x slopes (44), put $\lambda = kx$ in the front-pole margin of (42) (the front binds at small x). With $s_- = \sqrt{1 - x + 2kx} = 1 + \frac{1}{2}(2k - 1)x + \mathcal{O}(x^2)$, the four terms contribute $3s_- - 3 = (3k - \frac{3}{2})x + \frac{5}{2}x$, $-7\lambda = -7kx$, and $+s_- \lambda/(1 + \lambda) = kx + \mathcal{O}(x^2)$, so

$$8\pi R s_-(\sigma - p)|_{\lambda=kx} = (1 - 3k)x + \mathcal{O}(x^2), \quad (46)$$

whose zero is $k = \frac{1}{3}$: the exact threshold is $\lambda_{\text{DEC}} = x/3 + \mathcal{O}(x^2)$. Expanding (43) at $x \rightarrow 0$ with $\sqrt{16 - 14x} = 4 - \frac{7}{4}x + \mathcal{O}(x^2)$ gives $\tilde{\lambda}_{\text{DEC}} = [(24 - \frac{21}{2}x) - (24 - 35x)]/98 + \mathcal{O}(x^2) = x/4 + \mathcal{O}(x^2)$, hence the limiting ratio $\frac{3}{4}$.

For the ceiling inequality $\lambda_{\text{DEC}} < \frac{1}{2}(1 - x)$ on $(0, \frac{24}{25})$, use the pole margins $M_{\pm}(x, \lambda) = 3s_{\pm} - 3 + \frac{5}{2}x \pm 7\lambda \mp s_{\pm}\lambda/(1 \mp \lambda)$, $s_{\pm} = \sqrt{1 - x \mp 2\lambda}$ of (42), positive at $\lambda = 0$. At the ceiling $\lambda = \frac{1}{2}(1 - x)$ the rear margin has $s_+ = 0$, so $M_+ = \frac{1}{2} - x < 0$ for $x > \frac{1}{2}$, placing the rear zero strictly inside $(0, \frac{1}{2}(1 - x))$; on the front branch $s_- = \sqrt{2(1 - x)}$ and $M_- = \sqrt{2(1 - x)}(10 - 4x)/(3 - x) + 6x - \frac{13}{2} < 0$ iff $Q(x) := (13 - 12x)^2(3 - x)^2 - 32(1 - x)(5 - 2x)^2 > 0$, which holds on $(0, \frac{1}{2}]$ ($Q(0) = 721$, $Q(\frac{1}{2}) = \frac{201}{4}$, root-free by Sturm). The two branches cover $(0, \frac{24}{25})$, so $\lambda_{\text{DEC}} < \frac{1}{2}(1 - x)$; the bound is not slack (the rear zero meets the ceiling exactly at $x = \frac{1}{2}$ and the front zero exceeds it for $x \gtrsim 0.632$, the strict inequality being a property of the minimum), the worst ratio being $\lambda_{\text{DEC}}/[\frac{1}{2}(1 - x)] \approx 0.9931$ at $x \approx 0.631$. $\square$

The estimate is quoted only as such: past its front/rear crossover $x = 24/35$ it tracks the exact root to better than 1% for $x \lesssim 0.84$ (supremum 2.5% as $x \rightarrow 24/25$), and the exact λ_{DEC} is used

throughout. This frozen-shape axial diagnostic is a distinct object from the operative matched frontier $g(x)$ and not globally ordered with it (App. C): the matched Darmois shell shape-responds and avoids the rigid -7λ front penalty, so at low compactness the front-pole λ_{DEC} lies well below the matched rear-pole $g(x)$.

Remark 2 (Acceleration makes the rear pole *safer*: the frontier is not a redshift effect). The intuitive picture, that acceleration crushes the rear pole until the dominant energy condition fails there, is *false*. With the shape frozen, both the linear and the quadratic rear-pole margin coefficients are strictly positive for $x < \frac{3}{4}$ (Sturm, App. C), so acceleration alone makes the rigid rear pole *safer*. The rear pole is where *regularity* degrades first (the ceiling (40)) but not where *admissibility* does: the frontier $g(x)$ is forced by the global shape response of the matched shell, a nonlinear isometric-embedding constant, not a root of a local redshift condition.

The kinematic ceiling bounds the frontier from above. A matching rigorous bound holds from below, bracketing the frontier between the two.

Lemma 6 (Rigorous frontier lower bound). *Fix $x \in (0, \frac{24}{25})$, $s = \sqrt{1-x}$, and restrict to the sub-horizon strip $\lambda \leq \Lambda_h(x) := \frac{1}{4}(1-x)$. The matched Darmois shell of Theorem 2 has worst-observer surface-DEC margin bounded below, uniformly in the emission angle, by (in the dimensionless normalization $M_0 = 8\pi R \delta_0$, so that $8\pi R$ multiplies the raw margin Φ of (35), and A, C_{shape} are the correspondingly scaled coefficients)*

$$8\pi R \Phi(x, \lambda) \geq M_0(x) - A(x) \lambda - C_{\text{shape}}(x) \lambda^2, \quad (47)$$

where $M_0(x) = -(5s-1)(s-1)/(2s) > 0$ is the closed-form anchor margin (Lemma 3), $A(x) = \|(-m_1(x, \cdot))_+\|_\infty = \mathcal{O}(x^2)$ is the worst-angle linear reduction (quadratic onset: the exterior is locally Schwarzschild to $\mathcal{O}(a)$), and $C_{\text{shape}}(x)$ is a finite constant built from the inverse bound $\Gamma(x)$ below, the second-variation majorant $M_2(x)$ of the Darmois map, and the quantitative implicit-function radius $\delta_Q(x) > 0$. Consequently the frontier admits the lower bound

$$g(x) \geq c(x) := \min\left\{\Lambda_h(x), \frac{1}{2}\delta_Q(x), \frac{\sqrt{A(x)^2 + 4C_{\text{shape}}(x)M_0(x)} - A(x)}{2C_{\text{shape}}(x)}\right\} > 0, \quad (48)$$

positive because $M_0 > 0$, and, with the kinematic ceiling (40), brackets the frontier between a rigorous positive lower bound and the closed-form ceiling, $c(x) \leq g(x) \leq \frac{1}{2}(1-x)$.

Every constant is explicit up to one fixed round-sphere constant, the $H^s \hookrightarrow C^2$ Sobolev embedding constant that passes from the matched shape to the pointwise margin: $\Gamma(x)$ is closed-form and $M_2, \delta_Q, A, C_{\text{shape}}$ are built from it (App. C), so $c(x)$ is an evaluable curve rather than a mere positivity statement. Off the $\ell = 1$ gauge the block-diagonal inverse bound is

$$\Gamma(x) = \max\left\{\frac{1}{x}, \frac{1}{2s^3} + \frac{1}{2s}\right\}, \quad (49)$$

both entries from the inverse of the $\ell = 2$ minor (88), the $\ell = 0$ breathing block (80) contributing only the dominated $(1-x)/x$ and the $\ell \geq 2$ blocks only the spectral value $\frac{1}{4}$; Γ diverges at both endpoints, which is why c vanishes there ($c(0.3) \approx 1.7 \times 10^{-4}$). The bound is conservative by about three orders against the numerically mapped $\underline{g} \sim 0.2$ (App. C); the looseness is in the Cauchy majorant M_2 , not in Γ , and the sharp $g(x)$ remains a nonlinear isometric-embedding constant.

Proof, the full argument in App. C. On the strip the effective potential $f - 2aR \geq \frac{1}{2}(1-x)$ and the interior factor $1 - aR \geq \frac{3}{4}$ stay bounded away from zero; the analytic-continuation horizon of the exterior, $f - 2aR = 0$ at $\lambda = \frac{1}{2}(1-x)$, lies strictly outside it, so the Darmois functionals are real-analytic in the shell configuration on the strip. Split the shape into its finite $\ell \leq 2$ block and its $\ell \geq 3$ tail. The $\mathcal{O}(a)$ steering source is a pure dipole, so the $\mathcal{O}(a^2)$ junction defect is quadratic in that dipole and carries no $\ell \geq 3$ content; the finite block (cos-row $(-x/(1-x), -2s)$, tilt row 1, monopole $-x/s^2$, quadrupole minor $2x\sqrt{1-x}$, each nonzero on $(0, \frac{24}{25})$) is solved in closed form, and the residual feeding the tail is $\mathcal{O}(\lambda^3)$. On the tail the round-sphere rigidity operator $\Delta_{S^2} + 2$ is diagonal with $\ell \geq 3$ spectral gap 10, and its inverse in the working scale $H^{s-2} \rightarrow H^s$ ($s > 3$, so $H^s \hookrightarrow C^2$) has norm $\frac{13}{10}$, attained at $\ell = 3$; with the closed-form source derivatives the analytic implicit-function theorem yields a matched $q(\lambda)$ with a uniform inverse bound Γ on a quantitative radius $\delta_Q(x) > 0$ off the $\ell = 1$ gauge (App. D), and $\|q'\|, \|q''\|$ follow from Γ and the source derivatives. A second-order Taylor expansion of the matched margin bounds its nonlinear reduction by $C_{\text{shape}}(x)\lambda^2$ at every angle. The exact lapse (41) is affine in $\cos \vartheta$ and so carries no equatorial

a -dependence at all, and the anchor margin $M_0(x) > 0$ (which vanishes only at the endpoints $x \rightarrow 0, \frac{24}{25}$) is angle-independent, so the worst-angle matched margin stays positive on a positive λ -window at each interior x ; $M_0 > 0$ then gives (48). App. C carries the details; the single quantitative input beyond closed-form junction algebra is the second-variation modulus of the junction map on the H^s ball (equivalently the analytic-majorant radius of App. E). $\square$

Equation (40) is thus a rigorous, closed-form *upper* bound on the operative matched frontier $g(x)$, complemented from below by Lemma 6; the empirical envelope (39) is tighter but heuristic, and the exact value of $g(x)$ is a nonlinear global isometric-embedding constant (App. C). For a *general* accelerating shell (40) holds only as a quasi-local normal-form condition $2m_{\text{ql}}/R + 2aR < 1 + \varepsilon$, the exact areal combination being specific to the Kinnersley foliation.

The inequality (37) is a Buchdahl-type bound for *kinematics in form*: just as $2m/R \leq 8/9$ caps the static compactness of a fluid sphere [65], $a_{\text{max}}R \leq g(x)$ caps the proper acceleration of a positive-energy shell, achieved here with the energy conditions intact everywhere, in contrast to exotic-matter warp drives. Independently of the size of the admissible set, (37) is itself a positive-energy maneuver envelope.

9 The finite-duration maneuver and its worldtube

With the acceleration–compactness frontier established, we assemble the per-instant shells along a trajectory into a complete finite maneuver, and then lift that sequence of snapshots into a single time-evolved spacetime.

9.1 From the frontier to a complete maneuver

Theorem 2 is local in retarded time and the frontier (37) caps a single instant; assembled along a trajectory they give a *complete finite-duration maneuver*.

Theorem 3 (Admissible finite maneuver, per-instant quasi-static). *Fix $x_* < 24/25$ and a steering history $a(u)$ supported on $[u_0, u_f]$ with $a(u_0) = a(u_f) = 0$ and nonzero net rapidity $\Delta\eta = \int_{u_0}^{u_f} a \, du \neq 0$, the saturated Tsiolkovsky budget $m(u) = m_0 e^{-3 \int_{u_0}^u |a|}$ (15) (so $x(u) = 2m(u)/R$), and $\lambda(u) = |a(u)|R < c(x(u))$ throughout, with c the rigorous lower-bound frontier of Lemma 6. Then at every retarded cut $u \in [u_0, u_f]$ the matched quasi-static warpsheet is admissible: (i) the bulk obeys $n^2 \geq 0$ for all ϑ (closed form, the control law); (ii) the surface stress is Hawking–Ellis Type I with worst-observer surface DEC margin > 0 ; and (iii) the Rindler horizon never reaches Σ ($\lambda < 1$, closed form); the endpoints are the strictly admissible static anchor. The net rapidity $\Delta\eta \neq 0$ (by hypothesis) changes the Bondi velocity $v^i = P_B^i/P_B^0$ between the stationary (zero-acceleration) endpoints, $\Delta v \neq 0$ (covariantly $dP_B^\mu/du = \dot{m} v^\mu + m a^\mu$, the integrated $\ell = 1$ flux), a velocity change radiated as positive null momentum.*

Proof. These hypotheses assemble the per-instant results uniformly in retarded time: the anchor DEC window (Lemma 3), the frontier (37), and the Born-rigid (Rindler-horizon) ceiling $\lambda < 1$ (Sec. 14). A single-signed $a \geq 0$ (the saturating bump below) gives $\Delta\eta \neq 0$ and a net boost; a sign-changing round-trip with $\Delta\eta = 0$ returns to rest and steers no net velocity. Clauses (i) and (iii) are closed-form directly, and (ii) follows from the rigorous lower bound $c(x)$ of Lemma 6: $\lambda(u) < c(x(u))$ guarantees admissibility throughout. The construction is *per-instant quasi-static*: at each retarded cut the shell is the gauge-invariant finite- λ junction at the instantaneous $(x(u), \lambda(u))$ (App. C), with the dynamical $\mathcal{O}(\mu^2)$ corrections deferred to future work (App. F). Because the rocket radiates, $x(u) = 2m(u)/R$ decreases monotonically from its initial value x_* , so the whole path $(x(u), \lambda(u))$ stays inside the admissible window. Along such a bump profile $a(u)$ the worst-observer surface margin and the bulk minimum stay non-negative at every retarded step, the Bondi balance (12) closes identically, and the endpoints recover the static anchor. This closes, at the per-instant quasi-static level, the surface-DEC question that a static anchor and a perturbative existence theorem leave open: a finite burn that changes the velocity is admissible *throughout*, not only at its stationary endpoints. $\square$

9.2 From snapshots to a time-evolved spacetime

Theorem 3 establishes a *sequence* of instantaneous junctions. We now lift it to a statement about the *time-evolved* spacetime. The construction is exact on both sides at every retarded time (the exterior is the Kinnersley photon rocket for the full $m(u)$ and worldline, the interior is exactly flat); matched across a C^2 worldtube ($[h_{ab}] = 0$) it *would be* an exact, time-dependent distributional Einstein solution. Two questions remain: whether the per-cut matches assemble into that C^2 worldtube (Thm. 4), and whether its *exact* (fully retarded) surface stress is then admissible. This

requires the dynamical junction to be evaluated on a cut of *constant retarded time* (a constant- t slice mixes retarded times across the shell and is not the physical cut); the explicit retarded-time derivatives $(\dot{a}, \dot{m})$ then enter S_{ab} through the time-dependent extrinsic curvature.

Theorem 4 (Finite-duration worldtube: time-evolved admissible spacetime). *Fix $x_\star \in (0, 24/25)$. For a slow, fixed-axis (non-rotating, reflection-symmetric), sub-horizon burn ($\inf(f - 2aR) > 0$), the per-cut matched shells of Theorem 2 assemble into a single gauge-fixed matched timelike worldtube $\Sigma(\lambda, \varepsilon)$ (App. E). The assembled worldtube is C^2 for merely smooth burns and, for real-analytic $a(u)$, is real-analytic and closed to all amplitude orders, matching the exterior $a(u)$ throughout. The exact retarded surface stress S_{ab} is continuous in (λ, ε) , where $\lambda = aR$ and ε is the burn rate in a C^2 norm of $(m(u), a(u))$, reducing at $(\lambda, \varepsilon) = (0, 0)$ to the static anchor with strict Hawking–Ellis Type I surface DEC margin $\delta_0 > 0$ (Lemma 3). Hence there exist $\lambda_\star, \varepsilon_\star > 0$ such that for $\lambda < \lambda_\star$, $\varepsilon < \varepsilon_\star$ the matched shell is Type I with strict surface DEC throughout the burn: a time-evolved admissible accelerating spacetime, unique modulo the $\ell = 1$ rigid-translation gauge, not a sequence of snapshots.*

Remark 3 (Structure of the proof). The proof consists of three logically independent parts, which we separate here because the theorem is not an analytic evolution result.

(i) *Rigid determination.* On a cut of constant retarded time the shape is not free Cauchy data. The retarded shape velocity cancels identically from the induced-metric jump, so the constant- u cuts are *characteristic*: the energy row (83), though first order in ∂_u , does not propagate the shape but fixes it. At each order in the amplitude the shape multipoles are therefore determined elliptically by $a(u)$ through one and the same isomorphism, and the determination is an induction. The base case is the $\mathcal{O}(a^2)$, $\ell = 2$ obstruction, which cancels gauge-completely (App. E). At order n the linearized operator is unchanged (its principal angular part is always $\Delta_{S^2} + 2$, Lemma 7, and its $\ell \leq 1$ Schur block is always the non-degenerate cos-row and tilt determinant (77)), while the source is built solely from orders $< n$. Since $\Delta_{S^2} + 2$ is self-adjoint with kernel exactly the $\ell = 1$ translations, it is surjective off $\ell = 1$: there is no cokernel, so no solvability condition arises at any order, and the residual $\ell = 1$ component is absorbed by the rigid-translation gauge (Remark 1). This is the content of the phrase “closed to all amplitude orders.”

(ii) *The finite burn.* Existence over the burn is an ordinary differential equation in a Banach space: the solution map of the energy row gives $Y_u = G(Y; \lambda, u)$ with G locally Lipschitz on $C_{\text{ax}}^{2,\alpha}(S^2)$ (90) and no derivative lost, and Picard–Lindelöf with the static anchor as datum yields the worldtube on the closed interval, with a Grönwall bound $\|Y(u)\|_{C^{2,\alpha}} \leq C(\lambda + \varepsilon)$ uniform on it. The duration T is not produced by the theorem and is not limited by it: T is the prescribed duration of the maneuver $a(u)$, an input. What depends on T is the smallness threshold, $\varepsilon_\star$ being non-increasing in T and degrading as $\mu_\star^{-p}$ toward the rear-pole horizon. No Nash–Moser iteration and no loss of regularity occur.

(iii) *Analyticity.* For real-analytic $a(u)$ the amplitude series converges, by an Ovsyannikov analytic majorant [70, 71] on a scale of Banach spaces. Cauchy–Kovalevskaya enters only here, as a convergence majorant that raises the regularity of the worldtube from C^2 to real-analytic. It is not used as an evolution theorem and does not produce the existence; dropping it costs the analyticity and nothing else.

This is the openness of the strict (Type-I eigenvalue) DEC margin under a continuous perturbation of an exact admissible solution, the dynamical counterpart of Theorem 2 now carrying the retarded $\dot{m}, \dot{a}$ terms. The per-cut shape match $[h_{AB}] = 0$ is unconditional at small amplitude (the static elliptic problem of Lemma 4 at each cut); it is the compatibility identity (84) that upgrades the admissible snapshots to a single worldtube. Its leading $\mathcal{O}(a^2)$ ($\ell = 2$) obstruction cancels gauge-completely, and for real-analytic burns the all-orders closure follows from the order-independent per-cut isomorphism that rigidly determines the shape order-by-order (App. E). This all-orders closure is of the worldtube *assembly* (the compatibility identity), not of admissibility, which stays perturbative in the amplitude (the reach set by $\lambda_\star, \varepsilon_\star$). The *fixed-axis* restriction is on the time-evolved *assembly*, not on the steering direction: at small amplitude a matched shell exists per retarded cut for an *arbitrary* acceleration three-vector by $SO(3)$ -equivariance (Cor. 6), and the optimal-control reading of Sec. 12.1 is fully three-dimensional. Its analytic content is the constant-cut elliptic reduction of Lemmas 7–8, closed by a Grönwall bound in u once the time-row solvability (83) holds, with the multipole solve of the Supplement (Sec. S) extending it to finite amplitude; the exterior’s Ricci tensor keeps the rank-one null-dust form $R_{ab} \propto \ell_a \ell_b$ along the burn, and the junction reduces to the static anchor as $a \rightarrow 0$. Two limits fix its reach. First, it is *perturbative*: small acceleration and slow burns, not the saturating amplitude. Second, the leading dynamical term is the outgoing mass-loss flux, which *tightens* the

surface DEC margin through the dense wall core ($d(\text{margin})/d\mu < 0$ where the lapse deepening dominates), exactly as the $\mathcal{O}(\mu)$ back-reaction of Sec. 10.2 finds (App. F), so the fully dynamical frontier is strictly tighter than $g(x)$; the saturating-amplitude, all-angle reach is left open (Sec. 14). The analytic route is a gauge-fixed coercive estimate $\|\delta Y\|_{H^{s+2}} \leq C \|D_Y F \delta Y\|_{H^s}$ for the linearized constant-retarded-cut junction $F := [h_{ab}] = 0$, propagated in u . The one structural question is the principal angular (Douglis–Nirenberg) symbol of the reduced map; we resolve it.

Lemma 7 (Elliptic angular symbol on constant retarded cuts). *After Schur-eliminating the finite-dimensional $\ell \leq 1$ lapse/tilt gauge block (non-degenerate at the anchor, its $\ell = 1$ determinant nonzero by the closed-form cos-row (77) and its $\ell = 0$ Jacobian $\partial_R[\sqrt{f}] \neq 0$) the principal angular symbol of the reduced first-junction map on the shape Y is that of the static infinitesimal-bending operator $\Delta_{S^2} + 2$ (App. D), i.e. $|\xi|_{S^2}^2$, noncharacteristic for $\xi \neq 0$. In the retarded direction, by contrast, the constant- u cut carries no principal ∂_u symbol and is characteristic: the shape is rigidly determined order by order, not Cauchy-evolved (App. E). The retarded data enter only at subordinate angular order: on a cut of constant retarded time the dynamical components $h_{uu} = g_{\mu\nu} \partial_u X^\mu \partial_u X^\nu$ and $h_{uA} = g_{\mu\nu} \partial_u X^\mu \partial_A X^\nu$ carry the retarded derivative Y_u (and $\dot{m}, \dot{a}$) at angular order 0 and ≤ 1 respectively ($\partial_u X$ carries no angular derivative of Y and $\partial_A X$ at most one) while the order-two angular content $\partial_\Omega^2 Y$ (the Δ_{S^2} of the bending reduction) enters only the shape block $[h_{AB}]$, with coefficients that the time dependence of $(m(u), a(u))$ shifts but whose leading angular order it does not raise. Hence the order-two angular principal part is block-diagonal and identical to the static one; Y_u is a transport term in u , not a principal-order coupling.*

The principal symbol alone is not a well-posedness statement; we close the estimate. First, the Schur elimination must survive the burn, not only the anchor.

Lemma 8 (Finite-rate Schur non-degeneracy). *The $\ell \leq 1$ lapse/tilt Schur block of the dynamical junction stays non-degenerate at finite burn rate: there is $\varepsilon_0(x_\star) > 0$ such that $\det J_{\text{dyn}}(x, \varepsilon) \neq 0$ for $\varepsilon < \varepsilon_0$, so the Schur elimination of Lemma 7 remains valid throughout the burn, not only at the anchor.*

Proof. On a constant-retarded cut the block is a smooth $\mathcal{O}(\varepsilon)$ perturbation of its static value, so $\det J_{\text{dyn}}(x, \varepsilon) = \det J_0(x) [1 + \mathcal{O}(\varepsilon)]$ with $\det J_0(x)$ the static $\ell = 1$ lapse/tilt determinant. The cos-row lapse derivative $\partial_{A_{t1}}[h_{\tau\tau}]^{\text{cos}} = -2\sqrt{1-x}$ (77) and the unit tilt row $\partial_{w_1}[h_{\tau\theta}]^{\text{sin}} = 1$ are nonzero, so $\det J_0(x) \neq 0$ on $(0, 24/25)$ and the perturbed determinant is nonzero for small ε . $\square$

By Lemmas 7–8 the reduced constant- u junction is an over-determined elliptic system in the angles whose principal part is the static $\Delta_{S^2} + 2$, with the retarded Y_u of subordinate *differential* order and the $\ell \leq 1$ Schur block invertible throughout the burn. This reduces the persistence to the solvability of the higher-mode retarded time-rows (83) for Y_u : on a characteristic cut those $\ell \geq 2$ rows *fix* Y_u algebraically at each order rather than propagating it, so the shape is rigidly determined and no Nash–Moser iteration is needed; the single retarded derivative reappears only in the convergence of the amplitude series, controlled by the analytic majorant below. We isolate this determinacy as condition (83) and establish it in App. E for a fixed-axis burn.

Proof of Theorem 4. The proof (the constant-cut elliptic reduction, the fixed-axis time-row solution operator, the $\mathcal{O}(a^2)$ compatibility computation, and the Grönwall closure) is given in Appendix E. $\square$

The single quantity the argument leaves implicit is the *explicit* size of the ball $(\lambda_\star, \varepsilon_\star)$ as the rear-pole horizon is approached, $\mu_\star = \inf(f - 2aR) \rightarrow 0$ (the saturating-amplitude reach), set by the round-sphere elliptic constants of (76) and mapped numerically in the Supplement (Sec. S).

10 Finite-thickness wall and back-reaction

10.1 Thickening with admissible matter

The thin shell is the $\ell \rightarrow 0$ limit of a finite-thickness wall. To avoid the smoothing pathology of Ref. [23], the wall must be a *physically modeled* matter layer rather than an arbitrary smoothstep. Two microscopic models satisfy the dominant energy condition by construction. Collisionless (Einstein–Vlasov) matter has stress-energy $T_{ab} = \int f p_a p_b (d^3p/p^0)$ that is DEC for $f \geq 0$ [72] and interpolates a Minkowski cavity to the radiation exterior without a vacuum gap; but a *purely tangential* ($p_r = 0$) Vlasov layer requires its constituents on circular timelike orbits, which exist only outside the photon sphere, $2m(r)/r < 2/3$, so the kinetic realization of the wall below is limited to that window. Anisotropic *elastic* matter [73] carries tangential stress without circular orbits and is the appropriate microscopic model at the higher compactness of the frontier (the Supplement). This choice is required: a *generic* smooth (radial-tension) thickening fails, and the failure is closed-form.

Proposition 6 (Radial-tension thickening violates the dominant energy condition). *A static Kerr–Schild wall $g_{\mu\nu} = \eta_{\mu\nu} + (2M(r)/r)\ell_\mu\ell_\nu$ with smoothstep mass profile $M(r)$ has, from the exact curvature, the anisotropic stress*

$$\rho = \frac{M'(r)}{4\pi r^2}, \quad p_r = -\rho, \quad p_t = -\frac{M''(r)}{8\pi r} \quad (50)$$

The radial block saturates the dominant energy condition ($p_r = -\rho$), while the tangential margin $\rho - |p_t| < 0$ violates it wherever

$$r |M''(r)| > 2 M'(r). \quad (51)$$

The violation is forced at the matter–vacuum edge, independent of thickness or compactness, and for a thin transition $\delta \ll r$ fills the whole wall. The strict obstruction is this static tangential channel; the u -dependent (generalized-Vaidya) graft only saturates it.

Proof. At the cavity edge $r = r_-$ the C^1 match to the flat ($M \equiv 0$) interior gives $M(r_-) = M'(r_-) = 0$, so $\rho(r_-) = 0$; the DEC there requires also $p_t(r_-) = 0$, i.e. $M''(r_-) = 0$. A generic (C^1 , not C^2) profile has $M''(r_-) \neq 0$ and violates the DEC *exactly* at the edge, $\rho - |p_t| = -|M''(r_-)|/8\pi r_- < 0$; a higher-smoothness match with $M''(r_-) = 0$ merely displaces the violation infinitesimally off-edge: were (51) to fail on a right-neighborhood $(r_-, r_- + \epsilon]$ then $|M''| \leq 2M'/r \leq (2/r_-)M'$ there, and a Grönwall estimate from $M'(r_-) = 0$ would force $M' \equiv 0$, contradicting $M' > 0$ across the transition; so the DEC fails somewhere in every right-neighborhood of the edge, where $\rho = M'/4\pi r^2$ is still vanishingly small while M'' has turned on. Either way the source–vacuum boundary cannot satisfy the DEC at every compactness; for a thin transition $\delta \ll r$ the violation (51) is deep across the wall, $r|M''|/2M' \sim r/2\delta \gg 1$ wherever $M'' \neq 0$ (a smoothstep’s single interior inflection, where $M'' = 0$, is the lone escaping point). The u -dependent (generalized-Vaidya) version $M(u, r) = \bar{m}(u)\chi(r)$ adds the outgoing flux in a *single* slot of the $(\hat{u}, \hat{r})$ block ($T^{\hat{r}}_{\hat{u}} \neq 0$, $T^{\hat{u}}_{\hat{r}} = 0$), making it a *defective* Jordan block with a repeated real eigenvalue and invariant discriminant $\Delta = (\rho_h + p_r)^2 - 4q_h^2 = 0$ (the field equations force $\rho_h + p_r = 2|q_h|$ for a pure outgoing null flux): it is Hawking–Ellis *Type II*, DEC-saturating, not Type IV (an apparent Type-IV assignment reflects the eigenvalue-splitting degeneracy of the defective block, the Type I $\leftrightarrow$ II boundary discussed in App. A). (This is distinct from the vorticity-sourced Type IV of a *velocity*-carrying smooth warp shell [23], whose tilted flow circulates energy; the radial-tension wall here is twist-free and merely defective.) $\square$

Corollary 7 (The boundary cost is the $1/\delta^2$ edge law). *For a smoothstep profile $M(r) = m S((r - r_-)/\delta)$ of width δ matched C^1 to the flat cavity ($S(0) = S'(0) = 0$, $S''(0) \neq 0$), the worst-observer surface-DEC margin at the inner edge is, from (50),*

$$(\rho - |p_t|)|_{r \rightarrow r_-^+} = -\frac{m |S''(0)|}{8\pi r_- \delta^2} < 0, \quad (52)$$

diverging as $\delta \rightarrow 0$ at every compactness: the boundary cost of the radial-tension edge is an exact $1/\delta^2$ law.

Proof. Insert $M(r) = m S((r - r_-)/\delta)$ into the Kerr–Schild surface stress (50). The radial tension is $p_r = -\rho$ with $\rho = M'(r)/(4\pi r^2)$, and the tangential pressure carries the second derivative, $p_t \supset -M''(r)/(8\pi r)$. At the inner edge $S(0) = S'(0) = 0$, so $M(r_-) = 0$ and $M'(r_-) = 0$, while $M''(r_-) = m S''(0)/\delta^2 \neq 0$. Hence $\rho \rightarrow 0$ but $|p_t| \rightarrow m |S''(0)|/(8\pi r_- \delta^2)$, and the margin $\rho - |p_t|$ takes the stated value, which diverges to $-\infty$ as $\delta \rightarrow 0$ for every x . $\square$

Proposition 6 with Corollary 7 is the analytic form of the boundary cost of Ref. [23]: the obstruction is the radial tension $p_r = -\rho$ forced by the Kerr–Schild profile, not thickening per se. A *tangential-pressure* wall (radial pressure $p_r \rightarrow 0$ at both edges, the load carried by the tangential pressure fixed by anisotropic hydrostatic equilibrium, $p_t = \frac{r}{2}\rho\Phi'$ with $\Phi' = m/[r(r - 2m)]$) has the worst-observer pointwise margin $\rho - p_t = \rho[1 - m/(2(r - 2m))]$, which is positive (Hawking–Ellis Type I, strict DEC) *across the whole wall* iff the local compactness satisfies $2m(r)/r < 4/5$. Such a wall is source-consistent ($G^a_b/8\pi$ reproduces $(\rho, 0, p_t)$), is strictly DEC-admissible across the wall in the *static* limit for local compactness $2m(r)/r < 4/5$, and recovers the thin-shell anchor (σ_0, p_0) as $\ell \rightarrow 0$. The three admissibility windows nest by matter model, $2/3$ (collisionless, photon-sphere-limited) $< 4/5$ (anisotropic fluid/elastic) $< 24/25$ (distributional thin shell): the limit widens as the model supports the tangential stress under a progressively weaker kinematic constraint, each carrying the wall to higher compactness. The $4/5$ result is *static and local*, whereas

an accelerating shell carries an effective rear-pole redshift $x_{\text{eff}} = x + 2ar$ (Sec. 8, Eq. (60)). The static thick wall therefore covers the accelerating frontier only in the lower-compactness window where $x + 2aR < 4/5$. At higher compactness the accelerated rear pole exceeds $4/5$, and the admissible object is the thin shell, whose wider window $24/25$ the accelerated rear pole still respects; an accelerated thick-wall DEC theorem is left to future work. Within that window the distributional shell is the $\ell \rightarrow 0$ limit of admissible matter precisely where the construction operates.

10.2 The radiative wall

We now graft the photon-rocket radiation *through* the wall, making it itself radiative; strict admissibility survives. Model the radiating wall as the two-component stress

$$T_{ab} = T_{ab}^{\text{wall}} + \mu \ell_a \ell_b, \quad T_{ab}^{\text{wall}} = \rho u_a u_b + p_t \gamma_{ab}, \quad p_r = 0, \quad (53)$$

the $p_r = 0$ tangential-pressure wall above plus a *separately conserved* outgoing null-dust flux $\mu \ell_a \ell_b$ ($\mu \geq 0$) streaming through it: a generalized-Vaidya interior $m(u, r) = \bar{m}(u) \chi(r)$ with $\chi(0) = 0$, so the radiation fills the wall with no beam through the center and no matter $\rightarrow$ vacuum edge. For the idealized fixed-background split (53) at constant μ (with $p_r = 0$), the radial Hawking–Ellis block has discriminant

$$\Delta = (\rho + p_r)(\rho + p_r + 4\mu) = \rho(\rho + 4\mu) > 0, \quad (54)$$

so it stays Hawking–Ellis Type I: superposing the radial null dust does *not* analytically re-trigger the Type IV (rotating-energy) failure a heat-flux model would. For the conserved generalized-Vaidya realization $m(u, r) = \bar{m}(u) \chi(r)$ the exact curvature carries an outgoing-null amplitude $n^2 > 0$ streaming through the wall, sourced by $\dot{m}$, vanishing at the center, and conserved ($\nabla_a T_{ab} = 0$). That ramp, however, carries $p_r \neq 0$, whose *static* radial-tension channel already violates the DEC at the matter–vacuum edge (Prop. 6); its $(\hat{u}, \hat{r})$ block is the defective Hawking–Ellis Type II (DEC-saturating, $\Delta = 0$) of that proposition, *not* Type IV (the Supplement), so it serves only to exhibit the through-wall flux, not as the admissible object. Strict DEC admissibility (Type I) across the whole material wall is instead the property of the idealized $p_r = 0$ split (53) above (discriminant (54)) and, with the metric allowed to respond, of the *separately* constructed adiabatic $p_r = 0$ Eddington–Finkelstein wall of App. F; the pure-null exterior *saturates* DEC (Type II, as outgoing null dust must).

We close the back-reaction at the adiabatic level (App. F): realizing the radiating wall as the exact $p_r = 0$ tangential wall in outgoing Eddington–Finkelstein form with a slowly radiating mass $m(u)$ gives a self-consistent Einstein solution ($\nabla_a T_{ab} = 0$), Type I, whose *curvature-derived* worst-observer DEC margin stays strictly positive across the whole frontier regime over the directly tested radiation amplitudes (App. F). With the metric allowed to respond, the outgoing energy *flux* (the quantity the DEC constrains) *decreases* the margin through the dense wall core, $d(\text{margin})/d\mu = 1 - \frac{r}{2}\rho \partial\Phi'/\partial\mu < 0$ (the analytic sign of Prop. 9, neutralizing only in the $\rho \rightarrow 0$ edge skin), rather than raising it, while leaving it strictly positive; the radiating wall therefore remains admissible (Fig. S4d). A frozen-background calculation gives the opposite, looser sign; that lemma and the controls isolating it are collected in App. F. The residual is the time-dependent $\mathcal{O}(\mu^2)$ term.

The finite-thickness wall stays compatible with the accelerating frontier where $x + 2aR < 4/5$, the thin shell remaining admissible above (Fig. S4). The energy-condition obstruction of Ref. [23] is thereby removed: the radiative wall is strictly admissible, and the generalized-Vaidya geometry exhibits an explicit metric carrying the required through-wall flux. A self-consistent matter realization (DEC by construction) is available: collisionless Einstein–Vlasov matter below the photon sphere ($2m/r < 2/3$) and anisotropic elastic matter [73] at the higher compactness of the frontier; a source-consistent anisotropic equilibrium of the tangential-pressure wall, and the kinetic/elastic boundary, are given in the Supplement.

10.3 The finite maneuver is thick-wall realizable

The finite-duration maneuver of Theorem 3, established above for the distributional thin shell, is, at the per-instant adiabatic level, the $\ell \rightarrow 0$ limit of an admissible *radiating tangential-pressure thick wall* throughout the burn, *provided* the rear-pole effective compactness stays in the thick-wall window, $x_{\text{eff}}(u) = x(u) + 2\lambda(u) < 4/5$ for all u . This is not implied by the thin-shell frontier $\lambda < \underline{g}(x)$ at high compactness, but it holds for the worked saturating burn of the Supplement (whose operating point keeps the rear-pole $x_{\text{eff}} < 4/5$ throughout; the rocket radiates, $x(u)$ falling while $\lambda(u)$ rises). At every retarded step the static tangential-pressure wall at $x_{\text{eff}}(u)$ is then

strictly DEC-admissible across its width (Fig. S3c), and the radiating version is curvature-derived Hawking–Ellis Type I with worst-observer DEC margin > 0 and $\nabla_a T_{ab} = 0$ (the Supplement). The through-wall radiation stays well inside the back-reaction window of Sec. 10.2, so the *saturating* (minimum-fuel) burn, not merely a gentle one, is thick-wall realizable at the per-instant adiabatic level. The fully anisotropic accelerated thick wall, a dipole-deformed metric carrying both the steering and the through-wall flux, remains future work.

11 The radiating equilibrium is linearly unstable

Admissibility is not stability, and the two are decoupled: the restoring force can be increased without reducing the dominant-energy margin, and yet the equilibrium is unstable, because the flux itself anti-damps. We state the result for the mode the model resolves.

Proposition 7 (Anti-damping of the radiating radial mode). *Consider the spherically symmetric Minkowski|Vaidya proxy of the radiating shell (interior Minkowski, outgoing Vaidya exterior with luminosity $L = -m_{,u} \geq 0$, barotropic wall of equation-of-state slope $\eta = dp/d\sigma$), and keep the mass-history channel $\delta m = m_{,u} \delta u$ that a radial perturbation opens. The radial mode obeys the Poisson–Visser oscillator $\delta \ddot{R} + 2\kappa_d \delta \dot{R} + \Omega^2 \delta R = 0$ with Ω^2 and κ_d given in closed form by (97), and: (i) the restoring force is free. A radiating equilibrium exists for $L < \frac{1}{2}s(1-s)(5s-1)$, and stiffening the wall past $\eta_{\text{crit}}(x)$ (98) makes $\Omega^2 > 0$ while leaving the surface-DEC margin unchanged, that margin being junction-fixed and independent of η ; the stiffening is subluminal ($\eta < 1$) for every $x < x_\eta \simeq 0.844$, covering the whole mapped operating range (only in the sliver $x_\eta < x < 24/25$, still inside the surface-DEC window, would strict stability demand a superluminal equation of state). (ii) the flux destabilizes. On that oscillatory branch $\kappa_d < 0$ at first order in L : the outgoing radiation anti-damps the radial mode, so the radiating equilibrium is linearly unstable, not stable. (iii) a finite burn survives it. The growth rate is of order the luminosity, $|\kappa_d| = \mathcal{O}(L)$, so the amplification $\exp \int_0^T |\kappa_d| d\tau$ over a burn is bounded by the fuel budget (15), which bounds $\int L d\tau$; away from the marginal wall a finite maneuver therefore does not destabilize the mode, though the equilibrium is not asymptotically stable. The angular and shape modes of the full accelerating shell are outside this radial proxy and remain open.*

Proof. The Israel radial equation for the two faces, with the outgoing flux entering the exterior $K^{\hat{r}}_{\hat{r}}$ and the retarded shift $\delta m = -L \delta u$ retained, reduces the coupled $(\delta R, \delta \dot{R}, \delta m)$ system to the stated oscillator with the closed-form coefficients $\Omega^2 = D/(4s^2)$ and $\kappa_d = -LN/[2D(1-s)s^3]$ of (97) (App. I carries the two faces). For (i), the margin $\sigma_0 - p_0$ is fixed by the junction at given (x, L) and does not involve η , so raising η moves D without moving the margin; $D > 0$ is $\eta > \eta_{\text{crit}}$ with η_{crit} the root (98), subluminal below x_η . For (ii), on the oscillatory branch $D > 0$, and the threshold $D = 0$ exceeds the zero of N by $(s+1)(s-1)^2/[2s(3s+1)(s^2+6s+1)] > 0$, so $N > 0$ there and $\kappa_d = -LN/[2D(1-s)s^3] < 0$. For (iii), $|\kappa_d| \leq CL$ with C finite once $D \geq D_0 > 0$, and $\int_0^T L d\tau = m_0 - m(T) \leq m_0$ by (15). $\square$

Two consequences follow. The margin $\sigma_0 - p_0$ is η -independent, so stiffening deepens the restoring well at no dominant-energy cost; the anti-damping, sourced by the radiation the drive must emit, is intrinsic to a *radiating* equilibrium and bounded by the same fuel budget that sets the maneuver cost. The realized tangential-pressure wall of Sec. 10 is the marginal one $\eta_{\text{real}} = \eta_{\text{crit}}$; the rear-pole redshift $x \rightarrow x + 2\lambda$ raises η_{crit} and moves it onto the barrier side, so a stiffer, still-admissible wall is *required* to keep $\Omega^2 > 0$ (App. I, with the timescale estimate in the Supplement). Stiffening past η_{crit} leaves the junction-fixed surface-DEC margin unchanged and stays subluminal throughout $x < x_\eta \simeq 0.844$, so we take the stiffened wall as the operating configuration.

12 Minimum-radiation maneuvers as optimal control

The exterior is left *exactly* Kinnersley by the excision, so the Bondi news $\mathcal{N}(u, \Omega)$ of the matched spacetime equals that of the bare photon rocket and *vanishes* for dipole steering: Bonnor [28] established that the Kinnersley photon rocket is exactly non-radiative, and Damour [58] reconciled this with linearized wave-generation theory (for the purely dipolar emission the leading $1/r$ waves from the accelerated mass cancel exactly against those of the momentum-carrying radiation field), so dipole steering is *gravitational-wave silent* while higher-multipole steering reintroduces a news contribution and an additional energy-loss channel. The construction turns warp-drive design into a control problem, and makes the *saturation* clause of Theorem 1 precise: among news-silent maneuvers realizing a prescribed proper-acceleration history $a(u)$ (which, with the budget below, fixes ΔP_B^i), which radiates the least? (Optimizing the acceleration history itself at fixed ΔP_B^i is the free-profile problem of the scope note below.) The *state* is the Bondi four-momentum $P_B^\mu(u)$

(equivalently the worldline kinematics) together with the shell compactness $x(u)$; the *control* is the angular radiation profile, i.e. the mass-loss rate $\dot{m}$ and the multipoles of $(\ln P)_{,u}$; the *dynamics* are Eq. (12); and the *constraints* are the bulk positivity $n^2(u, \Omega) \geq 0$ and the surface admissibility $\Phi(\lambda, \mu) \geq 0$ of Theorem 2 and frontier (37).

A natural objective is the total radiated energy $J = m(u_0) - m(u_f) = \int_{u_0}^{u_f} (-\dot{m}) du$, minimized over the angular control profiles that realize a prescribed proper-acceleration history $a(u)$. Expanding the angular emission in Legendre modes, the conserved momentum couples *only* to the $\ell = 1$ (dipole) mode by S^2 orthogonality, whereas every mode feeds the fuel monopole $-\dot{m}$.

Theorem 5 (Minimum-radiation steering). *Restrict to gravitational-wave-silent maneuvers (vanishing Bondi news, $\mathcal{N} \equiv 0$) within the asymptotically flat Robinson–Trautman pure-radiation class of Sec. 4. These are exactly the dipolar emissions (13): vanishing news forces the conformal factor (6) to $\ell \leq 1$, $P = b(u) + c(u) \cos \vartheta$ in the momentary rest frame, hence the emission (8) to a pure monopole+dipole, for which bulk positivity $n^2 \geq 0$ is equivalent to the control law (14). Among all such news-silent maneuvers realizing a prescribed proper-acceleration history $a(u)$ over $[u_0, u_f]$ (which, with the budget below, fixes the integrated thrust $\int m a du$, hence ΔP_B^i), the radiated energy J is minimized by saturating the control law pointwise, $-\dot{m} = 3m|\alpha|$ (any isotropic mass loss above this minimum raises J without adding thrust), with minimum $J = m_0(1 - e^{-3 \int |\alpha| du})$ [the Tsolkovsky budget (15)]; optimizing the acceleration history itself over all profiles achieving a given ΔP_B^i is the free-profile problem (Theorem 6).*

Proof, the full argument in App. G. That the news-silent class is exactly dipolar is classical: the purely dipolar photon rocket has vanishing news (Bonnor [28], with Damour’s [58] linearized reconciliation), and conversely von der Gönna and Kramer [59] proved the Kinnersley rocket is the *unique* no-news solution of the regular *axisymmetric* Robinson–Trautman pure-radiation class, with Cornish [74] removing the axisymmetry restriction (a Robinson–Trautman pure-radiation metric is news-free iff it is a Kinnersley photon rocket for a point mass accelerating *arbitrarily*), so the dipolar restriction holds for a three-dimensional steering history, not only the axisymmetric one. On the dipole class the controls are $(\dot{m}, \alpha)$ and positivity $n^2 \geq 0$ is the affine-in- $\cos \vartheta$ inequality (14), binding at the forward pole. The control Hamiltonian $\mathcal{H} = (-\dot{m}) + p_\mu \dot{P}_B^\mu - \oint \chi(\Omega) n^2 d\Omega$, with Bondi-flux costates p_μ [Eq. (12)] and a Karush–Kuhn–Tucker multiplier field $\chi \geq 0$ enforcing positivity, gives by the Pontryagin maximum principle [75, 76] a constraint-saturating (single active-constraint) optimum: any slack $-\dot{m} > 3m|\alpha|$ could be reduced at fixed $a(u)$, lowering J , so the optimum saturates $n^2 = 0$ at the forward pole, $-\dot{m} = 3m|\alpha|$. Integrating $d \ln m / du = -3|\alpha|$ gives the stated J . The full costate derivation is in App. G. $\square$

Higher multipoles are excluded on two grounds (App. G): an $\ell \geq 2$ emission carries energy to infinity without contributing to the $\ell = 1$ momentum (S^2 orthogonality, “energy without momentum”), and it switches on the Bondi news. The *bare* program “minimize fuel at fixed thrust subject only to $n^2 \geq 0$ ” is formally optimized not by the dipole but by a singular rear-pole spike; that profile is excluded because it carries all multipoles and is not a regular news-free exterior, the physical input being the von der Gönna–Kramer converse that a news-free emission has $\ell \leq 1$. In the broader gravitational-wave-allowed class the trade-off is set by the exact news energy J_{news} , a positive-definite quadratic form on the $\ell \geq 2$ amplitudes (Lemma 12, App. G); since an $\ell \geq 2$ mode radiates it quadratically while saving fuel only linearly, for a sufficiently weighted gravitational channel the optimum is the dipole. The *unconditional* statement is Theorem 5 (the news-silent class), where $J_{\text{news}} = 0$ forces $\ell \leq 1$ and the dipole is the only momentum-carrying mode.

Theorem 5 fixes a *given* proper-acceleration history $a(u)$ and optimizes only the mass-loss; the *bang-bang* is in that one control. The complementary *free-profile* problem (the least radiation to achieve a prescribed change of velocity) is solved geometrically, and turns warp-maneuver design into geodesy on the velocity space.

12.1 Minimum-radiation steering as a hyperbolic geodesic

The kinematic state of the news-silent rocket is (ρ, v) with $\rho = \log m$ and v^μ the unit worldline four-velocity on the *velocity hyperboloid* $\mathbb{H}^3 = \{v : v \cdot v = -1, v^0 > 0\} \cong SO^+(3, 1)/SO(3)$, so $P_B^\mu = e^\rho v^\mu$. Along any news-silent maneuver the proper acceleration magnitude is the hyperbolic speed, $|a| = \sqrt{g_{\mathbb{H}^3}(\dot{v}, \dot{v})}$, and the saturated control law (14) gives $\dot{\rho} = -3|a|$, so

$$\frac{m_f}{m_0} = e^{-3L_{\mathbb{H}^3}[v]}, \quad L_{\mathbb{H}^3}[v] = \int |a| d\tau = \text{hyperbolic length of } v(\tau). \quad (55)$$

The natural *additive* cost is the log-fuel $C = \log(m_0/m_f) = 3L_{\mathbb{H}^3}$; the radiated energy $J = m_0(1 - e^{-3L_{\mathbb{H}^3}})$ is a monotone transform of it (App. G.1).

Theorem 6 (Free-profile minimum-radiation maneuver). *Among news-silent maneuvers within the Robinson–Trautman pure-radiation (Kinnersley) class, steering the drive between stationary Bondi frames with four-velocities v_0, v_1 , the minimum log-fuel (equivalently the minimum radiated mass) is*

$$C_{\min} = 3 d_{\mathbb{H}^3}(v_0, v_1) = 3 \operatorname{arccosh}(-v_0 \cdot v_1), \quad \frac{m_f}{m_0} = e^{-3 d_{\mathbb{H}^3}(v_0, v_1)} \quad (56)$$

attained by the hyperbolic geodesic of $\mathbb{H}^3$ (the single one-parameter boost from v_0 to v_1) with the saturated Damour dipole realizing $|a| = |\dot{v}|_{\mathbb{H}^3}$ at each retarded cut. Any non-geodesic velocity path costs strictly more, the boost geodesic being the unique minimizer between fixed endpoints on the Hadamard manifold $\mathbb{H}^3$ (Cartan–Hadamard).

Proof. By Theorem 5 the news-silent optimum saturates the control law at every cut, so by (55) the log-fuel is $C = 3L_{\mathbb{H}^3}[v]$, three times the hyperbolic length of the velocity path $v(\tau)$ on $\mathbb{H}^3$. Minimizing C between fixed endpoints is therefore minimizing hyperbolic length. Now $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$ is complete, simply connected, and of constant negative curvature, hence a Hadamard manifold, so by Cartan–Hadamard with Hopf–Rinow the minimizer exists, is the unique geodesic joining v_0 to v_1 , and every other path is strictly longer. That geodesic is the one-parameter boost subgroup carrying v_0 to v_1 , of length $d_{\mathbb{H}^3}(v_0, v_1) = \operatorname{arccosh}(-v_0 \cdot v_1)$, which gives (56); the mass ratio follows by exponentiating (55). $\square$

This is a left-invariant control problem on $SO^+(3, 1)$, equivalently Riemannian geodesy on $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$, the Pontryagin principle reproducing (56) by Lie–Poisson reduction [77, 78] (App. G.1). The content is not the classical control theory but that bulk positivity and news-silence (Damour [58], von der Gönna–Kramer [59]) collapse the steering cost to a single hyperbolic norm with the *universal* constant 3: the rectilinear Tsiolkovsky budget (15) is the boost geodesic through the origin ($d_{\mathbb{H}^3} = \Delta\eta$), and “a turn costs strictly more than the collinear figure” (Sec. 14) is the geodesic triangle inequality on $\mathbb{H}^3$, made exact.

The radiation cone and the factor 3. The factor 3 is the $\frac{1}{2}$ dipole projection of the news-silent emission. For the $\ell \leq 1$ class $4\pi n^2 = b_0 + \vec{b} \cdot \hat{\Omega}$, positivity $b_0 \geq |\vec{b}|$ together with the moments $Q^0 = b_0 = -\dot{m}$ and $\vec{Q} = \vec{b}/3$ (thrust $|\vec{Q}| = m|a|$, from $\oint \hat{\Omega}^i \hat{\Omega}^j d\Omega = \frac{4\pi}{3} \delta^{ij}$) gives $-\dot{m} \geq 3m|a|$, the control law (14); the collimated ideal rocket saturates the universal floor 1 (Cor. 2, App. G.1).

Reachability and the moving frontier. Because the boost geodesics reach every point of $\mathbb{H}^3$, every velocity is reachable with positive remaining mass $m_f = m_0 e^{-3\eta} > 0$ (the light speed $\eta \rightarrow \infty$ needs $m_f \rightarrow 0$ and is not reached): the drive is relativistically controllable in velocity. The reachable set in Bondi *four-momentum* is by contrast constrained (the radiated $P_B(u_0) - P_B(u_f)$ must be future causal) so not every future-timelike P_B^f is reachable. The *pure* minimum-radiation cost $3L_{\mathbb{H}^3}$ is geometric (independent of how fast the path is traversed), so its optimum is the *unconstrained* geodesic: one may always burn arbitrarily slowly, and the acceleration frontier $a_{\max} R \leq g(x)$ (37) never binds it. The frontier becomes active once the maneuver is also *time-constrained* (a bounded-duration or minimum-time burn, equivalently a bound on proper acceleration): then $|\dot{v}|_{\mathbb{H}^3} R \leq g(x(u))$ caps the hyperbolic speed, and because the rocket radiates, $x(u) = 2m(u)/R$ decreases so the ceiling $\frac{1}{2}(1 - x(u))$ widens along the burn (the realized $g(x(u))$ following numerically). The minimum-time maneuver is then a *state-constrained* geodesic with constraint-active (maximal-acceleration) arcs $|\dot{v}|_{\mathbb{H}^3} R = g(x(u))$, made precise in Theorem 8 (App. G.1): the bounded-control trajectory-optimization problem warp design reduces to, with explicit interior/boundary switching and transversality. The central structural fact is the *length reduction* $x(u) = x_0 e^{-3\ell}$, fuel = 3ℓ in hyperbolic arclength ℓ (both bare objectives factor through ℓ , hence single-arc); multi-arc switching requires an objective that breaks this reduction (a dwell/effort penalty), for which the frontier binds early (large x , small g) and releases as the rocket radiates (x falls, g rises).

13 Comparison with the literature

We now place the construction against the prior no-go and positive-energy literature.

13.1 Relation to the Santiago–Schuster–Visser theorem

Reference [3] proves that *generic warp drives violate the NEC*. Its hypotheses define a warp drive through the Natário prescribed-shift ansatz: unit lapse and flat spatial slices, with the flow encoded

entirely in the shift. Under those hypotheses the Eulerian observers form a zero-vorticity geodesic congruence, and the Raychaudhuri argument forces the violation. Note what the theorem does *not* assume: it explicitly tolerates Israel–Lanczos–Sen thin-shell distributional curvature and an asymptotically Schwarzschild exterior, so a non-vacuum exterior is by itself no escape. The radiative warps fall outside the hypotheses for two independent reasons. First, we do not prescribe the metric, and the Robinson–Trautman exterior has neither unit lapse nor flat spatial slices, so its Eulerian congruence is neither geodesic nor zero-vorticity and the Raychaudhuri step never begins. Second, the drive radiates to $\mathcal{I}^+$, so it is not a compactly supported modification of Minkowski at all. The matter we install (positive null radiation) satisfies the NEC by Proposition 3. The theorem remains valid under its own hypotheses; the present construction lies outside them. How positivity is established matters as much as that it is. Ref. [3] shows that earlier “positive-energy” claims inspected only the comoving Eulerian observer, whereas the energy conditions quantify over *all* timelike and null observers; our bulk and shell margins are computed observer-robustly over the boost manifold, so the binding test is not concealed by a frame choice, the failure mode the theorem exposes. The only place the NEC could fail is the shell, and Theorem 2 establishes that it does not on an open admissible set.

13.2 Structural soundness beyond the energy conditions

Barzegar, Buchert, and collaborators [17, 16] stress that energy-condition violation is not the only pathology of warp spacetimes: a physical drive must also have a well-defined asymptotic structure and covariant kinematics, and prescribing a metric and reading off whatever stress-energy the field equations return can manufacture sources with no interpretation as physical matter. The worldtube-first construction sidesteps this by installing interpretable matter region by region (real outgoing null dust outside, vacuum cavity, *Hawking–Ellis Type I* shell; Sec. 2, Prop. 3), with velocity and acceleration the Bondi-frame components of the geometrically defined four-momentum $P_B^\mu(u)$ (Cor. 5), not coordinate artifacts. The recent no-go theorems of Ref. [16] and the realizations of Buchert and Frackowiak [18] (which also report instability, consonant with our stability caveat of Sec. 14) bear on the restricted, metric-first (prescribed-shift) class and lie outside the worldtube-first hypotheses. One of those theorems requires explicit comment. Barzegar, Buchert, and Vigneron prove that *R-Warp models violate the DEC* [16]: an R-Warp model has vanishing ADM energy, $E_{\text{ADM}} = 0$, whence the positive-energy theorem forces it to be Minkowski, so any nontrivial member must violate the DEC. Their argument is correct, and it does not apply here, because its first step rests on a hypothesis the present construction does not satisfy. An R-Warp model is defined by the conjunction of their restrictions R1, R2, R3b, R4 (and R6); the vanishing of E_{ADM} follows from R3b, spatial flatness, its proof being the substitution $h_{ij} = \delta_{ij}$ into the ADM surface integral. Our spatial slices are *not* flat: they carry the Schwarzschild-like mass aspect of the exterior, and the drive has $E_{\text{ADM}} = m > 0$ strictly (Lemma 1), the positive mass permitted for a DEC-satisfying, non-Minkowski, asymptotically flat spacetime. The construction also fails R2, the constant (unit) lapse and geodesic slicing: the exterior lapse (41) is non-constant and the shell is not comoving with a geodesic foliation. With R3b relaxed and $m > 0$, the implication $E_{\text{ADM}} = 0 \Rightarrow \text{Minkowski} \Rightarrow \text{DEC-violation}$ no longer applies, and the positive-energy theorem instead accounts for the positivity of the mass. Two further requirements of Ref. [16] are satisfied here: that the energy conditions of a thin shell be evaluated on the *induced* junction stress-energy rather than a bulk Einstein tensor (they are, on the Lanczos tensor S_{ab} throughout, Lemma 10), and that a metric below C^2 be treated distributionally rather than pointwise (it is, by the Israel formalism). Their theorem on non-vacuum spacetimes admitting a vanishing-mean-curvature foliation is the Hamiltonian constraint read at $K = 0$: the constraint then reads $16\pi E = {}^{(3)}R - |K|_h^2 - 2\Lambda$, so its hypothesis ${}^{(3)}R < 2\Lambda + |K|_h^2$ is *identically* the statement $E < 0$. Since $E = T_{ab}n^a n^b$ for the unit normal n of any slice, the weak energy condition forces $E \geq 0$ for the Eulerian observers of *every* foliation; here it holds in the exterior (Prop. 3), trivially in the vacuum cavity, and strictly on the shell for all $x \in (0, 1)$ (Lemma 3; distributionally, on the Lanczos tensor S_{ab} , as Ref. [16] itself requires). The escape is thus independent of the foliation chosen. The spacetime is free of closed causal curves on the finite maneuver slab, the Rindler horizon staying off the cavity for $\lambda = aR < 1$; eternal acceleration is not claimed (Sec. 14).

A further structural objection to the superluminal bubbles is met here rather than avoided. Krasnikov [21], and Everett and Roman [22], observed that the crew of an Alcubierre-type bubble cannot causally *control* it: the bubble wall lies outside the causal future of the ship, so the region that must be manipulated to steer is unreachable from inside, and the maneuver has to be arranged in advance along the whole trajectory. Here the shell is a *timelike* worldtube in causal contact with the passenger worldline Γ at every retarded cut, and the steering input, the

acceleration history $a(u)$, acts on it through matter the passengers carry and modulate. The maneuver is therefore causally controllable from the interior, which makes the optimal-control formulation of Sec. 12 a statement about a drive rather than about a prescribed geometry. The construction thus lies outside the structural no-go results as well as the null-energy one.

13.3 Quantum inequalities do not apply

Ford–Roman–type quantum inequalities [6, 79, 5] bound the magnitude and duration of *negative* energy densities; the radiative warpsheet contains none (bulk and admissible shell stress-energy are positive pointwise), so the quantum-inequality no-go that rules out Alcubierre and Krasnikov [21] is inapplicable, independently of the debated fundamental status of the classical energy conditions [80].

13.4 The acceleration gap

Table 1 positions the construction. Bobrick–Martire and Fuchs *et al.* exhibit positive-energy shells, but only at constant velocity. The soundness of these constant-velocity shells is itself contested: Barzegar, Buchert, and Vigneron [16] (Error 18 and App. B) argue that the Fuchs *et al.* shell does not solve the Einstein equations, the Tolman–Oppenheimer–Volkoff system not being solved in the stated context, and attribute the same defect to Bobrick–Martire, following Santiago *et al.* [3]. We take no position on that dispute. The construction below is an exact solution, a Kinnorsley exterior joined by a Lanczos junction whose balance laws hold in closed form, so the objection does not arise here. To our knowledge, the radiative warpsheet is the first *compact, subluminal, matched flat-cavity* construction to *accelerate* by a covariant, matter-derived recoil, with the *observer-robust dominant* energy condition in bulk and shell and a closed-form steering law, filling the gap those papers identify [9, 10]. The closest membrane-based proposal is Huey [19], which likewise accelerates from an extrinsic-curvature jump, and evaluates the energy conditions on the *induced* surface stress-energy. It is the one prior construction that accelerates positive-energy matter. Huey gives *both* a sub-luminal and a super-luminal model, and the sub-luminal one satisfies the *dominant* energy condition, established on the junction tensor over all future-pointing causal observers; the super-luminal one satisfies the null and weak conditions but violates the dominant condition at the outer branes. The distinction from the present work is therefore not the energy condition but *compactness*: Huey’s branes extend infinitely in the directions parallel to the wall, which he identifies as the chief obstacle to realism and as an unsatisfying way to evade the superluminal no-go theorems. Carrying no Bondi four-momentum, such a wall is not constrained by Theorem 1. The present construction is compact, asymptotically flat, subluminal, matched to a flat cavity, and satisfies the balance in the closed form $-\dot{m} \geq 3m|a|$. Fuchs *et al.* [10] point to radiation- and momentum-transfer steering (mass shed as focused emission, with ADM/Bondi four-momentum [81] as the governing constraint) as an open direction beyond simple photon rockets; the present work carries out that program, supplying the explicit matched positive-energy metric, the conservation law that sets its cost, and the dominant-energy-admissible saturating instance. The recent positive-energy constructions of Rodal and Garattini–Zatrimaylov sharpen the same distinction (Sec. 1); the table below and its footnotes record where each sits.

13.5 What is new

The ingredients (photon rockets [26, 28, 58, 29], Israel junctions [31, 67], radiating and thin shells [83, 84], and the Bobrick–Martire taxonomy [9]) are individually classical; the novelty is their synthesis into the compact, accelerating, positive-energy drive of Sec. 1, set against its nearest neighbors in Sec. 13 and Table 1. The thrust is sourced by physical radiation recoil, which distinguishes the construction from the other classical accelerating exact solution, the C-metric [85], whose uniform acceleration is supplied by a conical-singularity strut reaching to infinity rather than by emitted radiation.

14 Limitations and scope

Subluminal. The construction is causal and subluminal; it is not a faster-than-light proposal.

Not reactionless. It is a rocket: it expends mass as radiation, with the budget (15).

Felt acceleration. Passengers feel proper acceleration, not the zero- g free fall of an idealized Alcubierre interior; but in a flat Minkowski cavity the *tidal* tensor vanishes, so there is no differential stretching. This is the warp feature of Definition 1, a tidally protected passenger laboratory, and it locates the construction as the accelerating member of the positive-energy

Table 1. Where the radiative momentum warps the shell. “Obs.-robust DEC est.?” is whether the pointwise dominant energy condition has been *established* over all timelike and null observers, in bulk *and* shell; “not est.” means it was not evaluated over all observers in that work, *not* that it fails, and “fails” is entered only where NEC violation forces it. “Accel. (covariant)?” is a matter-derived change of a covariantly defined state of motion; “FTL?” records only whether a superluminal *coordinate* configuration is permitted. The discriminator is the conjunction of an observer-robust bulk-and-shell DEC with a covariant, matter-derived acceleration on a *compact* body (the last four columns jointly); “Compact?” is decisive, since a non-compact wall carries no Bondi four-momentum and is not constrained by Theorem 1. The per-entry qualifications are set out in Sec. 13: [†]Rodal (predominantly positive density; null, weak, and dominant conditions not globally satisfied); [‡]Garattini–Zatrimaylov (de Sitter bubble; weak and null conditions up to an averaged divergence, at a background-fixed velocity); [¶]Ref. [16] disputes that the Bobrick–Martire and Fuchs *et al.* shells solve the Einstein equations, the entries recording what those works report (Fuchs *et al.* use Warp Factory [82]; the margins here are analytic Hawking–Ellis Type I); [§]Huey’s sub-luminal model satisfies the DEC and the super-luminal one violates it, both on *infinite* branes.

	Bulk NEC	Obs.-robust DEC est.?	Accel. (covariant)?	Compact?	Mechanism	Energy	FTL?
Alcubierre [1]	violated	fails	no	yes	prescribed shift	negative	yes
Natário [7]	violated	fails	no	yes	prescribed shift	negative	yes
Van Den Broeck [8]	violated	fails	no	yes	prescribed shift	negative	yes
Bobrick–Martire [9]	satisfied [¶]	not est.	no	yes	static shell	positive	no
Lentz [13, 3, 15]	fails	fails	no	yes	plasma soliton	negative	yes
Fuchs <i>et al.</i> [10]	satisfied [¶]	yes (num.)	no	yes	constant- <i>v</i> shell	positive	no
Rodal [11]	reduced [†]	not est.	no	yes	constant- <i>v</i> shift	mostly positive	no
Garattini–Zatrimaylov [12]	averaged [‡]	not est.	no	yes	comoving bubble (dS)	positive	n/a
Huey [19]	satisfied	sub-lum.: <i>yes</i> [§]	yes	no	membrane jump	positive	both
This work	satisfied	yes	yes	yes	positive null flux	positive	no

shell family. The cavity is Born-rigid, so its Rindler horizon must stay off the shell: $\lambda = aR < 1$. This is the weakest of the amplitude bounds, the kinematic ceiling (40) being strictly tighter.

Energy cost. The exact budget (15) is $m_f/m_0 = e^{-3 \int |a| du}$; for a *rectilinear* boost the integrated proper acceleration equals the net rapidity, $\int |a| du = \Delta\eta$, and the cost is severe: $m_f/m_0 = e^{-3\Delta\eta}$, so even mildly relativistic boosts radiate the bulk of the rest mass (the worked saturating burn of the Supplement radiates $\approx 51\%$ of its rest mass for a rapidity gain of only 0.24), and approaching the speed of light ($\Delta\eta \rightarrow \infty$) sends $m_f/m_0 \rightarrow 0$. A *steering* history that changes direction has $\int |a| du \geq |\Delta\eta|$ (the integrated rapidity arc exceeds the net rapidity, as boosts do not add as vectors), so a turn costs strictly more than the collinear figure. The cost is steep, but *positive and finite*: no negative energy and no quantum-inequality violation appear at any point. The barrier is the size of the budget, the sense in which warp propulsion is here an engineering problem.

Local in retarded time. The solution is a finite maneuver on a *u*-slab; global, eternal existence is not claimed, and $m(u) > 0$ on the slab by (14) (Prop. 4).

Positivity for the distributional shell. The Bondi-mass step of Theorem 1 needs only the sign of the junction surface term, which the surface DEC fixes (Lemma 10), and the construction has $m_B = m(u) > 0$ directly. At spatial infinity the scalar ADM mass of the C^0 shell is non-negative by the published low-regularity positive-mass theorem [48], which is all Corollary 3 uses; future-causality of the full ADM four-momentum follows if the shell meets the corner hypotheses of the Lorentzian distributional theorem [51]. The *Bondi* refinement at $\mathcal{I}^+$ is left open (and is not needed for the bound). On an asymptotically null slice the natural route is a Ludvigsen–Vickers propagation of the Witten spinor inward from $\mathcal{I}^+$ [45, 46]; closing it requires three inputs an outward parallel transport does not supply. First, the transverse spinor jump at the shell must lock onto the extrinsic-curvature jump $[K_{ab}]$, so that the corner term $\oint_C \bar{\lambda} [\tilde{\nabla}^\perp \lambda]$ combines with the Israel term into the DEC-non-negative surface contribution rather than a residual indefinite one. Second, the propagated spinor must retain the asymptotic constancy and peeling fall-off that define the Bondi charge; boundedness alone does not. Third, the shell must not focus the null congruence into a caustic before $\mathcal{I}^+$. These are the precise missing analytic inputs, and the refinement remains open.

Existence. Theorem 2 is unconditional in the small-amplitude regime, its matching closing analytically on the round-sphere rigidity operator (Lemma 4). The acceleration–compactness frontier is bracketed between a rigorous lower bound and the closed-form ceiling (Lemma 6); the tighter finite-amplitude value is mapped numerically in the Supplement.

Dynamical existence. Theorem 4 lifts admissibility to a time-evolved spacetime for slow, fixed-axis (non-rotating, reflection-symmetric) sub-horizon burns, the shape rigidly

determined order-by-order through the constant-cut compatibility identity (84). At small amplitude the matched *shell* exists per retarded cut for an *arbitrary* acceleration axis (the $SO(3)$ -equivariant $\ell = 1$ closure, Cor. 6), and the finite-amplitude results are computed per cut in the momentary axisymmetric rest frame, covering any planar maneuver in full and a rotating one cut by cut; assembling the per-cut shells into a single *rotating*-axis worldtube is left open (the Conclusion). The leading retarded correction, the outgoing mass-loss flux L , *tightens* the thin-shell surface DEC margin, which carries the term $-L/(8\pi R s^2)$ and so is strictly decreasing in L (97); the fully dynamical frontier is therefore tighter than the quasi-static $g(x)$.

Thickening. A generic *radial-tension* smoothstep wall reproduces the source–vacuum boundary cost, whereas the *tangential-pressure* wall is strictly DEC-admissible across its whole width for local $2m/r < 4/5$, recovers the thin-shell anchor, is a self-consistent anisotropic equilibrium, and stays admissible once outgoing null radiation streams through it (Sec. 10, App. F). Its $\mathcal{O}(\mu)$ back-reaction is closed at the adiabatic level, but the saturating burn has an order-unity ωR , so the fully time-dependent $\mathcal{O}(\mu^2)$ correction is not parametrically negligible there and remains open.

Linear stability. The realized tangential-pressure wall sits on the Poisson–Visser marginal curve $V''(R) = 0$, and under the rear-pole acceleration redshift it moves onto the barrier side, so a slightly *stiffer*, still-admissible wall is *required* (not merely available) to keep the restoring force positive ($\Omega^2 > 0$); the stiffening costs nothing in the (junction-fixed) surface DEC margin and stays subluminal throughout $x < x_\eta \simeq 0.844$ (App. I). In the spherically symmetric Vaidya model, keeping the mass-history channel $\delta m = m_{,u} \delta u$ that a radial perturbation opens, the outgoing flux $L = -m_{,u}$ weakly anti-damps the radial mode ($\kappa_d < 0$ on that branch, the Supplement), so the radiating equilibrium is linearly unstable rather than stable: away from the marginal wall the $\mathcal{O}(L)$ growth is finite-burn-bounded by the fuel budget, so the radial mode is not destabilized on finite maneuvers, though it is not asymptotically stable. The semiclassical instability that afflicts superluminal warp drives [86] does not arise here: the drive is subluminal and horizon-free on the finite maneuver slab, so there is no horizon to seed it. The angular modes of the full accelerating shell and the DEC-respecting finite-thickness wall remain open.

15 Conclusion

We have inverted the warp-drive design problem. Rather than prescribing a metric (and inheriting exotic matter) or a static source (and forgoing acceleration), we prescribe the passenger worldtube, accept positive null radiation as the exterior, and solve for the shell. The exterior is an exact photon rocket whose energy conditions reduce to $n^2 \geq 0$, and steering obeys the positive-energy control law $-\dot{m} \geq 3m|a|$. Admissible accelerating shells exist at small amplitude about the static Schwarzschild–Minkowski anchor, their matching closing by an implicit-function argument on the round-sphere rigidity operator with the $\ell = 0$ monopole absorbed by the radial breathing mode. An acceleration–compactness frontier caps the proper acceleration, below the rigorous closed-form ceiling $\frac{1}{2}(1 - x)$.

The organizing principle is model-independent: no asymptotically flat, dominant-energy drive with a confined material source can change its velocity without radiating four-momentum to null infinity (Theorem 1). This flux balance is classical Bondi–Sachs; our contribution is to read it as a no-reactionless-steering principle, the general-relativistic counterpart of the relativistic rocket, and to read it directly from the curvature across the Kinnersley rocket, static Schwarzschild, and Vaidya. Reactionless steering is therefore impossible, and the velocity reading is exact between stationary Bondi frames and, for the closed-form Kinnersley rocket, instantaneous (Prop. 1). Our construction is the explicit positive-energy instance realizing this law for an accelerating warp drive, with the minimum-radiation maneuver the gravitational-wave-silent Damour dipole. Acceleration is paid for by positive radiation, not negative energy. Within the compactness window $x + 2aR < 4/5$ the thin shell is realized as the limit of an admissible *radiating* tangential-pressure wall that stays dominant-energy-admissible under its own $\mathcal{O}(\mu)$ back-reaction, and the worked saturating burn (the Supplement) lies inside this window throughout, so that burn itself, not merely its endpoints, is realized by an admissible radiating wall at the adiabatic $\mathcal{O}(\mu)$ level, the fully time-dependent $\mathcal{O}(\mu^2)$ dynamics remaining open (Secs. 10.2, 10.3).

A steering maneuver is in principle observable: the mandatory exhaust is a transient, beamed, strictly outgoing positive-energy flux whose saturating angular pattern (a single rear lobe vanishing at the forward pole) is a sharp signature, connecting to proposals to detect emissions from

positive-energy warp bubbles [87] and complementing the gravitational-wave burst computed for warp-drive containment failure [88].

Beyond the principal contributions listed in Sec. 1, the following supporting results are established analytically:

- the closed-form control law and the small-amplitude existence closure on the round-sphere rigidity operator (Lemma 4, App. D);
- the universal rocket bound (Cor. 2): the radiated four-momentum is a causal vector, so mass loss dominates momentum gain in the rest frame;
- the radiating-wall lapse response (Theorem 7), the back-reaction sign density-conditional in the dense core (Prop. 9);
- the radial-tension thickening obstruction with its $1/\delta^2$ boundary-cost law (Prop. 6, Cor. 7);
- the gravitational-wave silence of the Damour dipole, analytic from the exact Kinnersley news, the peeling analytic by the generalized Goldberg–Sachs theorem for the algebraically special exterior (App. H);
- maneuver design as a state-constrained geodesic-control theorem on $\mathbb{H}^3$, universal within the news-silent Robinson–Trautman class (Theorems 8, 6);
- the radiating equilibrium is linearly *unstable*, the outgoing flux anti-damping the radial mode ($\kappa_d < 0$), with growth of order the luminosity and bounded over any finite burn by the fuel budget (Sec. 11).

The acceleration–compactness frontier is trapped in a rigorous analytic band between the lower bound $c(x)$ (Lemma 6) and the closed-form ceiling; only its sharp finite-amplitude value $\underline{g}(x) \sim 0.2$ remains a numerical refinement, not an open matter of rigor. The axial frozen-shape diagnostic $\lambda_{\text{DEC}}(x)$, a distinct object from the operative frontier $g(x)$, is *proved* (Sturm) to lie strictly below the same ceiling yet within 0.7% of it ($\lambda_{\text{DEC}}/[\frac{1}{2}(1-x)] \approx 0.9931$ at $x \approx 0.631$).

Several open directions remain, with scope set out in Sec. 14. The fixed-axis assembly closes to all amplitude orders through the constant-cut compatibility identity (84), but the *rotating*-axis worldtube closes only through first order in the rotation rate about the round anchor (Lemma 11); whether the boost–rotation non-commutativity leaves a residual $\ell \geq 2$ magnetic-parity defect, the odd-parity analog of (84), is open. The distributional Bondi-mass positivity is closed at spatial infinity, its refinement at $\mathcal{I}^+$ open. Beyond the analytic $\mathcal{O}(\mu)$ radiating-wall theorem (Theorem 7), the non-adiabatic $\mathcal{O}(\mu^2)$ shell dynamics, the DEC-respecting finite-thickness wall, and the angular stability modes of the full accelerating shell remain open, while the flux-coupled radial mode stays bounded over any finite burn (the Supplement) and the de Sitter budget closes to all orders (Lemma 2). None of these bears on Theorem 1 or the exact exterior: steering a warp drive is a problem of energy budget, not of negative energy.

Data availability

The code that reproduces the numerical results of the Supplement is openly available at https://github.com/anindex/world_tube.

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A Photon-rocket conventions and curvature

We use the null tetrad aligned with $\ell_a = -(\mathrm{d}u)_a$ and the Robinson–Trautman conventions of Refs. [26, 29, 58]. The cleanest route to Eq. (8) is the Kerr–Schild form of the photon rocket on a flat seed,

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2m(u)}{r} \ell_\mu \ell_\nu, \quad (57)$$

with ℓ the retarded outgoing null direction from an accelerated worldline $z^\mu(u)$ (u its proper time) and $r = -(X - z) \cdot \dot{z}$ the retarded distance; an arbitrary conformal factor $P(u, \zeta, \bar{\zeta})$ with $K = 1$ but *not* of boost form does not solve the field equations. For an accelerated worldline with proper acceleration a along a fixed axis ($\vartheta = 0$), evaluated in the momentary rest frame of the cut ($\eta = 0$): (i) the Ricci scalar vanishes; (ii) $T_{ab} = (n^2/r^2)\ell_a \ell_b$ is exactly rank-one null; and (iii) $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$, a *pure dipole* (in that rest frame) of amplitude exactly $3m|a|$, maximal at the rear (exhaust) pole $\vartheta = \pi$ and vanishing at the forward pole at saturation; at finite rapidity the fixed-frame pattern is its aberration. Positivity for all ϑ is therefore equivalent to the control law (14), and the observer-robust margins are non-negative iff $n^2 \geq 0$ (Proposition 3). Exact null dust is algebraically Hawking–Ellis Type II (a defective null eigenvector) on the Type I $\leftrightarrow$ II boundary; its frame-invariant content is the rank-one null structure above, dominant-energy-admissible precisely when $n^2 \geq 0$.

The admissibility test the paper uses throughout is the pointwise dominant energy condition read over *all* observers, not a single comoving frame.

Lemma 9 (Observer-robust dominant energy condition). *A symmetric stress tensor T_{ab} satisfies the dominant energy condition for every future-directed timelike observer, equivalently $-T^a{}_b u^b$ is future-causal for all future-timelike u^a . For a Hawking–Ellis Type I tensor [25, 84] this holds exactly when the energy density dominates each principal pressure, $\rho \geq |p_i|$; adding a non-negative rank-one null flux ($n^2 \geq 0$, the Type II case on the Type I $\leftrightarrow$ II boundary carried here) preserves it, the dominant-energy tensors forming a convex cone; Type III and Type IV tensors, carrying a spacelike or superluminal energy flux, violate it. The binding margin is thus the minimum over the full boost orbit, not the value in one frame.*

Proof. The eigenvalue criteria are the dominant-energy specialization of the Hawking–Ellis types [25, 84]. The Type I bound $\rho \geq |p_i|$ is Lemma 10 applied in the eigenframe (there for the surface stress, verbatim for a bulk Type I tensor), and the non-negative null flux is the rank-one case of Proposition 3, whose sum with a dominant-energy Type I core again satisfies the condition because $-T^a{}_b u^b$ is additive in T_{ab} and future-causal vectors form a cone; Type III and IV carry a spacelike or superluminal energy flux and fail the weak condition already. The bulk and surface instances used in the text are therefore established here in closed form. $\square$

B Junction algebra and the balance laws

For the static shell, $K_{ab}^\theta = \sqrt{f}/R$ and $K_{ab}^\tau = f'/(2\sqrt{f})$ on each side ($f_- = 1$, $f_+ = 1 - 2m/R$); the Lanczos equation (10) then gives Eqs. (22)–(23) *exactly*, and

$$8\pi R(\sigma_0 - p_0) s = \frac{1}{2}(1 - s)(5s - 1), \quad s = \sqrt{1 - x}, \quad (58)$$

positive iff $1/5 < s < 1$, i.e. the surface-DEC window $x < 24/25$ of Eq. (25). The moment identities (29) follow from the angular structure of n^2 alone: with $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$,

$$\oint n^2 \mathrm{d}\Omega = -\dot{m}, \quad -\oint n^2 \cos \vartheta \mathrm{d}\Omega = ma, \quad (59)$$

(the dipole using $\oint \cos^2 \vartheta \mathrm{d}\Omega = 4\pi/3$), so the monopole of the radiated flux is the luminosity $-\dot{m}$ and the dipole gives the thrust $\mathrm{d}P_B^z/\mathrm{d}u = -\oint n^2 \cos \vartheta \mathrm{d}\Omega = ma$, reproducing the Bondi budget (12). These are moments of the amplitude n^2 , the Bondi flux density at $\mathcal{I}^+$; the corresponding *local* flux crossing Σ at finite radius, $T_{N\hat{0}}^+ = (n^2/R^2)(\ell_a N^a)(\ell_b e_0^b)$, carries the angle-dependent normal and redshift factors of the exact lapse (41), so its finite-radius moments differ from (29) at $\mathcal{O}(\lambda)$ and are not used. No consistency condition need be imposed to relate them, because the balance laws (27)–(28) are not extra postulates but the contracted Gauss–Codazzi (projected Bianchi) identities for a timelike hypersurface. With N^a the unit spacelike normal and $\bar{K}_{ab}$ the mean extrinsic curvature, the distributional Einstein equations give $D_b S^b{}_{\hat{a}} = -[T_{N\hat{a}}]$ (tangential) and $S^{ab} \bar{K}_{ab} = -[T_{NN}]$ (normal): the standard *timelike*-shell relations of Israel [31] (Poisson [32], §3.7–3.8). The surface stress is *not* conserved precisely because the crossing flux $[T_{N\hat{a}}] \neq 0$ sources it. Here the interior is a vacuum cavity ($T_{ab}^- = 0$), so $[T_{N\hat{a}}] = T_{N\hat{a}}^+$ and $[T_{NN}] = T_{NN}^+$: the

radiation is *emitted at* Σ (the regularized rocket nozzle), not transmitted through it from the cavity, and the induced surface energy flux $S^{\hat{0}\hat{a}}$ balances the lateral recoil. (We use “Darmois” for the first junction condition $[h_{ab}] = 0$, continuity of the induced metric, which fixes $R(u, \Omega)$; the second fundamental form jumps, and the Lanczos S_{ab} carries that jump, so a shell is present.)

Surface dominant energy and the junction term. Step (iii) of Theorem 1 uses one property of the distributional shell: that it adds a non-negative term to the Witten–Nester spinor expression for the Bondi four-momentum.

Lemma 10 (Surface dominant energy and the junction term). *Let S_{ab} be a Hawking–Ellis Type I surface stress, with surface rest frame u^a , $S_{ab} = \sigma u_a u_b + \sum_A p_A e_{\hat{A}a} e_{\hat{A}b}$ (σ the surface energy density, p_A the principal surface pressures). Then $S_{ab} V^a W^b \geq 0$ for every pair of future-directed causal V^a, W^a if and only if the surface dominant energy condition $\sigma \geq \max_A |p_A|$ holds. Consequently, integrating the Sen–Witten identity across Σ on an asymptotically null slice, the distributional shell contributes the single boundary term $8\pi \oint_{\Sigma} S_{ab} \xi^a J^b dA$, the positive coefficient 8π being that of the distributional Israel junction (the surface stress is the codimension-one part of $G_{ab} = 8\pi T_{ab}$, so it enters the Sen–Witten matter term with the same sign as bulk stress-energy), with ξ^a the asymptotic time translation and $J^a = \bar{\lambda} \gamma^a \lambda$ the Witten–spinor Dirac current (both future-directed causal) which is non-negative whenever Σ is admissible.*

Proof. Write $\gamma_V = -u \cdot V > 0$, $\gamma_W = -u \cdot W > 0$ and the spatial parts $V_{\perp}^A = e_A \cdot V$, $W_{\perp}^A = e_A \cdot W$; for causal V, W , $-V \cdot V = \gamma_V^2 - |V_{\perp}|^2 \geq 0$ gives $|V_{\perp}| \leq \gamma_V$ and likewise for W . Then $S_{ab} V^a W^b = \sigma \gamma_V \gamma_W + \sum_A p_A V_{\perp}^A W_{\perp}^A$, and by Cauchy–Schwarz $|\sum_A p_A V_{\perp}^A W_{\perp}^A| \leq (\max_A |p_A|) |V_{\perp}| |W_{\perp}| \leq (\max_A |p_A|) \gamma_V \gamma_W$, so $S_{ab} V^a W^b \geq (\sigma - \max_A |p_A|) \gamma_V \gamma_W \geq 0$. Conversely, if $\sigma < |p_B|$ for some B , the future-null pair $V = u + e_{\hat{B}}$, $W = u - \text{sgn}(p_B) e_{\hat{B}}$ gives $S_{ab} V^a W^b = \sigma - |p_B| < 0$. The Sen–Witten boundary integrand is $S_{ab} \xi^a J^b$ with ξ, J future-causal. Since S_{ab} is tangential to the timelike Σ ($S_{ab} N^b = 0$), only the tangential projections $\xi_{\parallel}, J_{\parallel}$ enter the contraction, and the projection of a future-causal spacetime vector onto a *timelike* hypersurface is itself future-causal in the induced Lorentzian 3-metric; the bound above, applied to $\xi_{\parallel}, J_{\parallel}$, gives the non-negativity. $\square$

The matched shell of Theorem 2 is Type I with strictly positive surface-DEC margin, so its boundary term is strictly positive; the flat cavity, being vacuum, contributes none. This sign is the only property of the distributional junction that the Bondi-mass positivity step of Theorem 1 requires; the spinor existence and decay across the C^0 junction that a full distributional positive-Bondi-mass theorem at $\mathcal{I}^+$ would also establish is left open (Sec. 14).

C The dipole response and the frontier

The dipole coefficients σ_1, p_1, q_1 follow from the Darmois condition $[h_{ab}] = 0$ for the shell shape $R(u, \Omega)$, a nonlinear isometric-embedding problem on S^2 : the induced 2-metric of the anisotropic exterior must match that of the Rindler-accelerated flat cavity. Linearizing about the anchor, the steering enters only through the $\ell = 1$ sector of $(\ln P)_{,u} = -a \cos \vartheta$, which by S^2 orthogonality decouples from the monopole, reducing the embedding to a gauge-invariant linear per-mode system matched against the exact Kerr–Schild exterior. The matched surface stress is Hawking–Ellis Type I across the admissible set (no Type IV transition) with worst-observer surface DEC margin strictly positive on an open subinterval of $2m/R < 24/25$, so Theorem 2’s neighborhood is non-empty. Two features fix the frontier (Sec. 8): the exterior is locally Schwarzschild to $\mathcal{O}(a)$, so $\sigma_1 = \mathcal{O}(x^2)$, and the recoil flux enters the margin only as q^2 ; the linear envelope $g_{\text{lin}}(x) = (\sigma_0 - p_0)/(\sigma_1 - p_1)|_{\vartheta=\pi} \sim \mathcal{O}(1/x)$ is therefore wide, and the finite-amplitude frontier $g \sim \mathcal{O}(10^{-1})$ is an $\mathcal{O}(\lambda^2)$ effect where the recoil flux and the nonlinear shape response re-enter.

Anchor solvability. The anchor linearization DF_0 is the round-sphere normal operator $\Delta_{S^2} + 2$ on the shape, a bounded isomorphism off the $\ell = 1$ translation gauge (proved in App. D); it supplies the quantitative implicit-function radius $\delta_Q(x) > 0$ and the uniform inverse bound $\Gamma(x)$ on which Lemma 6 builds the rigorous lower bound $c(x) > 0$. The remaining sharp *numerical* value $g(x) \sim 0.2$ across $x \in [0.1, 0.7]$ (hardest near the effective horizon $f - 2aR \rightarrow 0$ at high x) is deferred to the Supplement. The frontier is set by the shape anisotropy $\sigma - p$ at the rear pole (the recoil flux decouples, Sec. 8).

Why the rear pole, in closed form. On the symmetry axis the retarded structure is exactly solvable: for a point a coordinate distance b behind the source the areal radius is $r = b(1 - ba/2)$, the conformal factor $w^2 = 1 - 2ar$, and the squared gradient of the areal radius is

$$|\nabla r|^2 = 1 - \frac{2m}{r} \mp 2ar = f \mp 2ar \quad (60)$$

($-$ at the rear, $+$ at the front), matching the exact Kerr–Schild metric; equivalently the exact retarded inverse metric $g^{rr} = 2H = f + 2ar \cos \vartheta$ is $f - 2ar$ at the rear pole ($\vartheta = \pi$) and $f + 2ar$ at the front, so the rear pole carries effective compactness $x + 2ar$ and the effective horizon $f - 2ar \rightarrow 0$ is approached there. This deepening closes the *regularity* ceiling at the rear (Prop. 5) but not the dominant-energy frontier. The exterior extrinsic curvatures are closed-form, $K^{\hat{\tau}\hat{\tau}+} = (m/r^2 - a)/\sqrt{f - 2ar}$, $K^{\hat{\theta}\hat{\theta}+} = \sqrt{f - 2ar}/r$, with the flat-Rindler interior giving $K^{\hat{\tau}\hat{\tau}-} = -a/(1 - ar)$, $K^{\hat{\theta}\hat{\theta}-} = 1/r$, and the rear-pole margin $\sigma(\pi) - p(\pi) = -\frac{1}{8\pi}([K^{\hat{\tau}\hat{\tau}}] + 3[K^{\hat{\theta}\hat{\theta}}])$. The frame term $[K^{\hat{\tau}\hat{\tau}}]$ is closed-form and nearly λ -independent; the constant $g(x)$ is set by the $\mathcal{O}(\lambda^2)$ shape response in $[K^{\hat{\theta}\hat{\theta}}]$, mapped numerically ($\underline{g} \sim 0.2$). With the shape frozen both rear-pole margin coefficients are positive on $x < \frac{3}{4}$ (Sturm, below), so acceleration alone makes the rigid pole *safer* and the frontier is forced by the global shape response, not the low-order jets.

Route to a rigorous lower bound on the frontier. The frozen-shape rear-pole margin is closed-form through the jumps above, $8\pi R M_{\text{frozen}}(x, \lambda) = -R([K^{\hat{\tau}\hat{\tau}}] + 3[K^{\hat{\theta}\hat{\theta}}])$, and its retarded expansion at fixed compactness is

$$8\pi R M_{\text{frozen}}(x, \lambda) = M_0(x) + c_1(x)\lambda + c_2(x)\lambda^2 + \mathcal{O}(\lambda^3), \quad M_0(x) = -\frac{(5s-1)(s-1)}{2s}, \quad (61)$$

with $s = \sqrt{1-x}$, $M_0 = 8\pi R(\sigma_0 - p_0) > 0$ the anchor margin, and the closed forms $c_1(x) = -1 + 7/s + (5x-6)/(2s^3)$, $c_2(x) = -1 + \frac{5}{2s^3} - \frac{3x}{4s^5}$ (from the exact axis $[K]$ -jumps). Both are strictly positive on $x < \frac{3}{4}$, by Sturm’s theorem in each case: $2s^3c_1 = -2s^3 + 9s^2 - 1$ and $4s^5c_2 = -4s^5 + 13s^2 - 3$ each have no root on $s \in [\frac{1}{2}, 1]$, and both are positive there (at $s = \frac{1}{2}$ they equal 1 and $\frac{1}{8}$). The two-term expansion itself converges only below the effective-horizon branch point $\lambda < \frac{1}{2}(1-x)$ (Prop. 5), which the computed $g(x)$ respect. With both coefficients strictly positive, at the computed amplitudes $\lambda = g(x) \lesssim 0.2$ the rigid pole is strictly *safer* than the anchor, so the surface-DEC closure at $g(x)$ is driven by the *global shape response*, not the frozen-shape redshift (the rigid margin itself closes only near the kinematic ceiling for $x \gtrsim 0.6$). This establishes the frozen expansion in closed form: the rigid pole is safer at low order.

The *front* pole closes the complementary gap. In the axis-static reduction of the exterior ($u = t - \int dr/h$, $h = f + 2aR$ at the front), matching the rigidly accelerated flat interior (Rindler pole redshift $RK^{\hat{\tau}\hat{\tau}-} = +\lambda/(1+\lambda)$) across $r = R$ gives the rigid front-pole margin $8\pi R s_-(\sigma - p) = 3s_- - 3 + \frac{5}{2}x - 7\lambda + s_-\lambda/(1+\lambda)$ with $s_- = \sqrt{1-x+2\lambda}$, which *falls* with λ and vanishes at $\lambda_{\text{DEC}}(x)$, the smallest positive root of (42). Neglecting the $\mathcal{O}(\lambda)$ interior redshift gives its static-interior estimate (43),

$$\tilde{\lambda}_{\text{DEC}}(x) = \frac{6\sqrt{16-14x} - |24-35x|}{98} \approx \lambda_{\text{DEC}}(x), \quad (62)$$

whose absolute value locates the estimate’s front/rear crossover with no branching condition: the front pole binds for $x \lesssim 24/35$ and the rear beyond (for the exact threshold the crossover sits at $x \simeq 0.63$). The interior redshift is an $\mathcal{O}(\lambda)$ term (its coefficient is the -1 in c_1 of (61)), raising the front-binding threshold from $\tilde{\lambda}_{\text{DEC}} = x/4 + \mathcal{O}(x^2)$ to $\lambda_{\text{DEC}} = x/3 + \mathcal{O}(x^2)$ near $x \rightarrow 0$ (the limiting ratio $\frac{3}{4}$ of (44)) and leaving the rear-binding value within 1% for $x \lesssim 0.84$, the relative error rising monotonically to its supremum $41/40 - 1 = 2.5\%$ as $x \rightarrow 24/25$. Both frozen thresholds still vanish linearly as $x \rightarrow 0$; what removes that decay is the shape response, not the interior term. This rigid front threshold lies well below the matched frontier at low compactness ($\underline{g} \approx 0.2$ there), and the discrepancy is precisely the shape response: a rigid *spherical* shell incurs the linear -7λ front penalty against the front-back-asymmetric exterior, whereas the matched Darmois shell shape-responds (its dipole slope $\sigma_1 - p_1 = \mathcal{O}(x^2)$, below) and does not, so the front pole lifts clear and the operative frontier binds at the rear. The poles bind before any off-pole angle at leading order. The exact off-axis lapse is $2H = f + 2ar \cos \vartheta$ (41), affine in $\cos \vartheta$ with positive slope, so g^{rr} is strictly larger off-axis and regularity is pole-dominated *exactly*, at every angle and to all orders in a . The $\mathcal{O}(\lambda)$ correction to the s -weighted worst-observer margin $8\pi R \sqrt{h}(\sigma - p)$ is the pure dipole $\lambda \cos \vartheta (3/s_0 - 7)$, $s_0 = \sqrt{1-x}$, extremal at $\vartheta = 0, \pi$, so the frozen-shape margin is controlled by

the two poles; the leading-order dipole fixes the binding angle, not which pole (λ_{DEC} of Lemma 5 being their exact, all-orders-in- λ value). The full-angle finite-amplitude margin receives the $g_{u\vartheta}$ tilt and the $\mathcal{O}(a^2)$ shape ρ_2 at the same order, and stays numerical. The frozen-shape threshold $\lambda_{\text{DEC}}(x)$ (Eq. (62) its static-interior estimate) is therefore a rigorous *frozen/rigid-shape* axial diagnostic (a front-pole, spherical-match threshold), a distinct object from the operative frontier $g(x)$ and not globally ordered with it: at low compactness the front-pole λ_{DEC} lies well below the matched rear-pole $g(x)$, while near the effective-horizon regime the rigid rear threshold rises toward the kinematic ceiling. The two lie at different binding poles and for different shells, so (62) is not a bound on the operative frontier; the sole rigorous closed-form member of the matched frontier hierarchy remains the kinematic ceiling $\frac{1}{2}(1-x)$ (Prop. 5).

The *matched* worst-observer margin Φ , by contrast, binds at the rear pole and *falls* with λ , the matched dipole $\sigma_1 - p_1 > 0$ reducing it. The $\mathcal{O}(\lambda)$ dipole seeds are constrained but not fully fixed: the frame tilt is closed-form, $w_1(x) = x/(2s)$, the one reparametrization invariant of the linear junction, while the redshift and radial seeds (A_{t1}, ρ_1) form a one-parameter rigid-translation gauge orbit fixed only by the $\mathcal{O}(\lambda^2)$ centering and the all- ℓ shape response. This is the gauge-structural reason the frontier constant lies beyond every finite-mode truncation. Split Φ as

$$8\pi R \Phi(x, \lambda) = M_0(x) - b_1(x) \lambda - \mathcal{D}_{\text{sh}}(x, \lambda), \quad b_1(x) = 8\pi R(\sigma_1 - p_1)|_{\pi}, \quad \mathcal{D}_{\text{sh}}(x, \lambda) \geq 0, \quad (63)$$

with $M_0 = 8\pi R(\sigma_0 - p_0) > 0$ the anchor margin, b_1 the matched linear slope, and $\mathcal{D}_{\text{sh}}$ the nonlinear shape-response reduction. We now prove the rigorous lower bound of Lemma 6 by bounding $\mathcal{D}_{\text{sh}}$ uniformly in the angle.

Proof of Lemma 6. Work on the sub-horizon strip $\lambda \leq \Lambda_h(x) = \frac{1}{4}(1-x)$. There the rear effective potential $f - 2aR = 1 - x - 2\lambda \geq \frac{1}{2}(1-x)$ and the interior acceleration factor $1 - aR \geq \frac{3}{4}$ are bounded away from zero, so the extrinsic-curvature jumps $[K_{ab}]$, and hence the worst-observer margin $m(\vartheta; q, \lambda)$, are real-analytic functions of the shell configuration $q = (Y, A_t, w)$ (shape, lapse and tilt seeds) and of λ , with closed-form derivative suprema on the strip.

(i) *The matched branch is quantitatively controlled.* By the implicit-function theorem of App. D, the Darmon system $F(q, \lambda) = 0$ has, off the $\ell = 1$ translation gauge, a solution branch $q(\lambda)$ with $q(0) = 0$ and a uniform inverse bound $\|F_q^{-1}\|_{C^{0,\alpha} \rightarrow C^{2,\alpha}} \leq \Gamma(x)$ over a quantitative implicit-function radius $\lambda \leq \delta_Q(x)$, positive by continuity from the invertible anchor operator, Γ assembled from the round-sphere Schauder constant C_S (76) (shape sector), the $\ell = 0$ breathing slope $-x/s^2$ (80), and the $\ell = 1$ cos-row and tilt block (77). Differentiating $F(q(\lambda), \lambda) = 0$ once and twice,

$$q' = -F_q^{-1} F_\lambda, \quad q'' = -F_q^{-1} (F_{\lambda\lambda} + 2F_{q\lambda} q' + F_{qq} [q', q']), \quad (64)$$

so with the closed-form source suprema $S_\lambda, S_{\lambda\lambda}, S_{q\lambda}, S_{qq}$ of the Darmon map on the strip, $\|q'\| \leq U_1 := \Gamma S_\lambda$ and $\|q''\| \leq U_2 := \Gamma(S_{\lambda\lambda} + 2S_{q\lambda} U_1 + S_{qq} U_1^2)$, both finite closed forms.

(ii) *Uniform quadratic bound on the reduction.* Throughout this step m, A, C_{shape} carry the dimensionless normalization of (47), i.e. $8\pi R$ times the raw margin, so their common anchor value is $M_0 = 8\pi R \delta_0$. The worst-observer margin $m(\vartheta; q, \lambda)$ is the pointwise minimum of the finitely many Hawking–Ellis Type I candidate margins $8\pi R(\sigma - |p_A|)$ (one per principal surface pressure); near the anchor the shell pressures are strictly positive and the eigenvalues simple, so each candidate is real-analytic in (q, λ) and all share the common anchor value M_0 . It suffices to bound each candidate, the minimum of the resulting lower bounds being again (47). Fix a candidate; Taylor’s theorem gives $M(\vartheta, \lambda) = M_0 + m_1(\vartheta; x)\lambda + r(\vartheta, \lambda)$, with $m_1(\vartheta; x) = \partial_\lambda M|_{\lambda=0}$ the signed linear coefficient (negative where the margin falls) and $|r| \leq \frac{1}{2}\lambda^2 \sup_{[0,\lambda]} |M_{\lambda\lambda}|$, and

$$M_{\lambda\lambda} = m_{\lambda\lambda} + 2m_{\lambda q} q' + m_{qq} [q', q'] + m_q q''. \quad (65)$$

Each candidate is a finite 2-jet of q and the embedding $C^{2,\alpha} \hookrightarrow C^2$ controls pointwise angular values, so each factor in (65) is bounded on the strip by the closed-form candidate-derivative suprema (written $M_{\lambda\lambda}, M_{\lambda q}, M_{qq}, M_q$ for their sup norms) together with U_1, U_2 , uniformly in ϑ :

$$C_{\text{shape}}(x) := \frac{1}{2} (M_{\lambda\lambda} + 2M_{\lambda q} U_1 + M_{qq} U_1^2 + M_q U_2) < \infty, \quad (66)$$

whence $|r(\vartheta, \lambda)| \leq C_{\text{shape}}(x) \lambda^2$ for all ϑ ; taking $A(x)$ and $C_{\text{shape}}(x)$ as the maxima over the finitely many candidates gives a single uniform bound. This is the shape-reduction bound, uniform in angle: it needs no separate proof that the rear pole is the binding direction.

(iii) *The frontier lower bound.* The bound needs only that the worst-angle linear reduction $A(x) = \|(-m_1(x, \cdot))_+\|_\infty$ and $C_{\text{shape}}(x)$ are *finite*, established in closed form by (i)–(ii); the

positivity of $c(x)$ does not rest on their size. A refinement fixes that size: the matched linear response is locally Schwarzschild, its rear-pole dipole slope $\sigma_1 - p_1 = \mathcal{O}(x^2)$ (App. C), and the recoil flux enters the margin only as q^2 , whose λ -derivative vanishes at the anchor, so $A(x) = \mathcal{O}(x^2)$, the linear envelope $M_0/A \sim 1/x$ is wide, and the binding is carried at $\mathcal{O}(\lambda^2)$. Combining, for every ϑ

$$m(\vartheta; x, \lambda) \geq M_0(x) - A(x)\lambda - C_{\text{shape}}(x)\lambda^2, \quad (67)$$

which is (47); any finite A, C_{shape} suffice, the right side being positive below the least positive root $(\sqrt{A^2 + 4C_{\text{shape}}M_0} - A)/(2C_{\text{shape}})$ (or M_0/A if $C_{\text{shape}} = 0$), and intersecting with $\lambda \leq \min\{\Lambda_h(x), \frac{1}{2}\delta_Q(x)\}$ (the factor $\frac{1}{2}$ keeping the branch strictly inside the contraction ball) gives $c(x)$ (48), positive since $M_0 > 0$. With the kinematic ceiling this brackets the frontier, $c(x) \leq g(x) \leq \frac{1}{2}(1-x)$. $\square$

The constants in closed form. The proof above needs only finiteness of $\Gamma, \delta_Q, A, C_{\text{shape}}$. We now evaluate them, so that $c(x)$ is an explicit curve.

The inverse bound. At the anchor the linearized Darboux map DF_0 is block-diagonal in the spherical harmonics, and each block has already been computed: the $\ell \geq 2$ shape blocks are the rigidity operator $\Delta_{S^2} + 2$ with eigenvalues $2 - \ell(\ell + 1)$, so their inverse multipliers are $|2 - \ell(\ell + 1)|^{-1} \leq \frac{1}{4}$ for $\ell \geq 2$; the $\ell = 0$ breathing block is the scalar $-x/s^2$ (80), with inverse $s^2/x = (1-x)/x$; the $\ell = 1$ block off the translation gauge is the cos-row and tilt pair $\text{diag}(-2s, 1)$ (77), with inverse multipliers $\max\{1/(2s), 1\}$; and the $\ell = 2$ minor (88) has inverse rows $(-1/x, 0)$ and $(1/(2s^3), -1/(2s))$, hence row-sum norm $\max\{1/x, 1/(2s^3) + 1/(2s)\}$. The norm of a block-diagonal inverse is the largest of the block inverse norms, and on $(0, \frac{24}{25})$ the last entry dominates the others, so

$$\Gamma(x) = \max\left\{\frac{1}{x}, \frac{1}{2s^3} + \frac{1}{2s}\right\}, \quad (68)$$

which is (49). The shape sector contributes only $\frac{1}{4}$ and never binds, so the Schauder constant C_S does not enter the *size* of Γ ; it is used to obtain the $C^{2,\alpha}$ regularity of the matched shape, not the bound. Γ diverges as $1/x$ when $x \rightarrow 0$ and as $1/(2s^3)$ when $x \rightarrow \frac{24}{25}$.

The analyticity radius and the second-variation majorant. On the axis the exterior lapse is the exact affine function $L = f(R) + 2\lambda R \cos \vartheta$ (41). Let $r_*(x)$ be the largest ϱ such that the polydisc $|\delta R| \leq \varrho R, |\delta \lambda| \leq \varrho$ keeps $|L| \geq \frac{1}{2}(1-x)$; explicitly $r_*(x)$ is the largest root of $x\varrho/(1-\varrho) + 2\varrho(1+\varrho) = \frac{1}{2}(1-x)$, which is positive on the whole window and shrinks as $x \rightarrow \frac{24}{25}$. Inside that polydisc the junction functionals are holomorphic (the only branch point of $\sqrt{L}$ and $1/L$ sits at $L = 0$, i.e. at $\lambda = \frac{1}{2}(1-x)$, which lies outside), and Cauchy's estimate on the polydisc bounds the second derivatives of the Darboux map,

$$M_2(x) := \frac{2 \max\{\text{NL}_A, \text{NL}_B\}}{r_*(x)^2} \geq \sup \|D^2 F\|, \quad (69)$$

where NL_A and NL_B are the suprema on the polydisc of the two rows carrying nonlinearity in the areal graph gauge: the seed-normalized lapse row (78), whose lapse is L , and the shape row, whose only second-order contribution is the quadratic embedding term $R'^2(1/L - 1) - Lw^2 + Y^2$ (the operator part $(\Delta_{S^2} + 2)Y$ being linear and contributing nothing to $D^2 F$).

That these two rows exhaust $D^2 F$ is not obvious, because a third quantity is nonlinear in the shape: the interior Rindler lapse $N = 1 + az$ of (42) depends on the shell's *height* z along the acceleration axis, and once the tangential reparametrization is freed, as Remark 1 requires, z is not $R \cos \vartheta$ but a quadrature of the shape. Writing the interior as a surface of revolution in the rigidly accelerated slice, the angular junction rows give $\rho = R \sin \vartheta$ and

$$z'^2 = \left(\frac{d(R \cos \vartheta)}{d\vartheta}\right)^2 + R'^2 \left(\frac{1}{L} - 1\right) - Lw^2, \quad (70)$$

so the integrand of NL_B is exactly the amount by which the rigid-angle height fails to solve (70). The resulting drift $\zeta := z - R \cos \vartheta$ is *second* order in the shape perturbation (its first variation vanishes identically at the anchor), so it enters the junction map only through $a\zeta$, at *cubic* order: it changes neither DF_0 nor $D^2 F(0)$, and on the polydisc it is bounded strictly below the NL_B term already charged. The majorant (69) is therefore complete, and Γ, δ_Q and $c(x)$ stand as written.

The quantitative implicit-function radius. Let $S_\lambda = 2x/s^2$ be the anchor λ -slope of the lapse row, and put $\kappa := 1 + 2\Gamma S_\lambda$. On the ball $\|q\| \leq \varrho_Q(\lambda) := 2\Gamma S_\lambda \lambda$ the Newton–Kantorovich contraction closes for

$$\delta_Q(x) = \min\left\{\frac{1}{4}(1-x), \frac{1}{2\Gamma M_2 \kappa}, \frac{2S_\lambda}{M_2 \kappa^2}, \frac{r_*}{\kappa}\right\} > 0, \quad (71)$$

the four conditions being, in order, that the branch remain sub-horizon, that the Newton map be a contraction ($\Gamma M_2 \|q\| \leq \frac{1}{2}$), that it map the ball into itself, and that the ball lie inside the polydisc of holomorphy. Finally, writing B_m^* for the supremum of the margin on that polydisc and $\varrho_\star := \varrho_Q(\delta_Q)$ for the corresponding ball radius,

$$A(x) = \frac{2B_m^* \max\{\varrho_\star, \delta_Q\}}{r_\star \delta_Q}, \quad C_{\text{shape}}(x) = \frac{4A(x)}{\delta_Q(x)}, \quad (72)$$

by the Schwarz lemma and Cauchy's estimate applied to $\Phi(\lambda) = m(\vartheta; q(\lambda), \lambda) - M_0$, valid for $\lambda \leq \delta_Q/2$.

Values. Substituting (68)–(72) into (48) gives

$$c(x) \approx \{8.5, 14, 17, 14, 2.0\} \times 10^{-5} \quad \text{at } x = \{0.1, 0.2, 0.3, 0.5, 0.7\}, \quad (73)$$

the quadratic root of (48) binding for $x \lesssim 0.7$ and δ_Q beyond, while the sub-horizon strip Λ_h never binds. These lie below the numerically mapped $\underline{g}(x) \sim 0.2$ of the Supplement by about three orders of magnitude, as a lower bound must. The conservatism is due entirely to the Cauchy majorant (69), since $\delta_Q \propto 1/M_2$ and $c \propto \delta_Q$ in this regime; (68) is sharp. A tighter majorant for M_2 , which requires the full nonlinear junction map rather than the two rows bounded above, would raise $c(x)$ without altering (48).

The quadratic onset that keeps $A = \mathcal{O}(x^2)$ small lifts $c(x)$ above a crude linear estimate; its overall scale is otherwise set by the round-sphere elliptic constant. The finite-amplitude reach $\underline{g}(x) \sim 0.2$ mapped numerically in the Supplement is larger, and the binding angle, closed-form at linear order (the matched dipole $\sigma_1 - p_1 > 0$ binding the rear pole), is at finite amplitude a real-analytic inequality decidable order-by-order but not elementary in $\cos \vartheta$.

The full multipole-frontier solve is detailed with the numerical implementation (the Supplement).

D The linearized Darboux map at the anchor: rigidity, Schauder bound, and the dipole block

Lemma 4 asserts that the linearized first-junction map DF_0 at the Schwarzschild–Minkowski anchor is a bounded isomorphism off the $\ell = 1$ translation gauge. We prove it here, addressing the two points the claim turns on: (i) that the shape sector reduces to the round-sphere rigidity operator $\Delta_{S^2} + 2$ in the *Schauder* (not merely L^2) topology that the C^2 shell embedding requires, and (ii) that the dipole ($\ell = 1$) steering source (which lies in the *kernel* of that operator) is absorbed by a non-degenerate finite-dimensional lapse/tilt/shape block, rather than being applied on the kernel.

Setup and harmonic split. The first junction condition $[h_{ab}] = 0$ on the timelike worldtube $\Sigma : r = R(\tau, \Omega)$ requires the induced 3-metric to agree from both sides. Linearizing about the anchor (exterior exactly Schwarzschild, interior exactly flat, Σ the round sphere of areal radius R_0 , $x = 2m/R_0$), the perturbation carries three fields on S^2 : the normal shape displacement $u(\Omega)$ of Σ , the lapse/redshift perturbation (the $\tau\tau$ block), and the foliation tilt (the τA block). Correspondingly $[h_{ab}] = 0$ splits into three sectors, the angular-angular $[h_{AB}]$ (*shape*), the time-time $[h_{\tau\tau}]$ (*redshift*), and the time-angle $[h_{\tau A}]$ (*tilt*), each expanded in spherical harmonics $u = \sum_\ell u_\ell$.

Shape sector, $\ell \geq 2$: rigidity and the Schauder estimate. A normal displacement u of $S_{R_0}^2$ preserves the first fundamental form to first order iff

$$(\Delta_{S^2} + 2)u = \mathcal{S}[u_{\text{tan}}, \delta g_{AB}^+], \quad (74)$$

the inhomogeneity $\mathcal{S}$ being fixed by the tangential embedding field and the $\mathcal{O}(a)$ angular anisotropy of the exterior induced metric; this is the classical infinitesimal-bending equation for the sphere [69]. Its homogeneous operator is the round-sphere rigidity operator, with kernel the $\ell = 1$ harmonics: a rigid translation $\mathbf{a} \cdot \hat{n}$ is an $\ell = 1$ normal displacement and $(\Delta_{S^2} + 2)Y_1 = 0$ (Cohn–Vossen/Weyl rigidity [69]). Its eigenvalues $2 - \ell(\ell + 1)$ have magnitude ≥ 4 for $\ell \geq 2$ (and 2 for $\ell = 0$), so $\Delta_{S^2} + 2$ is invertible on the $\ell \neq 1$ subspace. The bound that controls the matched shell is, however, *not* the L^2 spectral statement $\|(\Delta_{S^2} + 2)^{-1}\|_{L^2} = \frac{1}{2}$ but a Schauder estimate: as a second-order elliptic operator on the compact manifold S^2 , $\Delta_{S^2} + 2$ restricted to the closed subspace $\ell \neq 1$ (equivalently, on fields satisfying the gauge condition $P_1 u = 0$, with P_1 the L^2 -projection onto the $\ell = 1$ kernel) is an isomorphism $C^{2,\alpha}(S^2) \rightarrow C^{0,\alpha}(S^2)$,

$$\|u\|_{C^{2,\alpha}} \leq C_S \|(\Delta_{S^2} + 2)u\|_{C^{0,\alpha}}, \quad P_1 u = 0, \quad (75)$$

the constant C_S depending on the round metric, the Hölder exponent α , and the spectral gap [69]. The constant is *explicit in structure*. On the pure shape sector $\ell \geq 2$ the gap is $|2 - \ell(\ell + 1)| \geq 4$ (sharper than the $\frac{1}{2}$ off-kernel bound, which is the $\ell = 0$ value $|\mu_0|^{-1}$), so the L^2 inverse there is $\frac{1}{4}$; the elliptic gain $\|\Delta_{S^2} u\|_{L^2} \leq \frac{3}{2} \|(\Delta_{S^2} + 2)u\|_{L^2}$ (since $\ell(\ell + 1)/|2 - \ell(\ell + 1)| \leq \frac{3}{2}$ for $\ell \geq 2$) gives $\|u\|_{H^2} \leq \sqrt{\frac{1}{16} + \frac{9}{4}} \|(\Delta_{S^2} + 2)u\|_{L^2} = 1.5207 \|(\Delta_{S^2} + 2)u\|_{L^2}$, and absorbing the lower-order term in the closed-manifold Schauder estimate bounds

$$C_S \leq C_0(\alpha) (1 + 5.391 C_{\text{Sob}}), \quad (76)$$

with $C_0(\alpha)$ the round-sphere Schauder constant of $\Delta_{S^2} + 2$ and C_{Sob} the $H^2(S^2) \hookrightarrow C^0(S^2)$ embedding constant (finite since $2 > \frac{n}{2} = 1$), so C_S is reduced to tabulated round-sphere elliptic data. Estimate (75) yields the C^2 shell embedding, hence the continuity of $[K_{ab}]$ and of S_{ab} that Theorem 2 uses; it replaces the L^2 inverse bound quoted in Lemma 4. It is used for the *regularity* of the matched shape and not for the *size* of any bound: the inverse bound $\Gamma(x)$ of (68) is the maximum over the finitely many closed-form harmonic blocks, in which the $\ell \geq 2$ shape sector contributes the exact spectral value $\frac{1}{4}$ and never binds, so C_S does not enter Γ , C_{shape} , or $c(x)$.

Dipole sector, $\ell = 1$: the steering source is off the kernel. Because the $\ell = 1$ harmonic lies in $\ker(\Delta_{S^2} + 2)$, the shape sector alone cannot fix the dipole response; the full first-junction map, including lapse and tilt, must be used. For the axisymmetric steering source the three first-junction conditions (the constant and $\cos \vartheta$ parts of $[h_{\tau\tau}]$ and the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$) are affine in the three dipole unknowns (ρ_1, A_{t1}, w_1) (shape dipole, exterior lapse dipole, foliation tilt). The interior lapse is not among them: it is not a free seed but the Rindler value $N = 1 + aR_0 \cos \vartheta$ of the rigidly accelerated cavity, whose $\mathcal{O}(a)$ dipole is geometry, and which supplies the constant of the $\cos \vartheta$ row below. Only the zeroth-order normalization $[h_{\tau\tau}]^{(0)} = 0$ (fixing $A_{t0} = 1/\sqrt{1-x}$ against $N = 1$) is a gauge choice. The linearized map is *block-triangular*:

- the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$ is set by the foliation tilt with unit leading coefficient, $\partial[h_{\tau\vartheta}]^{\sin}/\partial w_1 = 1$, the remaining tilt couplings being $\mathcal{O}(a)$ (vanishing at the anchor): the tilt *decouples*;
- the residual 2×2 block $(\rho_1, A_{t1}) \times \{[h_{\tau\tau}]^{\text{const}}, [h_{\tau\tau}]^{\cos}\}$ carries the redshift Jacobian. Its $\cos \vartheta$ row is closed-form: the areal-shape match responds to the shape and lapse dipoles through

$$\frac{\partial[h_{\tau\tau}]^{\cos}}{\partial \rho_1} = -\frac{x}{1-x}, \quad \frac{\partial[h_{\tau\tau}]^{\cos}}{\partial A_{t1}} = -2\sqrt{1-x}, \quad (77)$$

both nonzero on $x \in (0, 1)$ (and underwritten by $\partial_R[\sqrt{f}] = m/(R_0^2 \sqrt{1-x}) \neq 0$, $1/\sqrt{1-x} \neq 0$).

The $\mathcal{O}(a)$ steering source has only two components: the $\cos \vartheta$ part of $[h_{\tau\tau}]$ and the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$. The constant part of $[h_{\tau\tau}]$ carries *no* $\mathcal{O}(a)$ source: a $\cos \vartheta$ shape feeds the $\ell = 0$ jump of the quadratic induced metric only through the $\langle \cos^2 \vartheta \rangle$ self-coupling, so $[h_{\tau\tau}]^{\text{const}} = \mathcal{O}(a^2)$. Because the $\cos$ row (77) and the tilt row ($\partial[h_{\tau\vartheta}]^{\sin}/\partial w_1 = 1$) are independent and nonzero for all $x \in (0, 1)$, the linearized junction is *surjective onto this two-dimensional source*: a matched accelerating shell exists at $\mathcal{O}(a)$ *in closed form, unconditionally*. The three dipole unknowns meet two rows, so one dipole direction is left free; it is the $\ell = 1$ shape amplitude ρ_1 , which displaces the shell's center. This is the residual rigid-translation gauge, not a second steering knob: the steering is set by a , and ρ_1 is fixed by centering the interior embedding on Γ , consistently with the momentum balance.

The lapse jump on each side is built by seed-orthogonalization. The shell's unit timelike tangent is the seed time direction ξ , Gram-Schmidt-orthogonalized against the shell's spatial tangent $\partial_\vartheta \Phi$, so the induced lapse on each side is

$$h_{\tau\tau} = g(\xi, \xi) - \frac{(g(\xi, \partial_\vartheta \Phi))^2}{g(\partial_\vartheta \Phi, \partial_\vartheta \Phi)}, \quad (78)$$

the orthogonalization denominator carrying the $1/(1-x)$ and $1/\sqrt{1-x}$ redshift factors. The exterior seed is $\xi^+ = A_t \partial_t$ with $g(\xi^+, \xi^+) = -2HA_t^2$. The interior seed is the *boost* Killing field of the Born-rigid cavity, not the inertial translation ∂_T : the two sides must share one shell time τ , and the exterior is stationary, so the interior generator is the one whose orbits are the shell's, with $g(\xi^-, \xi^-) = -N^2$ and $N = 1 + aR_0 \cos \vartheta$ the Rindler lapse. Expanding (78) to $\mathcal{O}(a)$, the exterior $h_{\tau\tau}^+ = -2HA_t^2$ against the interior $-N^2$, with the dipole shape $R(\vartheta) = R_0(1 + a\rho_1 \cos \vartheta)$, the

exterior redshift seed $A_t = (1 - x)^{-1/2} + aA_{t1} \cos \vartheta$, and the rocket redshift $2H = f + 2ar \cos \vartheta$, and using only the anchor data $f(R_0) = 1 - x$, $R_0 f'(R_0) = x$, gives the $\cos \vartheta$ jump

$$\frac{1}{a} [h_{\tau\tau}]^{\cos} = \frac{(\rho_1 + 2)x + 2A_{t1}(1 - x)^{3/2}}{x - 1}, \quad (79)$$

whose derivatives are the $\cos$ -row (77). The additive $2x$ is the interior Rindler dipole: dropping it, as a static interior would, leaves the row's *derivatives* untouched but shifts the root A_{t1} , and with it the $\ell = 0$ elimination below.

The $\ell = 0$ monopole is absorbed by the breathing mode. At $\mathcal{O}(a^2)$ the ϑ -averaged jump $[h_{\tau\tau}]^{\ell=0}$ (the $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$ self-coupling of the $\cos \vartheta$ dipole) is *not* an obstruction: it is met by the radial breathing mode ρ_0 , a uniform change of areal radius, the same monopole mode that inverts the anchor redshift block. Its derivative along ρ_0 is nonzero,

$$\frac{1}{a^2} \frac{\partial [h_{\tau\tau}]^{\ell=0}}{\partial \rho_0} = -\frac{x}{s^2} \neq 0 \quad (0 < x < 1), \quad s := \sqrt{1 - x}, \quad (80)$$

so the single $\ell = 0$ monopole constraint is non-degenerate and is closed by the one $\ell = 0$ variable ρ_0 through this nonzero redshift slope; the round-sphere rigidity operator (74) carries the shape content only at $\ell \geq 2$, its $\ell = 0$ and $\ell = 1$ harmonics being the areal and translation gauge. That the breathing mode is *necessary*, not merely available, is sharp: freezing $\rho_0 = \rho_2 = 0$ and eliminating $A_{t1} = -(\rho_1 + 2)x/(2s^3)$, the root of the $\cos$ row (79), against the $\mathcal{O}(a^2)$ ϑ -average leaves the quadratic $x(4 - x)\rho_1^2 + 12x^2\rho_1 + 4(2x + 1) = 0$, whose discriminant

$$\Delta = 16x(x - 1)(9x^2 + 11x + 4) < 0 \quad \text{on } (0, 1) \quad (81)$$

is negative ($9x^2 + 11x + 4$ has negative discriminant, hence stays positive), so no real dipole alone closes the junction: the $\ell = 0$ jump lies in the range of ρ_0 and no one-dimensional cokernel survives.

Existence. The linearized first junction is thus block-diagonal in the harmonics and *index zero*, each block square with nonzero determinant. At $\ell \geq 2$ the shape defect is inverted by the Fredholm isomorphism $\Delta_{S^2} + 2$ (74)–(75) (with the $\mathcal{O}(a^2)$ $\ell = 2$ assembly closed by the non-degenerate energy–lapse minor (88)). At $\ell = 0$ the single monopole is closed by the single breathing variable ρ_0 (80). At $\ell = 1$ the shape defect *vanishes identically*, a rigid translation being an isometry (the Cohn–Vossen kernel), so it is not a constraint; the two finite rows $[h_{\tau\tau}]^{\cos}, [h_{\tau\vartheta}]^{\sin}$ are closed by the two seeds (A_{t1}, w_1) through the nonsingular $\cos$ -row/tilt block (77), leaving the $\ell = 1$ shape amplitude ρ_1 free: it *is* the $\ell = 1$ rigid translation, the residual gauge. Hence the map is a bounded isomorphism off the finite-dimensional $\ell = 1$ translation gauge, and the implicit function theorem yields, for every anchor $x \in (0, 24/25)$, a locally unique $C^{2,\alpha}$ matched accelerating shell for small $a > 0$: *Theorem 2 is unconditional at small amplitude*, with no residual scalar obstruction. Because the construction is $SO(3)$ -equivariant, the same conclusion holds per retarded cut for an arbitrary acceleration three-vector (Cor. 6), not only the axial case.

Existence by Lyapunov–Schmidt. The harmonic-sector facts above assemble into a nonlinear existence statement, not merely linearized solvability.

Proposition 8 (Local existence of the accelerating shell). *Fix $x_* \in (0, 24/25)$. Split the axisymmetric first-junction defect $[h_{ab}] = 0$ into the conformal shape-defect scalar $[h_{AB}]$ (its magnetic-parity component absent by axisymmetry), the redshift monopole $[h_{\tau\tau}]^{\ell=0}$, and the two finite dipole rows $[h_{\tau\tau}]^{\cos}, [h_{\tau\vartheta}]^{\sin}$. A rigid translation is an isometry of the round sphere, so the shape defect carries no $\ell = 1$ content ($[h_{AB}]^{\ell=1} \equiv 0$, the Cohn–Vossen kernel), and its $\ell = 0$ part is absorbed by the breathing mode below, so it enters through its $\ell \geq 2$ modes; write $C_{\ell \geq 2}^{0,\alpha}(S^2)$ for that range. The defect is then the map*

$$F : (C^{2,\alpha}(S^2)/\mathfrak{t}) \times \mathbb{R}^2 \times \mathbb{R} \longrightarrow C_{\ell \geq 2}^{0,\alpha}(S^2) \times \mathbb{R}^3, \\ F(Y, (A_{t1}, w_1), a) = ([h_{AB}]_{\ell \geq 2}, [h_{\tau\tau}]^{\ell=0}, P_{\ell=1}^{\cos}[h_{\tau\tau}], [h_{\tau\vartheta}]^{\sin}),$$

in the normal shape displacement Y modulo the three-dimensional rigid-translation gauge $\mathfrak{t} = \ker(\Delta_{S^2} + 2)$, the exterior-lapse and tilt dipoles (A_{t1}, w_1) , and the steering amplitude a (the interior lapse fixed by $[h_{\tau\tau}]^{(0)} = 0$; the $\ell = 1$ shape amplitude ρ_1 , the shell's center, is the residual rigid-translation gauge and is quotiented out with $\mathfrak{t}$). The tensorial S^2 map is the non-axisymmetric

case of Cor. 6. Then F is C^1 near the anchor with $F = 0$ there, and $DF(0)$ is block-diagonal in the harmonics and of index zero, each block square with nonzero determinant: the Fredholm isomorphism $\Delta_{S^2} + 2$ on the $\ell \geq 2$ shape (eigenvalues ≥ 4 in modulus); the single nonzero redshift-monopole slope $\partial_{\rho_0}[h_{\tau\tau}]^{\ell=0} = -x/s^2$ (80) closing the redshift monopole by the one breathing variable $\rho_0 = Y^{\ell=0}$ (the $\ell = 0$ shape defect being absorbed by the same mode, so it carries no separate constraint); and the nonsingular cos-row/tilt block (77) closing the two $\ell = 1$ finite rows by (A_{t1}, w_1) . Hence $DF(0)$ is a bounded isomorphism off the finite-dimensional $\ell = 1$ rigid-translation gauge, and the implicit function theorem yields a locally unique $C^{2,\alpha}$ matched shell with $[h_{ab}] = 0$ exactly, a branch parametrized by small $a > 0$, whose surface stress S_{ab} is C^1 in a ; hence the worst-observer margin of Theorem 2 satisfies $\Phi = \delta_0 + \mathcal{O}(a, \mu) > 0$ on an open neighborhood of the anchor and the accelerating warpsell is admissible there.

Proof. The C^1 regularity and the block structure of $DF(0)$ are the harmonic split above: the round-sphere rigidity isomorphism (74)–(75) on $\ell \geq 2$; the nonzero redshift-monopole slope $\partial_{\rho_0}[h_{\tau\tau}]^{\ell=0} = -x/s^2 \neq 0$ (80) closing the $\ell = 0$ redshift row by ρ_0 ; and, on $\ell = 1$, the identically-vanishing shape defect (Cohn–Vossen) together with the block-triangular dipole rows (77), so the two finite seeds (A_{t1}, w_1) close the two finite $\ell = 1$ rows, ρ_1 remaining free as the rigid-translation gauge. Each sector is square with nonzero determinant, so $DF(0)$ is a bounded isomorphism off the $\ell = 1$ translation gauge and the implicit function theorem yields a locally unique branch $(Y, A_{t1}, w_1)(a)$ directly, with no bifurcation-scalar rescaling. Continuity of S_{ab} in a then propagates the strict anchor margin δ_0 of Lemma 3. $\square$

This is the precise content of Lemma 4, and it makes Theorem 2 unconditional (perturbatively) for every small $a > 0$.

An arbitrary acceleration axis (proof of Cor. 6). The closure above is stated for an axisymmetric steering source, but at small amplitude it holds for an arbitrary acceleration *direction* by rotational covariance of the anchor.

Proof of Corollary 6. The linearized first-junction map at the static Schwarzschild–Minkowski anchor is the round-sphere operator $\Delta_{S^2} + 2$, whose $\ell = 1$ kernel is the three-dimensional space of rigid translations, and the block-triangular dipole map of Prop. 8 is $SO(3)$ -equivariant on that representation. The $\ell = 1$ steering source rotates with a^i , so aligning the axis with $\hat{z}$ reduces any direction to the axisymmetric case, whose $\ell = 0$ residual (entering at $\mathcal{O}(a^2)$ through $\langle \cos^2 \vartheta \rangle$) is absorbed by the radial breathing mode $\partial_{\rho_0}[h_{\tau\tau}]^{\ell=0} = -x/s^2 \neq 0$ (80), with no residual scalar obstruction on $(0, 24/25)$. $\square$

The finite-amplitude non-axisymmetric solve (where the $\ell \geq 2$ multipole response is no longer a single Legendre tower but a tensorial coupling on S^2) and the assembly of the per-cut shells into a single C^2 worldtube for a *rotating* acceleration axis remain future work (Sec. 14).

E Assembly of the time-evolved spacetime (proof of Theorem 4)

We prove Theorem 4: for a slow, fixed-axis, sub-horizon burn the per-cut admissible snapshots of Sec. 9.2 assemble into a single time-evolved C^2 worldtube, by the compatibility identity (84), closed to all orders below.

Proof of Theorem 4. After Schur-eliminating the $\ell \leq 1$ lapse/tilt gauge block (non-degenerate throughout the burn by Lemma 8, $\det J_{\text{dyn}} = \det J_0(x)[1 + \mathcal{O}(\varepsilon)] \neq 0$ with $\det J_0(x)$ the a -independent cos-row/tilt determinant (77)) the reduced first-junction condition on the normal shape $Y \in C^{2,\alpha}(S^2)$ reads

$$(\Delta_{S^2} + 2)Y = \mathcal{N}(Y, Y_u; \lambda, u) + \sigma(\lambda, u), \quad (82)$$

with steering source $\sigma = \mathcal{O}(\lambda)$ (the dipole row of Lemma 4) and $\mathcal{N}$ the subordinate part: by Lemma 7 the retarded derivative Y_u enters $\mathcal{N}$ at angular *differential* order ≤ 1 and the quadratic embedding nonlinearity at order ≤ 2 with the *same* leading symbol as the static case, so $\mathcal{N} : C^{2,\alpha} \times C^{1,\alpha} \rightarrow C^{0,\alpha}$ is smooth with $\mathcal{N}(0, 0; 0, u) = 0$, and $(\Delta_{S^2} + 2)^{-1} : C^{0,\alpha}(S^2) \rightarrow C^{2,\alpha}(S^2)$ is bounded off the $\ell = 1$ kernel (the Schauder estimate (76), gaining two angular derivatives). Equation (82) is the *shape* block: it fixes Y given (Y_u, λ, u) , but to propagate it as an evolution one must *solve for* Y_u , and that closure comes not from (82) but from the higher-mode retarded time-rows. Subordinate differential order is *not* lower harmonic degree: a term with one angular

derivative of Y_u still carries arbitrary $\ell \geq 2$ content, so after the $\ell \leq 1$ Schur elimination of Lemma 8 the higher-mode projection of the retarded junction

$$R_{\geq 2}(Y, Y_u; \lambda, u) := P_{\ell \geq 2}([h_{uu}], [h_{uA}])_{\text{red}} = 0 \quad (83)$$

is a system of equations in Y_u , not a consequence of (82). Three roles must be kept distinct. Closure, that (83) fixes Y_u with no cokernel and no loss of angular derivative, is established below by the characteristic, order-independent rigid determination and needs no Cauchy–Kovalevskaya estimate. Convergence of the resulting amplitude series, for real-analytic $a(u)$, then follows from the analytic majorant. And regularity of the assembled worldtube, for merely smooth burns, is the Picard–Lindelöf/Grönwall wrapper (90), applied only after closure. In this sense (83) is a per-cut algebraic compatibility condition on a characteristic cut, and the u -evolution below propagates already-determined data rather than integrating free Cauchy data. The persistence closes *provided* (83) is solvable for Y_u by a bounded, locally Lipschitz map $Y_u = G(Y; \lambda, u) : C^{2,\alpha}(S^2) \rightarrow C^{2,\alpha}(S^2)$ with no loss of angular derivative, the over-determined $[h_{uA}]$ row carrying no cokernel obstruction. Lemma 7 supplies the no-derivative-loss *budget* (the elliptic gain compensates the single Y_u derivative) and for a fixed-axis (non-rotating) burn the solvability itself follows by equivariance. An axisymmetric source about the burn axis makes the entire junction map $SO(2)$ -equivariant, so an admissible matched worldtube may be sought within the axisymmetric invariant subspace (existence, not uniqueness, is at issue): the shape $Y = Y(\vartheta) \in C_{\text{ax}}^{2,\alpha}(S^2)$ (∂_φ -free), the worldtube embedding $X^\mu(u, \vartheta)$ with no azimuthal motion ($\partial_u X^\varphi = \partial_\vartheta X^\varphi = 0$), and the interior slicing $T_- = T(u) + \beta(\vartheta)$ with tilt $B_\vartheta = \partial_\vartheta \beta$. On this subspace the time-angle defect $[h_{uA}]$ is a one-form with $[h_{u\varphi}] \equiv 0$ (the angular metric is diagonal and nothing moves in φ): in the Hodge split of axisymmetric one-forms into the gradient class $\partial_\vartheta(\cdot) d\vartheta$ and the axial class $\propto d\varphi$, only the gradient class is present on both the data and the unknown $B = d\beta$, so the axial (magnetic) sector (the sole candidate cokernel of $[h_{uA}]$) is empty and the gradient row is square. The two surviving rows are lower-triangular. In the constant-retarded graph gauge $r = R + Y$ the $-2 du dr$ term of (4) makes the exterior energy row carry the retarded weight $-2 Y_u$. The interior, however, advances with the shell proper time: with the re-slicing $T_- = T(u) + \beta(u, \vartheta)$ it contributes $+2s T_{-,u}$ to the same row, and the tilt row $[h_{u\vartheta}] = 0$ forces $s \beta_{,\vartheta} = Y_\vartheta$, hence $s T_{-,u} = Y_u$ up to the residual monopole gauge and, during the burn ($\dot{m} \neq 0$, hence $\dot{s} \neq 0$), an $\mathcal{O}(\lambda^2)$ angular source $-(\dot{s}/s)Y$ from the slow drift of s in $\beta = Y/s$. That source carries the shape Y , not its velocity Y_u : the shape velocity therefore *cancels identically* from $[h_{uu}]$, which collapses to a rigid algebraic relation for the shape (the drift entering as a known lower-order datum) rather than an evolution equation for Y_u . The cuts $u = \text{const}$ are *characteristic*: the shape is not free Cauchy datum but is fixed, order by order, by $a(u)$ and its retarded derivatives (Lemma 7); analyticity enters only to majorize the resulting recursion, Cauchy–Kovalevskaya serving as a convergence majorant and not as an evolution theorem. The polar row then reads $-(f + 2aR \cos \vartheta) \partial_\vartheta \beta = \Lambda(Y_u, Y; \lambda, u)$, with diagonal coefficient $\partial_{B_\vartheta} [h_{u\vartheta}] = -(f + 2aR \cos \vartheta)$ the interior timelikeness factor (60), bounded below by $\inf(f - 2aR) > 0$ through the sub-horizon regime and degenerating only at the rear pole $\vartheta = \pi$ as $f - 2aR \rightarrow 0$ (Prop. 5), and right side $\Lambda \in C_{\text{ax}}^{1,\alpha}$ the already-fixed Y_u at angular order ≤ 1 . Built from the pole-regular axisymmetric Y_u, Y , the datum Λ is a smooth axisymmetric one-form component and so vanishes at $\vartheta = 0, \pi$; dividing by the nonvanishing factor and integrating the first-order ϑ -equation, the integration restoring the one angular derivative ($\partial_\vartheta \beta \in C^{1,\alpha} \Rightarrow \beta \in C_{\text{ax}}^{2,\alpha}$), gives a β with $\partial_\vartheta \beta \rightarrow 0$ at the poles, unique once fixed by the gauge $\beta(0) = 0$ (an additive interior time shift). The triangular system thus has a unique, bounded, locally Lipschitz solution $(Y_u, B_\vartheta) = \mathcal{G}(Y; \lambda, u)$ on $C_{\text{ax}}^{2,\alpha}(S^2)$ with inverse norm $\lesssim (\inf(f - 2aR))^{-1}$ and no loss of angular derivative on Y_u , its Y_u -component the map G of (90) (here on the axisymmetric subspace). The one-form $B = B_\vartheta d\vartheta$ is closed automatically on the axisymmetric subspace ($dB = \partial_\vartheta B_\varphi - \partial_\varphi B_\vartheta = 0$) and pole-regular, so $B = d\beta$ with β a single-valued interior time tilt and the cavity stays exactly flat. It remains to show that the time-row solution operator $\mathcal{G}$ is consistent with the *full* first junction $[h_{ab}] = 0$, not only its time rows. Here the structure helps. The angular block carries *no* retarded derivative ($h_{AB} = g_{\mu\nu} \partial_A X^\mu \partial_B X^\nu$ involves only angular tangents) so on each constant- u cut $[h_{AB}] = 0$ is the *same* elliptic shape problem as the static existence theorem at the instantaneous $(m(u), a(u))$: the round-sphere rigidity reduction with the closed-form $\ell = 1$ steering closure (77) and the breathing-mode-absorbed monopole (80) (Lemma 4). The shape match therefore holds at every cut *unconditionally* at small amplitude, with $Y(u, \cdot) = Y_\star(u, \cdot)$ the per-cut matched shape; there is no shape *evolution* to propagate. What the dynamical problem adds is a *compatibility* between the two ways the retarded derivative enters: the burn rate Y_u fixed by the energy row $[h_{uu}] = 0$ (graph gauge $\partial_{Y_u} [h_{uu}] = -2$) must coincide with

the u -derivative $\partial_u Y_\star$ of the per-cut elliptic shape:

$$Y_u|_{[h_{uu}]=0} = \partial_u Y_\star. \quad (84)$$

This compatibility is *not* a contracted-Codazzi propagation: the contracted Codazzi balance (27) presupposes the full $[h_{ab}] = 0$ and so cannot establish it. Its precise content is that the $\ell \geq 2$ part of the retarded energy row $[h_{uu}]$ (which carries *no* free junction parameter once the shape, tilt, and $\ell \leq 1$ lapse/time rows are solved) vanishes identically, equivalently that the exterior-induced 3-metric on Σ is isometrically embeddable in the flat interior (a Gauss–Codazzi class-one consistency). *The candidate leading $\mathcal{O}(a^2)$ $\ell = 2$ obstruction cancels*: the angular block is discharged by the nonlinear isometric embedding below, which carries no solvability condition. The worry an obstruction would express is that the per-cut shape match fixes the quadrupole ρ_2 to a value *conflicting* with the $\rho_2^\star$ the energy row demands. As a gauge-fixed consistency check we exhibit the mechanism in the rigid-angle gauge $P = R$, $\Theta = \vartheta$ (in which the angular match does not itself close, the tangential reparametrization being supplied by the embedding); what survives the gauge choice, and all the argument uses, is that the $\mathcal{O}(a^2)$ shape sector is ρ_2 -free. Write the shell areal radius on the momentary-rest cut as

$$R(\vartheta) = R_0 [1 + a\rho_1 \cos \vartheta + a^2 \rho_2 P_2(\cos \vartheta)], \quad (85)$$

matched at equal areal radius to the flat interior and the Schwarzschild exterior $f(R) = 1 - 2m/R$. The two angular induced-metric jumps are then

$$[h_{\varphi\varphi}] = R^2 \sin^2 \vartheta - R'^2 \sin^2 \vartheta \equiv 0, \quad [h_{\vartheta\vartheta}] = \left(R^2 + \frac{R'^2}{f(R)} \right) - (R^2 + R'^2) = R'^2 \frac{1 - f(R)}{f(R)}, \quad (86)$$

the shared areal R^2 canceling in each ($R' = dR/d\vartheta$). Since $R' = -R_0 a \rho_1 \sin \vartheta + \mathcal{O}(a^2)$ carries the quadrupole only at $\mathcal{O}(a^2)$, $R'^2 = R_0^2 a^2 \rho_1^2 \sin^2 \vartheta + \mathcal{O}(a^3)$ is ρ_2 -free at $\mathcal{O}(a^2)$, whence

$$[h_{\vartheta\vartheta}]^{(a^2)} = R_0^2 a^2 \rho_1^2 \sin^2 \vartheta \frac{x}{1-x}, \quad \partial_{\rho_2} [h_{\vartheta\vartheta}]^{(a^2)} = \partial_{\rho_2} [h_{\varphi\varphi}] = 0, \quad (87)$$

with $x = 2m/R_0$; the quadrupole first reaches the jump at $\mathcal{O}(a^3)$ (the Kerr–Schild anisotropy). The shape sector therefore imposes *no* $\ell = 2$ constraint, so no shape-fixed value can conflict with $\rho_2^\star$; the quadrupole is left to the retarded energy row, whose static $\ell = 2$ part carries it with nonzero slope $\partial_{\rho_2} [h_{uu}]_{\ell=2} = -x$ and is met at $\rho_2 = \rho_2^\star$. This removes the one mechanism by which the leading-order obstruction could survive, and the remaining $\ell = 2$ rows close it completely. Collect the four $\mathcal{O}(a^2)$ $\ell = 2$ first-junction rows in the unknowns $(Y_2, \rho_2, A_{t2}, w_2)$: the shape bending Y_2 , the areal quadrupole ρ_2 , the exterior-lapse quadrupole A_{t2} , and the regular tilt quadrupole w_2 in $\beta = a w_1 P_1 + a^2 w_2 P_2$. The round-sphere rigidity row $(\Delta_{S^2} + 2)Y_2 = \mathcal{S}_2$ fixes the bending (eigenvalue -4 at $\ell = 2$, the ρ_1^2 source $\mathcal{S}_2$ being ρ_2 -blind by the computation above); the polar tilt row $[h_{u\vartheta}]_{\ell=2}$ carries w_2 ; the retarded energy row $[h_{uu}]_{\ell=2}$ carries ρ_2 with slope $-x$ and is *free* of the lapse normalization A_{t2} ; and the seed-normalized lapse row $[h_{\tau\tau}]_{\ell=2}$ (78) *does* carry A_{t2} . In the column order $(Y_2, \rho_2, A_{t2}, w_2)$ the four rows decouple into three independent blocks (with $s = \sqrt{1-x}$): the rigidity row fixes the bending Y_2 through the nonzero $\ell = 2$ eigenvalue -4 , the polar tilt row fixes w_2 , and the energy–lapse minor fixes the areal and lapse quadrupoles (ρ_2, A_{t2}) ,

$$\partial_{(\rho_2, A_{t2})} ([h_{uu}], [h_{\tau\tau}])_{\ell=2} = \begin{pmatrix} -x & 0 \\ -x/s^2 & -2s \end{pmatrix}, \quad \det = 2x\sqrt{1-x} \neq 0, \quad (88)$$

non-degenerate because the retarded energy row carries ρ_2 with slope $-x$ and is *free* of A_{t2} while the seed lapse row (78) carries A_{t2} : the two are independent conditions, not one row in two gauges. Each $\ell = 2$ unknown is thus fixed by a distinct nonzero block, so the $\ell = 2$ rows are non-degenerate on the whole physical range $x \in (0, 1)$ (the 24/25 window elsewhere is the independent surface-DEC threshold, not a determinant degeneracy). This minor is built from the row *slopes* alone, so it is independent of the asserted $\mathcal{O}(a^2)$ retarded source: the energy row’s ρ_2 -independent piece (the slow-burn graph-gauge retarded input, whose all-orders justification is the compatibility identity of Sec. 14) shifts the *value* $\rho_2^\star$ but neither the non-degeneracy nor the non-obstruction. Solving the minor gives the quadrupole

$$\rho_2 = \rho_2^\star = \frac{2(\rho_1^2 x - 2\rho_1 x - x + 1)}{3x}, \quad (89)$$

together with the (Y_2, A_{t2}, w_2) it determines. This value is gauge-dependent bookkeeping, not a physical prediction: it depends on the $\ell = 1$ amplitude ρ_1 , which is not fixed by the junction, since

ρ_1 is the residual rigid-translation gauge, the shell's center, fixed by centering the interior embedding on Γ . A different centering shifts ρ_1 and, through it, ρ_2^* ; the non-degeneracy (88) and the absence of an obstruction are independent of that choice. The candidate leading $\mathcal{O}(a^2)$ $\ell = 2$ obstruction to the compatibility identity (84) therefore *cancels*, gauge-completely, for the slow fixed-axis burn: the bending and the areal, lapse, and tilt quadrupoles are each fixed by a distinct independent row, with no residual $\ell = 2$ obstruction. This settles the absence of the leading obstruction and the nondegeneracy of the $\ell = 2$ rows. The all-orders identity (84) is established next *not* by extrapolating this $\mathcal{O}(a^2)$ computation but by the square-count and order-independent-isomorphism argument that follows: the per-cut first junction is a fixed isomorphism at *every* order, so (84) closes by rigid determination rather than by iterating the leading cancellation.

The natural objection to an all-orders claim is that the *reduced* constant- u system is over-determined in appearance (Lemma 7), so that a nonlinear source at high order might project onto a cokernel. It cannot, for the following reason. In its *unreduced* form the axisymmetric first junction is *square*: with intrinsic coordinates (u, ϑ, φ) , the unknown functions are the exterior shape $R(u, \vartheta)$ together with the interior embedding into the flat cavity, its cylindrical radius $\rho(u, \vartheta)$, its axial height $z(u, \vartheta)$, and its time $T(u, \vartheta)$, and axisymmetry leaves exactly four rows, one per unknown:

- $[h_{\varphi\varphi}] = 0$ fixes $\rho = R \sin \vartheta$ algebraically;
- $[h_{\vartheta\vartheta}] = 0$ fixes the height z by the quadrature (70), whose one constant of integration is an axial translation of the interior, that is the $\ell = 1$ rigid-translation gauge itself;
- $[h_{u\vartheta}] = 0$ fixes the interior clock tilt, and it is here that the shape velocity cancels;
- $[h_{uu}] = 0$, stripped of that velocity, fixes the shape R itself, algebraically.

No row is left over, so the apparent over-determination of the reduced system is an artifact of the reduction, not a property of the junction. The count alone does not prove surjectivity, a system of differential and quadrature relations not being decided by counting functions; what removes the cokernel is that each block is discharged by a theorem that carries none. The first two rows are the nonlinear isometric embedding, which solves for *every* near-round induced metric of positive curvature (Nirenberg–Pogorelov, with Cohn–Vossen rigidity fixing the residual rigid motion), so it imposes no solvability condition at any amplitude; the last two are the order-independent characteristic block, whose invertibility off the $\ell = 1$ gauge is the fixed nondegenerate minor (77). The count shows there is nothing left over for those two theorems to miss, which is why the mechanism does not degrade with the order.

The reduced per-cut junction is an isomorphism whose structure is *order-independent*: its principal angular part is always the round-sphere operator $\Delta_{S^2} + 2$ (Lemma 7) and its $\ell \leq 1$ Schur block is always the nondegenerate cos-row/tilt determinant (77). On a constant retarded cut the cut is *characteristic*: the retarded energy row (83), first order in ∂_u , does not propagate the shape as free Cauchy datum but fixes it, so at each order in a the shape multipoles are *rigidly determined* by $a(u)$ through that fixed isomorphism, with no solvability condition, uniquely modulo the $\ell = 1$ rigid-translation centering. Hence (84) closes to all orders by rigid determination, for fixed-axis burns, needing no Cauchy–Kovalevskaya estimate to *close*; for real-analytic $a(u)$ the amplitude series additionally converges by an Ovsyannikov analytic majorant [70, 71], with an explicit derivative-loss budget: each order couples to the previous through a single retarded derivative ∂_u in the energy row (83), while the angular inversion $(\Delta_{S^2} + 2)^{-1}$ *gains* two angular derivatives (its $\ell \geq 3$ tail has norm $\frac{13}{10}$ as a map $H^{s-2} \rightarrow H^s$, the finite $\ell \leq 2$ block being solved by the nondegenerate Schur minors (77), (88)), so on a scale of Banach spaces $\{X_\varrho\}$ the recursion loses only analyticity radius ϱ and is majorized by a geometric series whose radius is set by the rear-pole distance $\mu_\star = \inf(f - 2aR)$; for merely smooth burns the C^2 worldtube follows from the Picard–Lindelöf persistence below. Equivalently, the exterior-induced 3-metric on Σ is isometrically embeddable in the flat interior at every order: the angular block is discharged by the nonlinear embedding (Nirenberg–Pogorelov/Cohn–Vossen), which carries no solvability condition, so the $\ell = 1$ residual that nonlinear sources generate at $\mathcal{O}(a^3)$ and beyond is the rigid-motion gauge (the shell's center on Γ), not an obstruction. The one item that remains open is the rotating-axis assembly. With (84) established (its leading $\mathcal{O}(a^2)$ obstruction shown to cancel above, its all-orders closure by the order-independent per-cut isomorphism), the per-cut shapes $Y_\star(u, \cdot)$ (each the unconditional static elliptic match) and the energy-row burn rate $Y_u = \partial_u Y_\star$ are consistent, so the per-cut admissible shells assemble into a single C^2 worldtube solving $[h_{ab}] = 0$ in full, and the contracted

Codazzi momentum balance (27) is the consequent stress balance of the matched solution (the absent magnetic-parity sector leaving it unobstructed). At the burn endpoints $a(u_0) = a(u_f) = 0$ the configuration is the static anchor, and the reduced junction stays nondegenerate there: the $\ell \leq 1$ lapse/tilt Schur determinant is the a -independent cos-row/tilt determinant (77) and the $\ell = 0$ monopole Jacobian is $-x/s^2 \neq 0$ (80), both nonzero at $a = 0$. There is thus no $\mathcal{O}(a)$ bifurcation to remove: the reduced Jacobian is *uniformly* nondegenerate through $a = 0$, so $\mathcal{G}$ and its Lipschitz constant extend continuously across the endpoints, $Y(u) \rightarrow 0$ as $a \rightarrow 0$, and the evolution (90) is uniformly Lipschitz (hence regular) on the *closed* interval $[u_0, u_f]$. This discharges (83) for a fixed-axis burn; the rotating-axis case, where the turning of the axis sources the magnetic-parity sector, is closed to first order in the rotation rate by Lemma 11 below, its residual sharpened there to a single possible $\ell \geq 2$ magnetic-parity defect from the boost-rotation non-commutativity, and otherwise left open (Sec. 14). Thereby (83) holds, its solution map $Y_u = G(Y; \lambda, u)$ making the persistence of the shape block (82) the explicit first-order Banach-space evolution

$$Y_u = G(Y; \lambda, u), \quad G : C_{\text{ax}}^{2,\alpha}(S^2) \rightarrow C_{\text{ax}}^{2,\alpha}(S^2) \text{ locally Lipschitz,} \quad (90)$$

uniformly in $u \in [u_0, u_f]$, with no derivative lost and *no* Nash–Moser iteration. Picard–Lindelöf in $C_{\text{ax}}^{2,\alpha}(S^2)$, with the static anchor $Y \equiv 0$ as datum at u_0 (where $a(u_0) = 0$), gives a unique C^1 -in- u solution $Y(u; \lambda, \varepsilon)$ on $[u_0, u_f]$ for λ, ε small; the C^2 burn data $(m(u), a(u))$ bootstrap it to C^2 in u . A Grönwall estimate gives $\|Y(u)\|_{C^{2,\alpha}} \leq C(\lambda + \varepsilon)$ uniformly on $[u_0, u_f]$, with C controlled by the Schauder constant C_S of (76) and the nondegenerate $\ell \leq 1$ Schur determinant (77), and degrading as $\mu_\star^{-p}$ ($\mu_\star = \inf(f - 2aR)$, $p \in \{1, 2\}$) toward the rear-pole horizon. Thus the embedding $X(u, \Omega)$ is C^2 across the worldtube, so $[K_{ab}]$ and the exact retarded surface stress S_{ab} are continuous in $(\lambda, \varepsilon, u)$. At $(\lambda, \varepsilon) = (0, 0)$, S_{ab} is the static anchor with strict Type I surface-DEC margin $\delta_0 > 0$ (Lemma 3); by continuity there are $\lambda_\star, \varepsilon_\star > 0$ with the margin $> \frac{1}{2}\delta_0$ for $\lambda < \lambda_\star$, $\varepsilon < \varepsilon_\star$ throughout the burn. $\square$

Slow-rotation extension. The fixed-axis restriction relaxes to first order in the rotation rate. Let the acceleration axis $\hat{a}(u)$ rotate at rate $\Omega = |\dot{\hat{a}}|$ with $\Omega R_0 \ll 1$.

Lemma 11 (Slowly rotating axis). *For a slowly rotating axis the leading $\mathcal{O}(\Omega)$ magnetic-parity (axial) source in the time-angle defect $[h_{uA}]$ lies entirely in the $\ell = 1$ rotation-generator subspace of S^2 and is removed by enlarging the time-row gauge with the rigid $SO(3)$ co-rotation of the interior frame ($Q^{-1}\dot{Q} = [\omega]$, ω the instantaneous angular velocity). No $\ell \geq 2$ magnetic cokernel is sourced at linear order in Ω about the round anchor, so the fixed-axis assembly of Theorem 4 extends to a single C^2 worldtube through linear order in Ω . This is not an all-orders rotating-axis existence theorem: the potentially unprotected magnetic-parity term arises from the co-rotation that clears the $\ell = 1$ magnetic row feeding back into the $\ell \geq 2$ sector, left open below.*

Proof. Pull the angular metric back by the co-rotating frame; the $du d\vartheta^A$ term is $R^2 q_{AB} K_\omega^B du d\vartheta^A$, which at the round anchor is $R_0^2 d(\omega \cdot \hat{q})$, a pure $\ell = 1$ axial one-form (the S^2 rotation Killing field); the co-rotation gauge contributes the same one-form with opposite sign, and they cancel. The apparent $\mathcal{O}(a\Omega)$ higher harmonic from the $\mathcal{O}(a)$ anchor shape is a gauge artifact of co-rotating only the round anchor: the rigid co-rotation of the whole matched cut returns it to the gauge image. The energy row $[h_{uu}]$ and the polar tilt row are unchanged at this order, so Lemmas 7–8 give the same bounded triangular solve, and the finite-dimensional rotation equation produces only a bounded frame rotation, not a secular elliptic mode. That the $\mathcal{O}(\Omega)$ rotation one-form carries no $\ell \geq 2$ magnetic projection is the Legendre identity $\int_{-1}^1 (1 - \xi^2) P'_\ell(\xi) d\xi = 2 \int_{-1}^1 \xi P_\ell(\xi) d\xi = \frac{4}{3} \delta_{\ell 1}$, immediate from integration by parts and the orthogonality of $P_1(\xi) = \xi$. The residual rotational coupling, where the co-rotation solving the $\ell = 1$ magnetic row can feed back into the $\ell \geq 2$ sector, is not controlled here; its structure is set out next. $\square$

Structure of the rotational residual. In the co-rotating frame the rotation enters the first junction through two channels only. The quadratic (centrifugal) term of the induced metric, $R^2 q_{AB} K_\omega^A K_\omega^B = R^2(|\omega|^2 - (\omega \cdot \hat{n})^2)$ with $\hat{n}$ the radial unit, is a *pure scalar*, so it feeds only the even redshift row $[h_{uu}]$ and by parity sources no axial part of $[h_{uA}]$: the even and magnetic sectors decouple at $\mathcal{O}(\Omega^2)$. The mixed $\mathcal{O}(a\Omega)$ term of the pure co-rotation gauge, the acceleration dipole times the rotation one-form $2a\rho_1 \cos \vartheta R_0^2 K_\omega$ (the difference between co-rotating the deformed cut and the round anchor), cancels in the jump, the gauge artifact already removed above. The $\ell = 1$ magnetic row itself is the intrinsic angular-momentum balance, the isotropic moment $\oint R^2(\delta_{ij} - \hat{n}_i \hat{n}_j) d\Omega = \frac{8\pi}{3} R^2 \delta_{ij}$ fixing the co-rotation rate ω ; solving it is not the same as setting the

co-rotation to the bare frame rotation, since the boost-frame drift beating the $\mathcal{O}(a)$ shape dipole carries a magnetic $\ell = 1$ mode the polar interior tilt cannot absorb, so the co-rotation rate departs from the frame rotation and that departure can re-enter the $\ell \geq 2$ magnetic sector. What is *not* established is whether this boost-rotation non-commutativity leaves a residual $\ell \geq 2$ magnetic-parity defect in $[h_{uA}]$. Such a defect could not be absorbed by any tangential drift: only the $\ell = 1$ rotations are conformal isometries of the round sphere, so an $\ell \geq 2$ tangential drift shears the angular metric, a shear the scalar shape response (a pure trace at the umbilical anchor) cannot cancel. Its absence is the magnetic-parity analog of the compatibility identity (84), and is left open.

F $\mathcal{O}(\mu)$ back-reaction of the radiating wall

On a *fixed* wall background the two-component stress (53) has worst-observer DEC margin $\rho_* - p_t$ with radial-block eigenvalue $\rho_* = \frac{1}{2}\rho(1 + \sqrt{1 + 4\mu/\rho})$ and p_t untouched by the radial null dust, so $d\rho_*/d\mu > 0$: on a frozen background the superposed null dust *raises* the margin (the discriminant $\Delta = \rho(\rho + 4\mu) > 0$ keeping it Hawking–Ellis Type I throughout, never Type IV). We test whether this frozen-background lemma survives the $\mathcal{O}(\mu)$ Einstein back-reaction of the radiation on the geometry, and resolve the sign density-conditionally: the first-variation formula and the lapse response are analytic, the sign fixed by the analytic threshold $\frac{r}{2}\rho \partial\Phi'/\partial\mu \gtrless 1$ (negative in the dense core, positive in the dilute skin); whether the realized wall crosses it is quantified in the Supplement.

Theorem 7 (Lapse response of the radiating wall). *For the exact tangential-pressure wall ($p_r = 0$, $\Phi' = m(r)/[r(r - 2m(r))]$, $p_t = \frac{r}{2}\rho \Phi'$) carrying an outgoing null-dust amplitude $\mu \geq 0$ (energy density, radial pressure, and flux all $= \mu$ in the static frame), let the metric respond (the Tolman–Oppenheimer–Volkoff lapse with $p_r = \mu$). Then the lapse deepens,*

$$\frac{\partial\Phi'}{\partial\mu} = \frac{4\pi r}{1-x} + \frac{1 + 8\pi r^2\mu}{(r-2m)^2} \frac{\partial m}{\partial\mu} \geq \frac{4\pi r}{1-x} \geq 4\pi r > 0, \quad (91)$$

uniformly on the local-compactness window $x = 2m/r < 4/5$.

Proof. With radial pressure $p_r = \mu$ the Tolman–Oppenheimer–Volkoff lapse gradient at radius r is $\Phi' = [m(r) + 4\pi r^3\mu]/[r(r - 2m(r))]$. Differentiate at fixed r . The explicit μ -dependence of the numerator contributes $4\pi r^3/[r(r - 2m)] = 4\pi r^2/(r - 2m) = 4\pi r/(1 - x)$. The implicit dependence through the enclosed mass contributes $(\partial\Phi'/\partial m)(\partial m/\partial\mu)$, and

$$\frac{\partial\Phi'}{\partial m} = \frac{r(r - 2m) + 2r[m + 4\pi r^3\mu]}{r^2(r - 2m)^2} = \frac{1 + 8\pi r^2\mu}{(r - 2m)^2} > 0,$$

while $\partial m/\partial\mu \geq 0$, since $m(r) = \int_0^r 4\pi r'^2 \rho_{\text{tot}} dr'$ is nondecreasing in the added energy density. Both contributions are therefore non-negative, and $1/(1 - x) \geq 1$ on $0 \leq x < 1$, giving (91). $\square$

The Type-I structure and the order of the flux effect are elementary. The two-component radial block $\begin{bmatrix} \rho + \mu & \mu \\ \mu & \mu \end{bmatrix}$ has discriminant $\Delta = \rho(\rho + 4\mu) > 0$ (54), so it stays Hawking–Ellis Type I, with eigen-density $\rho_* = \frac{1}{2}(\rho + \sqrt{\rho^2 + 4\rho\mu}) = \rho + \mu + \mathcal{O}(\mu^2)$: beyond the bare energy density μ the dust carries diagonally, the flux-coupling correction to ρ_* is $\mathcal{O}(\mu^2)$. The linear margin response thus has two pieces, the bare $+\mu$ of ρ_* competing with the *lapse* response of Theorem 7 acting through p_t , which signs as follows.

Proposition 9 (Density-conditional back-reaction sign). *The wall matter keeps $p_r = 0$, so its tangential pressure is set by its own anisotropic equilibrium $p_t = \frac{r}{2}\rho \Phi'$ with only the lapse responding to the added radiation energy; with $\rho_* = \rho + \mu + \mathcal{O}(\mu^2)$ the worst-observer surface-DEC margin's first variation is*

$$\left. \frac{\partial(\text{margin})}{\partial\mu} \right|_{\mu=0} = 1 - \frac{r}{2}\rho \frac{\partial\Phi'}{\partial\mu}, \quad (92)$$

with $\partial\Phi'/\partial\mu \geq 4\pi r/(1 - x) > 0$ the lapse response of Theorem 7. The sign is therefore analytic and density-conditional: $\partial(\text{margin})/\partial\mu < 0$ (outgoing radiation tightens the dominant energy condition) exactly where $\frac{r}{2}\rho \partial\Phi'/\partial\mu > 1$, and, because $p_t \propto \rho$, it returns to $+1$ as $\rho \rightarrow 0$.

Proof. On the tangential-pressure wall the worst-observer margin is $\rho_* - p_t$, with $p_t = \frac{r}{2}\rho \Phi'$. From the radial block above, $\rho_* = \rho + \mu + \mathcal{O}(\mu^2)$, so $\partial\rho_*/\partial\mu|_{\mu=0} = 1$. The wall matter keeps $p_r = 0$, so ρ itself does not respond at linear order and the only μ -dependence of p_t is through the lapse, giving $\partial p_t/\partial\mu = \frac{r}{2}\rho \partial\Phi'/\partial\mu$. Subtracting the two gives (92), and the stated threshold follows from $\partial\Phi'/\partial\mu > 0$ (Theorem 7). $\square$

The first-variation formula (92) and its threshold are analytic; whether the realized tangential-pressure wall crosses that threshold in its core is a quantitative question about a specific density profile, mapped in the Supplement (Sec. S), the pure-flux piece a further $\mathcal{O}(\mu^2)$ down.

Two self-consistent closures fail. (i) A generic generalized-Vaidya mass-ramp $M(u, r) = \bar{m}(u)\chi(r)$ carries $p_r \neq 0$, whose *static* radial-tension channel violates the DEC at the matter–vacuum edge (Prop. 6) as μ grows; its $(\hat{u}, \hat{r})$ block is the defective Hawking–Ellis Type II (DEC-saturating, $\Delta = 0$), *not* Type IV: the smoothstep pathology again. (ii) Feeding the null dust into a *static* constraint solve as a radial pressure $p_r = \mu\rho$ forces it into the wall’s hydrostatic equilibrium, which then demands large tangential *tension* $p_t < 0$ and fails DEC; but the through-wall radiation is a *flux* (radial momentum balanced by $\dot{m}$), not a static pressure the wall must support. Both controls show the $p_r = 0$ structure must be preserved.

The correct construction rewrites the exact $p_r = 0$ tangential wall $g = \text{diag}(-e^{2\Phi}, e^{2\Lambda}, r^2, r^2 \sin^2 \vartheta)$ in outgoing Eddington–Finkelstein form (the exact transformation $t = u + \int e^{\Lambda-\Phi} dr$),

$$ds^2 = -e^{2\Phi(r)} du^2 - 2e^{\Phi(r)+\Lambda(r)} du dr + r^2 d\Omega^2, \quad (93)$$

and lets it radiate *adiabatically*: a slow modulation $\omega R \ll 1$, $m(u) = m_0(1 - \epsilon \sin \omega u)$, so (Φ, Λ) acquire a slow u -dependence (linear in $m(u) - m_0$, exact to first order in the amplitude ϵ); the u -derivative generates a nonzero outgoing null flux ($n^2 > 0$) while the matter stays $p_r = 0$ in its rest frame. Reading the *curvature-derived* stress from the exact curvature, the metric is a self-consistent Einstein solution (satisfying the contracted Bianchi identity $\nabla_a T_{ab} = 0$) and stays Type I, its worst-observer DEC margin quantified numerically in the Supplement (Sec. S).

The self-consistent calculation *corrects* the sign of the fixed-background lemma: in this adiabatic family, with the metric allowed to respond, *outgoing* radiation *tightens* the dominant energy condition through the dense wall core ($d(\text{margin})/d\mu < 0$ where the lapse deepening dominates, $\frac{r}{2}\rho\partial\Phi/\partial\mu > 1$, Prop. 9; the sign neutralizes only in the $\rho \rightarrow 0$ edge skin), because DEC constrains the radiated energy *flux*, not a pressure. The radiating wall therefore remains admissible, but the back-reaction sharpens rather than relaxes the margin in the core (Fig. S4d); the frozen-background monotonicity was an artifact of treating $\mu\ell$ as separately conserved on a fixed geometry. This closes the back-reaction *at adiabatic linear order* for the present construction; the residual (the time-dependent terms beyond the slow-modulation expansion, formally higher order in $(\epsilon, \omega R)$) awaits a fully dynamical evolution. One piece of the $\mathcal{O}(\mu^2)$ correction is already signed: the flux-coupling channel alone tightens the margin, the eigen-density expanding as $\rho_* = \rho + \mu - \mu^2/\rho + \mathcal{O}(\mu^3)$, so any $\mathcal{O}(\mu^2)$ *relaxation* could arise only from the finite- ωR lapse response, which is the piece left to a dynamical evolution. The finite maneuver of Theorem 3, radiating a substantial fraction over a few light-crossing times, has an effective $\omega R \sim 0.25$, at the edge of the strict-adiabatic $\omega R \ll 1$ regime, so the adiabatic closure is itself quasi-static there and the $\mathcal{O}(\mu^2)$ dynamical terms are not parametrically negligible during a saturating burn.

G Optimal control and the geometric reading

This appendix supplies the momentum balance behind Sec. 3, the multipole form of the optimum of Sec. 12, and the geometric-control derivations behind Sec. 12.1.

Conservation law (Proposition 2). For the exact Kerr–Schild rocket, dP_B^μ/du can be obtained in two independent ways. Kinematically, differentiating the point-rocket value $P_B^\mu = m(u)v^\mu(u)$ (v^μ the unit worldline velocity) in u gives $\dot{m}v^\mu + m\hat{a}^\mu$, with $\hat{a}^\mu$ the unit acceleration direction ($\dot{v}^\mu = \hat{a}^\mu$, and $\hat{a} \cdot v = 0$). Dynamically, the radiated flux is $\oint n^2 \hat{\ell}^\mu d\Omega$ with n^2 read from the Einstein tensor (8); inserting the dipole $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$ and using the two elementary angular integrals $\oint \hat{\ell}^\mu d\Omega = 4\pi v^\mu$ and $\oint \cos \vartheta \hat{\ell}^\mu d\Omega = \frac{4\pi}{3} \hat{a}^\mu$ gives $\oint n^2 \hat{\ell}^\mu d\Omega = -\dot{m}v^\mu - ma\hat{a}^\mu$. The first route never uses the stress-energy and the second never uses the worldline momentum, and the two cancel identically, $dP_B^\mu/du + \oint n^2 \hat{\ell}^\mu d\Omega = 0$, in both the energy and the momentum components: the balance is a consequence of the field equations.

Multipole form of the optimum (Theorem 5). Writing $4\pi n^2 = \sum_\ell b_\ell P_\ell(\cos \vartheta)$, the moment identities read $-\dot{m} = b_0$ and $ma = -b_1/3$ (the thrust; $b_1 = -3ma < 0$), and bulk positivity is $\sum_\ell b_\ell P_\ell \geq 0$. On the news-silent class, which Theorem 5 identifies with the dipole class $\{b_0, b_1\}$, the profile is affine in $\cos \vartheta$ and therefore extremal at the poles, so positivity is $b_0 \geq |b_1|$; minimizing the fuel b_0 at the fixed thrust $|b_1| = 3m|a|$ gives $b_0 = |b_1| = 3m|a|$, the emission vanishing at the forward pole. This is the saturated control law, and integrating it reproduces the Tsolkovsky budget (15). Without the news-silence restriction the bare program “minimize b_0 at fixed b_1

subject to $\sum_\ell b_\ell P_\ell \geq 0$ is instead formally optimized by a rear-pole spike ($\langle \cos \vartheta \rangle \rightarrow -1$), a singular profile carrying all multipoles; it is excluded because by Lemma 12 the news functional is strictly positive on any $\ell \geq 2$ admixture, so no such profile is gravitational-wave silent.

Strict positivity of the news weights. In the broader gravitational-wave-allowed class the trade-off is governed by the exact news energy $J_{\text{news}} = \frac{1}{4\pi} \oint |\mathcal{N}|^2 d\Omega \geq 0$. For the Robinson–Trautman class the news is, to leading order in the $\ell \geq 2$ amplitudes, a linear functional of the $\ell \geq 2$ part of $(\ln P)_{,u}$ (Damour [58]), so

$$J_{\text{news}} = \sum_{\ell \geq 2} W_\ell |c_\ell|^2 + \mathcal{O}(c^3), \quad W_\ell > 0, \quad (94)$$

vanishing exactly on $\ell \leq 1$.

Lemma 12 (Strict positivity of the news weights). *To leading order the Bondi news is the rotationally covariant linear map $\mathcal{N} = Lc$ of the $\ell \geq 2$ emission multipoles $c = \{c_{\ell m}\}$, with L carrying the spin-2 raising ∂^2 of the Robinson–Trautman news in its angular part (Damour [58]). Rotational invariance makes L mode-diagonal, acting on each ℓ -component by the ∂^2 weight $\sqrt{(\ell-1)\ell(\ell+1)(\ell+2)}$ (a singular value, since ∂^2 raises spin 0 to spin 2), which is nonzero for every $\ell \geq 2$ and vanishes only for $\ell \leq 1$; hence L has trivial kernel on $\ell \geq 2$ (the linearized counterpart of the von der Gönna–Kramer [59]–Cornish [74] characterization $\mathcal{N} \equiv 0 \Leftrightarrow \ell \leq 1$). Therefore $J_{\text{news}} = \frac{1}{4\pi} \oint |\mathcal{N}|^2 d\Omega = \langle c, L^\dagger L c \rangle$ is positive-definite on the $\ell \geq 2$ amplitudes, with the single positive eigenvalue $W_\ell = (\ell-1)\ell(\ell+1)(\ell+2)w_\ell > 0$ per multipole, where $w_\ell > 0$ is the squared radial factor of L in that mode, positive precisely because L is injective on $\ell \geq 2$; this gives (94). Only the strict positivity of W_ℓ is used here, not its value.*

An $\ell \geq 2$ mode therefore radiates J_{news} quadratically in its amplitude while saving fuel only linearly, so for a sufficiently weighted gravitational channel the optimum is the dipole; the unconditional statement is Theorem 5, where $J_{\text{news}} = 0$ forces $\ell \leq 1$.

G.1 The geometric-control reading: hyperbolic norm, factor 3, and reachability

The cost is the hyperbolic length. The worldline four-velocity $v^\mu(\tau)$ is a unit timelike curve, $v \cdot v = -1$, hence a curve on the hyperboloid $\mathbb{H}^3$; its tangent $\dot{v}^\mu = dv^\mu/d\tau$ is the proper-acceleration four-vector a^μ , orthogonal to v , with magnitude $|a| = \sqrt{a_\mu a^\mu} = \sqrt{g_{\mathbb{H}^3}(\dot{v}, \dot{v})}$ the hyperbolic speed (the induced metric on $\mathbb{H}^3$ from $\eta_{\mu\nu}$ is the hyperbolic metric of curvature -1). Thus $\int |a| d\tau = L_{\mathbb{H}^3}[v]$ is exactly the hyperbolic arclength. Along a saturated news-silent maneuver $d \log m/du = -3|a|$ (with u the worldline proper time on each cut), so $\log(m_0/m_f) = 3L_{\mathbb{H}^3}[v]$, Eq. (55); minimizing the radiated mass $J = m_0(1 - e^{-3L_{\mathbb{H}^3}})$ is equivalent to minimizing $L_{\mathbb{H}^3}$, whose minimizer between fixed endpoints is the hyperbolic geodesic of length $d_{\mathbb{H}^3}(v_0, v_1) = \text{arccosh}(-v_0 \cdot v_1)$, Eq. (56). The geodesics of $\mathbb{H}^3$ are the one-parameter boost subgroups, so the optimal velocity history is a single boost and the optimal emission is the saturated Damour dipole tracking it; any cornered (multi-direction) path is strictly longer by the triangle inequality on the negatively curved $\mathbb{H}^3$. This is a left-invariant control problem on $SO^+(3, 1)$ (Riemannian geodesy on $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$): with $\Lambda(u) \in SO^+(3, 1)$ the boost frame, $\dot{\Lambda} = \Lambda \sum_i u_i K_i$ (K_i the boost generators) and cost $3 \int \|u\| d\tau$, whose Pontryagin/Lie–Poisson extremals are the boost geodesics [77, 78]. Note the *kinematic* (log-fuel) system is left-invariant, whereas the physical energy $\int (-\dot{m}) du = \int 3m\|u\| du$ carries the scalar m and is not literally a left-invariant length; the two share the same minimizers at fixed m_0 because J is monotone in $L_{\mathbb{H}^3}$.

The radiation moment cone and the factor 3. With $\hat{\ell}^\mu = (1, \hat{\Omega})$ the asymptotic null direction and $4\pi n^2 = b_0 + \vec{b} \cdot \hat{\Omega}$ the news-silent ($\ell \leq 1$) emission, the moments $Q^\mu = \oint n^2 \hat{\ell}^\mu d\Omega$ are, using $\oint d\Omega = 4\pi$, $\oint \hat{\Omega} d\Omega = 0$, $\oint \hat{\Omega}^i \hat{\Omega}^j d\Omega = \frac{4\pi}{3} \delta^{ij}$,

$$Q^0 = b_0 = -\dot{m}, \quad \vec{Q} = \frac{\vec{b}}{3}, \quad |\vec{Q}| = \frac{|\vec{b}|}{3} = m|a|, \quad (95)$$

the radiated momentum $\vec{Q}$ pointing along the exhaust and the thrust being the recoil $-\vec{Q} = m \vec{a}$; so $n^2 \geq 0$ for all Ω (i.e. $b_0 \geq |\vec{b}|$) is exactly $-\dot{m} \geq 3m|a|$, the control law (14). Dropping the news-silent restriction, an arbitrary $n^2 \geq 0$ lets Q^μ fill the future causal cone, $Q^0 \geq |\vec{Q}|$, with equality (factor 1) approached only in the collimated rear delta-emission limit; the broad dipole’s $\frac{1}{3}$ in $|\vec{Q}| = |\vec{b}|/3$ is the entire origin of the factor 3, a property of the news-silent Robinson–Trautman class, not a general-relativistic floor.

Reachability. In velocity space the geodesics reach all of $\mathbb{H}^3$, so any rapidity $\eta < \infty$ is attained with $m_f = m_0 e^{-3\eta} > 0$; the null boundary $\eta \rightarrow \infty$ needs $m_f \rightarrow 0$ and is excluded. In four-momentum space the constraint is $P_B(u_0) - P_B(u_f) = \int Q du \in \bar{V}_+$ (future causal), a necessary condition that is *not* satisfied by every future-timelike target, so the momentum-space reachable set is a proper sub-cone.

The state-constrained maneuver: Pontryagin structure. The bounded-control problem is posed on the state manifold $\mathbb{H}^3 \times (0, 1)$ with state (v, x) , control $u \in \mathbb{R}^3$ (the rest-frame boost rate, $\|u\|_{\mathbb{H}^3} = |a|$), saturated news-silent dynamics

$$\dot{v} = \left(\sum_i u_i K_i \right) v, \quad \dot{x} = -3x \|u\|_{\mathbb{H}^3}, \quad (96)$$

(K_i the boost generators), the *moving frontier* mixed state-control constraint $C(x, u) = R\|u\|_{\mathbb{H}^3} - g(x) \leq 0$ for any C^1 frontier g , and endpoints $v(u_0) = v_0$, $v(u_f) = v_1$. Because $\partial C / \partial u = R u / \|u\|_{\mathbb{H}^3} \neq 0$ on the active set, C is a *mixed* constraint of *order zero* (smoothed to $\frac{1}{2}(R^2\|u\|^2 - g^2)$ at $u = 0$); the identity is the *length reduction* $x(\ell) = x_0 e^{-3\ell}$, fuel = 3ℓ in hyperbolic arclength $d\ell = \|u\|_{\mathbb{H}^3} du$.

Theorem 8 (State-constrained multi-burn maneuver). *With the control Hamiltonian $\mathcal{H} = -L + \langle p_v, (\sum_i u_i K_i) v \rangle_{\mathbb{H}^3} - 3x p_x \|u\|_{\mathbb{H}^3} - \nu C(x, u)$ ($\nu \geq 0$ the KKT multiplier), the Pontryagin maximum principle gives the following necessary structure, globally sufficient for the bare objectives (ii). (i) The costate is Lie–Poisson, $\dot{P} = \text{ad}_{u \cdot K}^* P$, and on every arc the control aligns with the velocity-costate, so the velocity path is an $\mathbb{H}^3$ geodesic (a single boost direction) between switches. (ii) For the bare objectives $L = 3\|u\|_{\mathbb{H}^3}$ (min log-fuel) and $L = 1$ (min-time) the problem is single-arc: min log-fuel is the unconstrained geodesic, $C_{\min} = 3 d_{\mathbb{H}^3}(v_0, v_1)$ (56) (the frontier never binds), and min-time is the same geodesic traversed at the saturated frontier $\|u\|_{\mathbb{H}^3} = g(x)/R$ a.e. ($\nu > 0$), with $T_{\min} = \int_0^{d_{\mathbb{H}^3}} R/g(x_0 e^{-3\ell}) d\ell$ monotone in path length. (iii) Multi-arc optima arise only for an objective that breaks the length reduction (e.g. a quadratic dwell/effort penalty), as concatenations of interior geodesic arcs ($\nu = 0$, $\|u\| < g/R$) and boundary maximal-acceleration arcs ($\nu > 0$, $\|u\| = g/R$); the order-zero mixed constraint keeps the costates and $\mathcal{H}$ continuous across junctions (complementary slackness $\nu C = 0$), and the moving frontier binds early (large x , small g) and releases as the rocket radiates.*

Proof. The dynamics (96) are right-invariant on $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$ and the constraint C is of order zero, so the Pontryagin maximum principle applies in its mixed-constraint form [75, 76]. (i) Maximizing $\mathcal{H}$ over u at fixed costate aligns u with the velocity-costate, and right-invariance makes the costate equation the Lie–Poisson flow $\dot{P} = \text{ad}_{u \cdot K}^* P$; a curve whose tangent keeps a fixed boost direction is an $\mathbb{H}^3$ -geodesic, which gives the geodesic arcs between switches. (ii) For $L = 3\|u\|_{\mathbb{H}^3}$ the cost is $3 \int \|u\|_{\mathbb{H}^3} du = 3\ell$, the hyperbolic arclength, which is reparametrization-invariant and independent of $\|u\|$; the constraint therefore cannot lower it, $\nu = 0$, and the optimum is the unconstrained geodesic with $C_{\min} = 3 d_{\mathbb{H}^3}(v_0, v_1)$ (56), the Cartan–Hadamard minimizer. For $L = 1$ the cost is the elapsed time, strictly decreasing in $\|u\|$, so the maximum principle saturates the constraint almost everywhere, $\|u\|_{\mathbb{H}^3} = g(x)/R$ with $\nu > 0$; the path is still the same geodesic by (i), and integrating $d\ell = \|u\|_{\mathbb{H}^3} du$ along it with the length reduction $x(\ell) = x_0 e^{-3\ell}$ gives $T_{\min}$. (iii) Both bare objectives are homogeneous of degree one in $\|u\|$, which makes them single-arc; an objective that is not (a quadratic dwell or effort penalty) breaks the length reduction, and the standard order-zero junction conditions then permit concatenations of interior ($\nu = 0$) and boundary ($\nu > 0$) arcs, with ν continuous at the junctions. $\square$

This is the precise sense in which warp-maneuver design is bounded-control geodesy on the Lorentz group, whose state-constrained optima include multi-arc interior-geodesic/boundary-acceleration switching. Under a conformally stationary (de Sitter static-patch) drift $|W| < 1$ the *min-time* problem deforms to Zermelo navigation with $\mathbb{H}^3$ -geodesics replaced by Randers–Finsler geodesics (Sec. 12.1), with the kinematic ceiling tightened to $\lambda < \frac{1}{2}(1 - x - y)$, $y = \frac{\Lambda}{3} R^2$, by Lemma 2 (20). The state-constrained version (frontier $|\dot{v}|_{\mathbb{H}^3} R \leq g(x(u))$ with $x(u)$ falling as the rocket radiates) is the multi-burn trajectory-optimization problem of Sec. 12.1.

H Peeling of the asymptotic hypotheses

Theorem 1 assumes Newman–Penrose peeling [34, 35, 36]. For the Kinnersley exterior the hypothesis is analytic: the metric (4) is algebraically special (a Robinson–Trautman pure-radiation

spacetime whose repeated principal null direction $\ell = -(du)^\sharp$ is geodesic, shear-free, twist-free, and expanding) so by the generalized Goldberg–Sachs theorem for aligned pure radiation (the Ricci tensor being null and aligned with ℓ) $\Psi_0 = \Psi_1 = 0$ and the surviving Weyl scalars carry the standard falloff $\Psi_2 \sim r^{-3}$, $\Psi_3 \sim r^{-2}$, $\Psi_4 \sim r^{-1}$ in the asymptotic Bondi tetrad. The spacetime is Petrov type D: only $\Psi_2 \sim -m/r^3$ survives in the principal co-moving tetrad ($\ell = -(du)^\sharp$), the nonzero Ψ_3, Ψ_4 being the image of Ψ_2 under the null rotation to the accelerated Bondi frame (scaling as a and a^2 , hence vanishing in the static limit), with the Schwarzschild and Vaidya cases its static and straight-worldline specializations. The peeling exponents $\Psi_n \sim r^{-(5-n)}$ and the gravitational-wave silence of the dipole maneuver (the Bondi news $\mathcal{N} \equiv 0$, since ∂^2 annihilates the $\ell \leq 1$ dipole) are also mapped numerically in the Supplement (Sec. S).

I The radiating radial mode

This appendix carries the two faces behind Proposition 7: the closed-form oscillator coefficients (97) and the marginal equation-of-state slope (98).

Model the radiating shell as the spherical Minkowski | Vaidya proxy: interior Minkowski, outgoing Vaidya exterior $f = 1 - 2m(u)/r$, $m_{,u} = -L$, matched at $r = R(\tau)$ with a barotropic wall $p = p_0 + \eta(\sigma - \sigma_0)$. The exterior carries no acceleration, so $s = \sqrt{1-x}$ throughout (not the accelerated pole value $s_\pm = \sqrt{1-x \mp 2\lambda}$ of Sec. 8). The surface density is

$4\pi R\sigma = \sqrt{1 + \dot{R}^2} - \sqrt{f + \dot{R}^2}$, and the outgoing flux adds to the exterior $K^{\hat{r}}_{\hat{r}}$ the term

$L/[R\sqrt{f + \dot{R}^2}(\sqrt{f + \dot{R}^2} + \dot{R})^2]$. A radial perturbation shifts the retarded time, so the enclosed mass perturbs, $\delta m = m_{,u} \delta u = -L \delta u$; keeping this mass-history channel (freezing m leaves the sign gauge-dependent), linearizing the junction about the quasi-static equilibrium, and reducing the coupled $(\delta R, \delta \dot{R}, \delta m)$ system to the radial mode gives the non-autonomous Poisson–Visser oscillator $\delta \ddot{R} + 2\kappa_d \delta \dot{R} + \Omega^2 \delta R = 0$, to leading order in L and with $R_0 = 1$,

$$\Omega^2 = \frac{D}{4s^2}, \quad \kappa_d = -\frac{LN}{2D(1-s)s^3}, \quad \begin{aligned} D &= 12\eta s^3 + 4\eta s^2 + 3s^3 - s^2 - s - 1, \\ N &= 4\eta s^4 + 24\eta s^3 + 4\eta s^2 + s^4 + 6s^3 - 4s^2 - 2s - 1. \end{aligned} \quad (97)$$

Stiffening the wall keeps $\Omega^2 = D/(4s^2) > 0$ at no cost in the (still η -independent) surface-DEC margin $\sigma_0 - p_0 = (1-s)(5s-1)/(16\pi R s) - L/(8\pi R s^2)$, which stays positive for $L < \frac{1}{2}s(1-s)(5s-1)$, so a radiating equilibrium exists on that window. On the oscillatory branch $D > 0$ (positive restoring force) the threshold $D = 0$ exceeds the zero of N by $(s+1)(s-1)^2/[2s(3s+1)(s^2+6s+1)] > 0$, so $N > 0$ and $\kappa_d < 0$: the outgoing radiation *anti-damps* the radial mode (equivalently the oscillatory root of the characteristic polynomial has real part $-\kappa_d = \mathcal{O}(L) > 0$). For a wall held a fixed distance from marginal ($D \geq D_0 > 0$) the growth is finite-burn-bounded, $\exp \int |\kappa_d| d\tau$ set by the control law (14), so a finite maneuver does not destabilize the radial mode, though it is not asymptotically stable; near the marginal wall $|\kappa_d| \sim L/D$ grows and the bound degrades.

Solving $V''(R_0) = 0$ (equivalently $D = 0$) for the marginal equation-of-state slope gives

$$\eta_{\text{crit}}(x) = -\frac{3s^3 - s^2 - s - 1}{4s^2(3s+1)}, \quad s = \sqrt{1-x}, \quad (98)$$

so $\eta_{\text{crit}}(x) = \{0.021, 0.046, 0.079, 0.184, 0.427\}$ at $x = \{0.1, 0.2, 0.3, 0.5, 0.7\}$, and $\eta_{\text{crit}} = 1$ exactly at the root of $15s^3 + 3s^2 - s - 1 = 0$, that is at $x_\eta \simeq 0.844$: strict stability ($\Omega^2 > 0$) is reachable with subluminal ($\eta < 1$) wall matter for every $x < x_\eta$, covering the whole mapped operating range; only in the sliver $x_\eta < x < 24/25$, still inside the surface-DEC window, would strict stability demand a superluminal equation of state. The stiffening stays causal there and costs nothing in the junction-fixed surface-DEC margin.

Supplementary material

This Supplement collects the numerical results that support the analytical theory: the finite-amplitude frontier map, the worked finite-duration maneuver, the wall dominant-energy scans, and the frozen-background anchor-stability estimate. No theorem in the main text depends on it; the closed-form radiating-mode algebra behind Proposition 7 is in App. I.

Numerical frontier: the finite-amplitude bracket

We carry out the fully nonlinear matched Darmois solve carrying $\ell \geq 2$ multipoles: a finite- λ least-squares solve of the full $[h_{ab}] = 0$ with the areal shape $R(\vartheta) = R^0(1 + \sum_\ell a_\ell P_\ell)$, the lapse, and the foliation tilt on *both* sides expanded in Legendre/vector-harmonic modes. Under a successively refined multipole expansion the residual *falls geometrically with the multipole order*: with the matching frames held timelike (a physical constraint, since tilting a frame past null gives a spurious degenerate frame and a divergent surface stress), $\max|[h_{ab}]|$ falls to 10^{-7} – 10^{-9} by $L = 6$ for $x \lesssim 0.3$, and to $\sim 10^{-3}$ by $L = 6$ at $x = 0.5, 0.7$ (tightening further with $L = 8$ – 10), once the $\ell \geq 2$ shape and the *interior re-slicing* (lapse+tilt) modes are present; the heuristic envelope (39) is an independent (non-rigorous) comparison the computed values respect, not a proven bound (the rigorous upper bound is the kinematic ceiling, Prop. 5). The dipole-only mismatch of 1–10% is thus a missing-multipole artifact, not an obstruction: the residual falls under multipole refinement, *numerical evidence* that the accelerating photon-rocket exterior is Darmois-matchable to a flat cavity through a thin shell at finite amplitude (the rigorous, unconditional existence statement remains the small-amplitude Theorem 2). On the converged match the surface stress is Hawking–Ellis Type I and the worst-observer DEC margin is strictly *positive* at the computed amplitude $\lambda = \underline{g}(x)$ across the whole window $x \in [0.1, 0.7]$, binding at the rear pole and stable under multipole refinement: the converged margin runs $\{1.8, 9.1, 18, 21\} \times 10^{-3}$ at $x = \{0.1, 0.3, 0.5, 0.7\}$ (with $\max|[h_{ab}]| = 1.4 \times 10^{-7}$ at $x = 0.1$ and $\sim 10^{-3}$ at $x = 0.5, 0.7$), and gauge-invariant. The two computations are complementary. The observer-robust *dipole* scan locates the margin zero at

$$\begin{aligned} \underline{g}(x) &\simeq \{0.19, 0.20, 0.19, 0.14, 0.09\} \\ \text{at } x &= \{0.1, 0.2, 0.3, 0.5, 0.7\} \end{aligned} \tag{S1}$$

(to within the $\sim 5\%$ scan resolution; every value respects the envelope (39)); the *converged multipole* solve, evaluated at that same $\lambda = \underline{g}(x)$, finds the worst-observer margin still strictly *positive* (the same $\{1.8, \dots, 21\} \times 10^{-3}$ above), because the higher multipoles only relax the junction. The true margin zero therefore lies at $\lambda \geq \underline{g}(x)$, so $\underline{g}(x)$ is a numerical *lower bound* on the frontier $g(x)$ of (37), and the margin grows as $\underline{g}(x)$ falls with compactness. Together with the rigorous kinematic ceiling (Prop. 5) this *brackets* the frontier,

$$\underline{g}(x) \leq g(x) \leq \frac{1}{2}(1-x), \tag{S2}$$

a numerical lower bound $\underline{g}$ (supported by a Newton–Kantorovich estimate below) and a rigorous closed-form upper bound.

Newton–Kantorovich convergence estimate. A Newton–Kantorovich [89] existence proof at a computed amplitude $\lambda = \underline{g}(x)$ needs the inverse bound at the finite-amplitude approximate solution, not the anchor value $\beta_0 = \frac{1}{2}$ of Lemma 4. Using β_0 as an approximation, the Kantorovich product $h = K\beta_0^2\eta$ (with $\eta = \|F\|_\infty$ the converged residual and $K = \|F''\|$ the second-variation modulus) stays well below the $\frac{1}{2}$ threshold across $x \in [0.1, 0.7]$ ($h \leq 1.3 \times 10^{-5}$ for $x \leq 0.3$, $h \approx 6 \times 10^{-2}$ at $x = 0.5, 0.7$), the geometric decay of η in the multipole order doing the work. This is a convergence estimate, evidence that the multipole iteration contracts toward a solution at that amplitude, not a finite-amplitude existence theorem (which needs the finite-amplitude inverse bound and interval bounds on η, K , left open). Accordingly $\underline{g}(x)$ is a *numerical* lower bound: sharper than, but logically independent of, the *rigorous* lower bound $c(x)$ of Lemma 6. The analytic bracket $c(x) \leq g(x) \leq \frac{1}{2}(1-x)$ is rigorous at both ends (Lemma 6, Prop. 5); the numerical $\underline{g}(x) \in [c(x), g(x)]$ merely locates g inside it. As an internal consistency check, the shape-response modulus C_{shape} of Lemma 6, back-solved from the located margin zeros (S1), agrees to within a few percent with the value implied by the converged worst-observer margins, so $\underline{g}(x)$ sits self-consistently inside the analytic band.

A worked finite-duration maneuver

The dominant energy condition is a pointwise optimization over all observers. At the static anchor and along the symmetry axis this worst-observer margin is closed-form (Lemma 10, Eq. (62)); only

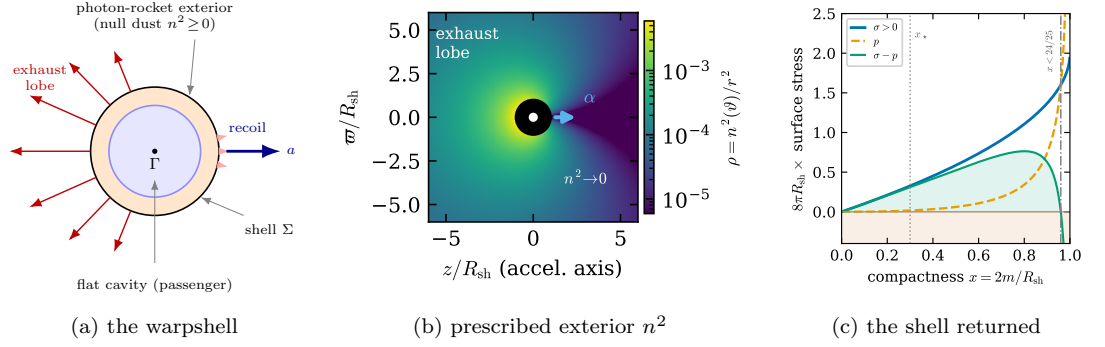


Figure S1. The worldtube-first warpsell: the schematic, the exterior it prescribes, and the shell the field equations return. (a) Schematic cross-section: a flat passenger cavity $\mathcal{M}_-$ is matched across a timelike shell Σ to the exact photon-rocket exterior $\mathcal{M}_+$, whose source is outgoing null dust with $n^2 \geq 0$; the exhaust is a forward-suppressed monopole-plus-dipole pattern whose dipole moment recoils the shell, giving the central worldline Γ a covariant acceleration a paid for by Bondi mass loss $-\dot{m}$, with no exotic matter. The design order is inverted: Γ and the admissible exterior matter are prescribed, and the Einstein equations *return* Σ . (b) The prescribed exterior null-dust density $\rho = n^2(\vartheta)/r^2$ at peak acceleration (a single rear dipole lobe trailing the drive, vanishing at the forward pole, with the cavity ($r < R$) field-free) is the *input*. (c) The shell the field equations then return is the *output*: ordinary matter, with a strictly positive surface energy density $\sigma(x) > 0$ and worst-observer surface dominant-energy margin $\sigma - p > 0$ out to the compactness window $x < 24/25$ ($8\pi R$ units; a representative operating point is marked).

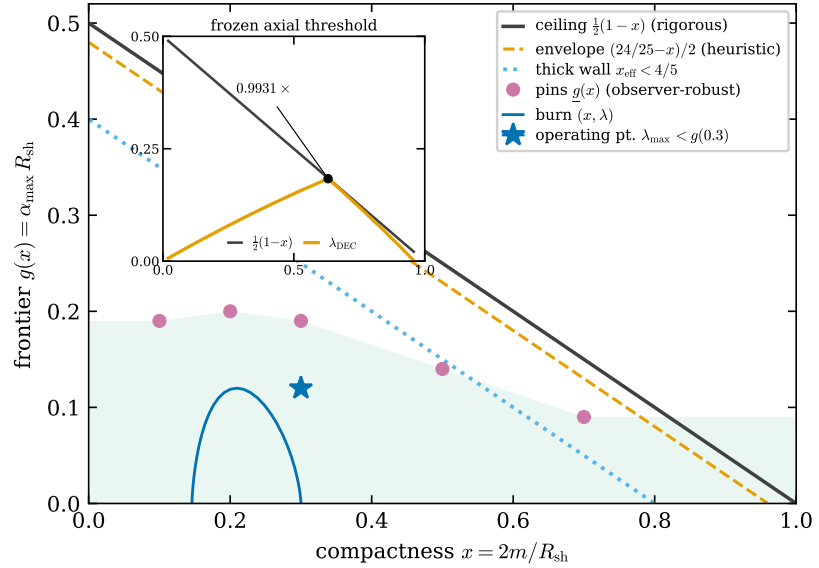


Figure S2. The acceleration-compactness frontier $g(x) = a_{\max} R$ versus compactness $x = 2m/R$, read by authority level: the kinematic ceiling $\frac{1}{2}(1-x)$ (Eq. (40), solid) is a rigorous, shape-independent *upper* bound; the computed values $\underline{g}(x)$ are observer-robust *lower* bounds at the sampled compactnesses (the faint shading merely interpolates them as a guide); the envelope $\frac{1}{2}(\frac{24}{25}-x)$ (Eq. (39), dashed) is heuristic; and the thick-wall realizability bound $\lambda < \frac{1}{2}(\frac{4}{5}-x)$ (dotted, $x_{\text{eff}} = x + 2\lambda < 4/5$) is the binding finite-thickness constraint. The saturating burn of Theorem 3 stays strictly inside every window at a representative operating point. The inset plots the frozen-shape axial threshold $\lambda_{\text{DEC}}(x)$ against the same ceiling: a distinct diagnostic from the matched pins (the two are not globally ordered), proved to lie strictly below the ceiling yet reaching 0.9931 of it near $x \approx 0.63$ (Lemma 5).

the finite-amplitude, all-angle margin of the deformed shell lacks a closed form, and there we evaluate it numerically over all timelike and null observers [24, 82], on the same footing as the boundary-cost result of Ref. [23]. The bulk (Sec. 5) and the static thin shell (Sec. 7.1) pass; a *generic* smoothstep Kerr–Schild thickening instead fails the pointwise DEC in the radial gradients of the transition, reproducing the boundary cost of Ref. [23] (Fig. S4), so the positive content rests on the distributional thin shell (Lemma 3, Theorem 2) and the carefully modeled tangential-pressure matter of Sec. 10, not an arbitrary profile.

All numerical curvature, energy-condition, and junction computations reported in this paper are carried out with WARPAX [24], a separate observer-robust energy-condition framework (following the approach of WarpFactory [82]), used here solely to study the present theoretical construction numerically; we claim no part of that framework as a contribution of this work. Its configurations and tolerances accompany the implementation referenced in the Data Availability statement.

We implemented a finite-thickness radiative warpsheet as a Kerr–Schild metric (57) with a smoothed mass profile $M(r, u) = m(u) S((r - (R - \delta/2))/\delta)$, S a smoothstep (flat cavity inside, $M = 0$; exact photon rocket outside, $M = m(u)$; a regularized radiating wall between) and scanned it with that observer-robust framework. The bulk behaves as designed (flat interior; exact null-dust exterior under steering). However, the *pointwise* dominant energy condition is *violated* in the radial gradients of the wall, worst at the cavity→matter inner edge, already in the static limit $\lambda = 0$ and at small x ; only the exact wall midpoint is marginally admissible (where the Kerr–Schild profile gives $p_r = -\sigma$, saturating DEC). This is precisely the *boundary cost* of Ref. [23], here reproduced in the radiative setting: a *generic smooth thickening does not inherit the admissibility of the distributional thin shell*. The Kerr–Schild radial-mass profile produces radial tension rather than the thin shell’s tangential pressure, so this demonstrates that one natural smooth realization fails, not a universal no-go. The u -dependent (generalized-Vaidya) version of this ramp does *not* produce a rotating-energy Type IV stress: its outgoing-flux block is a defective Hawking–Ellis *Type II*, DEC-saturating like the exterior null dust, and a numerical Type-IV label is the eigenvalue-splitting artifact of that defective null block (App. A). The strict obstruction is thus the *static* tangential-tension channel, not a dynamical Type IV. Section 10 shows the complementary positive result: the *tangential-pressure* wall (the right matter model) is strictly DEC-admissible across its whole width and recovers the anchor for $2m/R < 4/5$.

This wall is moreover a *self-consistent* anisotropic equilibrium, not merely a postulated stress: solving the constraint equations numerically for $(\rho, p_r=0)$ and reading back the sourced Einstein tensor reproduces the assigned $(\rho, 0, p_t)$ to $\sim 10^{-4}$, the anisotropic (Bowers–Liang) hydrostatic-equilibrium residual vanishes to $\sim 10^{-8}$, the stress is Type I with worst-observer DEC margin > 0 , and the surface reduction recovers (σ_0, p_0) to $\lesssim 1\%$ as $\ell \rightarrow 0$. This removes the “postulated single fluid” caveat at the static level. A collisionless Einstein–Vlasov realization [72] of the $p_r=0$ wall exists only outside the photon sphere ($2m/r < 2/3$, circular timelike orbits); between $2/3$ and the elastic limit $4/5$ the appropriate microscopic matter is anisotropic *elastic* [73], realizing the threshold ordering $2/3 < 4/5 < 24/25$ of Sec. 10 (Fig. S4b) and not constructed here. The radiative graft’s fully Einstein-sourced $\mathcal{O}(\mu)$ back-reaction (radiation streaming *through* the wall) is treated in App. F.

Complete finite-duration maneuver (Theorem 3). We integrate the bump steering $a(u) = a_{\max} \sin^2[\pi(u - u_0)/(u_f - u_0)]$ off the static anchor $x_* = 0.3$ with the saturated budget $m(u) = m_0 e^{-3 \int |a|}$ and $\lambda_{\max} = 0.12 < g(0.3)$. At every retarded step it evaluates the bulk dust $n^2(\vartheta)$ and the gauge-invariant finite- λ Lanczos surface stress (App. C), establishing $n^2 \geq 0$, a strictly positive worst-observer surface DEC margin (Type I), and $\lambda < 1$ throughout, with the Bondi moment identities $\oint n^2 d\Omega = -\dot{m}$ and $-\oint n^2 \cos \vartheta d\Omega = \dot{m} a$ closing to $\sim 10^{-7}$; the burn radiates $\approx 51\%$ of the rest mass for a rapidity gain 0.24 (Fig. S3), the velocity payoff against an ideal photon rocket being the *cube* of its mass ratio, the broad-dipole $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$ collimation penalty rather than a relativistic floor.

Linear stability of the anchor

We analyze the radial stability of the static Schwarzschild–Minkowski shell by the Poisson–Visser method [83, 84]; the radiating exterior has no autonomous potential, so acceleration enters only as a frozen-background redshift, closed by a timescale estimate. Promoting $R \rightarrow R(\tau)$, the junction reduces to $\dot{R}^2 + V(R) = 0$ with $V(R_0) = V'(R_0) = 0$ (the anchor is an equilibrium of the family $\sigma_0(R), p_0(R)$); stability requires $V''(R_0) > 0$, and V'' is affine in the equation-of-state slope $\eta \equiv dp/d\sigma|_0$. This result is exact: the *realized* tangential-pressure wall has $\eta_{\text{real}}(x) = (dp_0/dR)/(d\sigma_0/dR)$ equal to the marginal value η_{crit} at which $V''(R_0) = 0$, for all x : its

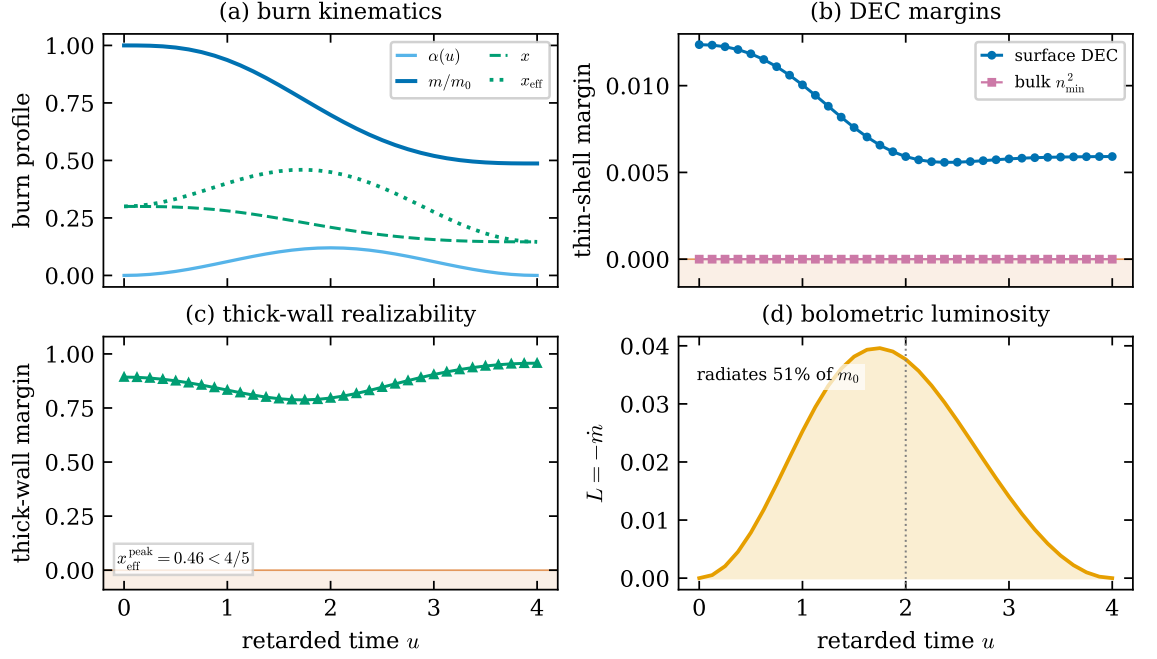


Figure S3. The complete finite-duration maneuver (Theorem 3) for one saturating burn: admissible three ways at every retarded time, at the mandatory cost of a radiated exhaust. (a) the kinematics $a(u)$, $m(u)/m_0$, $x(u) = 2m/R$, and rear-pole effective compactness $x_{\text{eff}} = x + 2\lambda$; (b) the worst-observer thin-shell surface DEC margin (Type I) and the bulk minimum n_{min}^2 , both above the shaded violating half-plane; (c) thick-wall realizability, the static tangential-pressure-wall margin at $x_{\text{eff}}(u)$ staying strictly positive ($\Leftrightarrow x_{\text{eff}} < 4/5$, Sec. 10.3); (d) the bolometric luminosity $L(u) = -\dot{m}$, the burn radiating $\approx 51\%$ of the rest mass to steer. The angular emission pattern $4\pi n^2(\vartheta) = -\dot{m} - 3ma \cos \vartheta$ is the minimum-radiation Damour dipole (Theorem 5).

equation of state is the tangent to the static equilibrium family, so a radial displacement is a zero mode (neutral stability). Stiffer walls ($\eta > \eta_{\text{crit}}$) sit in a strict potential well, softer walls on a barrier; the static anchor stress matches App. B and an independent numerical Israel solve to $\lesssim 5 \times 10^{-3}$. The frozen-background rear-pole redshift $x \rightarrow x + 2\lambda$ raises η_{crit} monotonically with $\lambda = aR$, so under acceleration the (marginal) realized wall moves onto the barrier side, $\eta_{\text{real}} < \eta_{\text{crit}}(\lambda)$. Its growth time follows from the first integral: $\dot{R}^2 = -V$ gives $\ddot{R} = -\frac{1}{2}V'(R)$, so a barrier ($V'' < 0$) e-folds at rate $\sqrt{|V''|/2}$, i.e. $\tau_{\text{efold}} = \sqrt{2/|V''|}$ with *no* factor 2π (the corresponding stable-well oscillation *period* is $2\pi\sqrt{2/|V''|} = 2\pi\tau_{\text{efold}}$, not the barrier e-fold). Evaluated at the realized accelerated wall, the e-fold is short: $\tau_{\text{efold}} \approx \{0.19, 0.36, 0.51, 0.78, 1.04\}$ at $x = \{0.1, 0.2, 0.3, 0.5, 0.7\}$ for $\lambda = 0.1$, in units of the light-crossing time. This τ_{efold} is measured in the *shell's* proper time, the first integral $\dot{R}^2 = -V$ being the Israel radial equation, while $\tau_{\text{burn}} = \Delta\eta/a$ is measured on Γ ; the cavity is flat with Γ at its center, so the two clocks differ only by the interior Rindler lapse $N = 1 + az$, that is at $\mathcal{O}(\lambda)$, and the comparison below is a leading-order one. The Tsiolkovsky burn is therefore faster than this static-anchor instability *only in a restricted corner*: at high compactness with a brief rapidity gain (e.g. $x = 0.7$, $\lambda = 0.1$, $\Delta\eta = 0.1$, admissible under the ceiling $\lambda < \frac{1}{2}(1 - x) = 0.15$: $\tau_{\text{burn}} \approx 1.0 < \tau_{\text{efold}} \approx 1.04$), whereas for $x \lesssim 0.5$ even the briskest burn has $\tau_{\text{burn}} \gtrsim \tau_{\text{efold}}$, so the marginal realized wall does not by itself guarantee that the burn completes before the (frozen-background) mode develops. Moreover $\tau_{\text{burn}} = \Delta\eta/a$ is the square-pulse *lower bound* on the duration; the $\sin^2$ burn of Theorem 3 ($\langle \sin^2 \rangle = \frac{1}{2}$) lasts $\sim 2\times$ longer, so even the high- x corner is not guaranteed safe without active stabilization. Two facts contain this. First, the realized mode is *neutral* at $\lambda = 0$ (zero growth rate): a static wall drifts only secularly, not exponentially. Second, a slightly *stiffer* wall, $\eta > \eta_{\text{crit}}$, sits in a strict potential well and is linearly stable, and this costs *nothing* in admissibility, because the surface dominant-energy margin $\sigma_0 - p_0$ is junction-fixed (a function of x alone) and *independent* of η , so the stiffer wall carries the identical strictly-positive margin. The required stiffening is moreover causal: the marginal slope $\eta_{\text{crit}}(x)$ (98) is closed-form and stays subluminal ($\eta < 1$) for every $x < x_\eta \approx 0.844$ (App. I), covering the whole mapped operating range; only in the sliver $x_\eta < x < 24/25$, still inside the surface-DEC window, would strict stability demand a superluminal equation of state. At the maneuver rear pole $x_{\text{eff}} \approx 0.46$ both the surface margin and the thick-wall across-wall factor $1 - m/[2(r - 2m)] \approx 0.79$ are bounded away

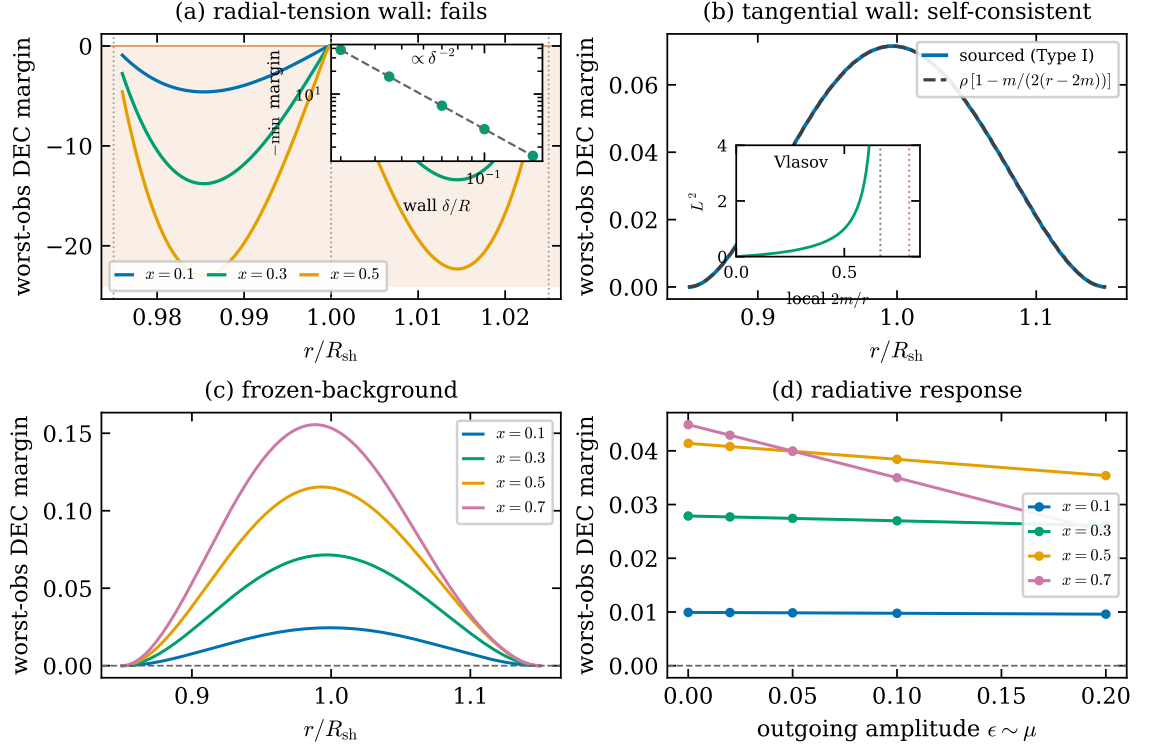


Figure S4. The wall dominant-energy conditions: the generic thickening fails, the engineered one holds. (a) *fails*: a generic smooth *radial-tension* thickening violates the pointwise DEC: the worst-observer margin across a smoothstep Kerr–Schild wall goes negative (shaded), worst at the cavity→matter inner edge; the inset shows the violation is a boundary cost that diverges as the wall sharpens, $\propto \delta^{-2}$ (Eq. (52)). The engineered *tangential-pressure* wall instead holds three ways. (b) *static* DEC: the metric-sourced margin (Type I) *coincides* with the closed-form single-fluid margin $\rho[1 - m/(2(r - 2m))]$: an equilibrium, not a postulated stress (inset: local compactness against the photon-sphere bound $2/3$ and the elastic bound $4/5$). (c) *frozen positive*: the across-wall margin is strictly positive and grows with x . (d) *back-reaction tightens*: with the metric free to respond, the curvature-derived margin versus the outgoing amplitude $\epsilon \sim \mu$ stays positive but the flux *tightens* the condition (slope < 0), reversing the frozen-background sign while preserving admissibility.

from zero, so a small stiffening to strict stability keeps strict DEC. Strict stability and strict dominant energy are thus decoupled, with no trade-off. The flux-coupled stability of the *radiating* shell (App. I) is a non-autonomous Poisson–Visser oscillator whose outgoing mass-loss $L = -\dot{m}$ *anti-damps* the radial mode ($\kappa_d < 0$ on the oscillatory branch (97)), so the radiating equilibrium is linearly unstable; away from the marginal wall the $\mathcal{O}(L)$ growth is finite-burn-bounded by the fuel budget (14), and Proposition 7 states this in the body. The interior is exactly flat: the tidal eigenvalues vanish, $K^\mu{}_\rho \equiv 0$ for $\lambda \in [0, 1)$, the Rindler lapse $1 - \lambda$ degenerating only as $\lambda \rightarrow 1$ (the Born-rigid Rindler-horizon bound [90, 91]).

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